

Information Theory and Inference

From Shannon entropy to Hamiltonian Monte Carlo — Lecture Notes

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1. About these notes

1.1. Scope

These are lecture notes for *Information Theory and Inference* (third-semester elective, Master in Physics of Data, University of Padova, instructor Prof. Michele Allegra). The course starts from Shannon's operational definition of information and ends with modern Markov-Chain Monte Carlo methods for high-dimensional Bayesian inference.

The two halves of the course share a single organising object: the *probability distribution*. In Part I we ask how much information such a distribution contains, and how many bits are needed to communicate outcomes over a noisy channel. In Part II we ask, given a small amount of data, how to *sample* from a posterior distribution we can evaluate only up to an unknown normalisation constant.

1.2. Reading the notes

Chapters 1–5 build up Shannon's toolbox: surprisal, entropy, KL divergence, mutual information, the two coding theorems (source and noisy-channel), and their continuous counterparts. Chapter 6 extends information to time series and introduces the Markov-chain theory that powers Part II. Chapters 7–8 develop Monte Carlo sampling: Metropolis, convergence and autocorrelation, Gibbs sampling with latent variables. Chapter 9 applies them to a real high-dimensional problem (the origin of grey-seal founders). Chapters 10–12 cover the alternatives and extensions: variational Bayes (and the variational autoencoder), Hamiltonian Monte Carlo, and approximate Bayesian computation for models whose likelihood can only be simulated.

Every chapter uses the styled callout environments (definition, theorem, proposition, lemma, corollary, example, remark, note, exercise, solution) so that key results can be scanned without reading the surrounding prose.

1.3. Notation

- X, Y, Z : random variables; x, y, z realisations; $\mathcal{X}$ their alphabet.
- $p_X(x) = P(X = x)$, $p(x, y)$: probability mass or density; $p(x | y)$ conditional.
- $H(X)$, $H(X, Y)$, $H(X | Y)$: entropy, joint entropy, conditional entropy.
- $I(X; Y)$: mutual information; $D(p \| q)$: Kullback–Leibler divergence.
- C : channel capacity; R : code rate.
- $\pi(\mathbf{x})$: target distribution in MCMC; $T(\mathbf{x} \rightarrow \mathbf{x}')$: transition kernel; $q(\mathbf{x}' | \mathbf{x})$: proposal.

1.4. Sources

These notes are a re-organised version of *Notes on Information Theory and Inference* by Prof. Michele Allegra (University of Padova, 2025), with additional material from the two standard textbooks:

- Thomas M. Cover & Joy A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- David J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge, 2003.

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Chapters 7–8 also draw on Steve Brooks, Andrew Gelman, Galin Jones & Xiao-Li Meng (eds.), *Handbook of Markov Chain Monte Carlo*, CRC Press, 2011.

Every chapter closes with its own *Sources* section naming the specific sections used.

Part I.

Part I: Information

2. Defining information

Information theory begins with a puzzle: what does it mean, quantitatively, that one message is *more informative* than another? The outcome of a random variable X is, by definition, uncertain — when we toss a coin we do not know whether it will land heads or tails. We can think of the information associated with X either as the *uncertainty* we have before the outcome is known, or, equivalently, as the *knowledge we gain* once the outcome is revealed.

In this chapter we derive a measure of information from first principles, by requiring it to satisfy a few intuitively appealing properties. Shannon's answer — the information content of an event is the logarithm of the reciprocal of its probability — turns out to be the *only* answer compatible with those properties. Averaging it over all outcomes gives the first central quantity of the course, the entropy of a probability space.

2.1. Events and probability spaces

We start in the most abstract setting.

Definition 2.1 (Probability space). A **probability space** is a triple $S = (\Omega, \mathcal{F}, P)$ where

- Ω is the **sample space**, the set of elementary outcomes;
- $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a σ -algebra of **events**. An event E is a set of outcomes, $E \subseteq \Omega$; the family $\mathcal{F}$ must be closed under union, intersection and complement;
- $P : \mathcal{F} \rightarrow [0, 1]$ is a **probability**, with (i) $P(\Omega) = 1$ and (ii) σ -additivity: for any countable family of pairwise disjoint events $\{E_i\}_{i \in I}$, $E_i \cap E_j = \emptyset$ for $i \neq j$,

$$P\left(\bigcup_{i \in I} E_i\right) = \sum_{i \in I} P(E_i).$$

For most of what follows we work with a finite alphabet $\mathcal{X}$ and take $\Omega = \mathcal{X}$, $\mathcal{F} = 2^{\mathcal{X}}$ (the power set). The atomic events are singletons $\{x\}$ with probabilities $p_X(x) = P(\{X = x\})$.

Example 2.1 (*Two fair coins*)

Toss two independent fair coins. The sample space is

$$\Omega = \{hh, ht, th, tt\},$$

where hh means “heads in the first toss and heads in the second toss”. The set of events is the whole power set, $2^4 = 16$ events:

$$\begin{aligned} \mathcal{F} = \{ & \emptyset, \{hh\}, \{ht\}, \{th\}, \{tt\}, \{hh, ht\}, \{hh, th\}, \{hh, tt\}, \{ht, th\}, \{ht, tt\}, \{th, tt\}, \\ & \{hh, ht, th\}, \{hh, ht, tt\}, \{hh, th, tt\}, \{ht, th, tt\}, \{hh, ht, th, tt\} \}. \end{aligned}$$

Since all outcomes are equiprobable, $P(E) = |E|/4$: one-outcome events have probability $1/4$, two-outcome events $1/2$, three-outcome events $3/4$, and $P(\Omega) = 1$.

2. Defining information

2.1.1. Events as answers

A useful way to think about events — and the one that leads naturally to information — is the following: **every event is a possible answer to the question “What was the outcome?”**. Answers can be more or less specific depending on how many outcomes they include.

- The four singletons $\{hh\}, \{ht\}, \{th\}, \{tt\}$ fully specify the outcome (“the outcome was hh ”).
- The six two-outcome events are less detailed: $\{hh, ht\}$ means “the outcome was hh or ht ” — equivalently, “the first coin gave heads”.
- The four three-outcome events are even vaguer: $\{hh, ht, th\}$ means “the outcome was *not* tt ”; it only excludes one possibility.
- The event $\Omega = \{hh, ht, th, tt\}$ is the trivial answer — we already knew that one of these happened.

So $\mathcal{F}$ contains every possible answer, from the most detailed to the least detailed, and each answer has its own probability.

2.1.2. Conditional probability and independence

Disjoint events ($E_1 \cap E_2 = \emptyset$) are *mutually exclusive* answers; non-disjoint events are *possibly compatible* answers. Any event E_2 splits into a part compatible with E_1 and a part incompatible with it,

$$E_2 = (E_1 \cap E_2) \cup (\bar{E}_1 \cap E_2).$$

If we know that E_2 occurred, then either E_1 occurred too, or it did not. The relative weight of the first case defines the conditional probability.

Definition 2.2 (Conditional probability and independence). For $P(E_2) > 0$, the **conditional probability** of E_1 given E_2 is

$$P(E_1 | E_2) \equiv \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{P(E_1 \cap E_2)}{P(E_1 \cap E_2) + P(\bar{E}_1 \cap E_2)}.$$

Two events are **independent** if the probability of the joint event factorises,

$$P(E_1 \cap E_2) = P(E_1) P(E_2),$$

or equivalently $P(E_1 | E_2) = P(E_1)$ and $P(E_2 | E_1) = P(E_2)$.

In the “answers” language: **independent events are complementary answers** — knowing one does not make the other more or less likely.

Example 2.2 (Conditioning can raise or lower a probability)

In the two-coin space:

- $E_1 = \{hh, ht\}$ (“first toss gave h ”) and $E_2 = \{th, tt\}$ (“first toss gave t ”) are mutually exclusive: $E_1 \cap E_2 = \emptyset$.
- $E_1 = \{hh, ht, th\}$ (“not tt ”) and $E_2 = \{hh, ht, tt\}$ (“not th ”) intersect in $\{hh, ht\}$. Then

$$P(E_1 | E_2) = \frac{P(\{hh, ht\})}{P(\{hh, ht, tt\})} = \frac{1/2}{3/4} = \frac{2}{3} < P(E_1) = \frac{3}{4}.$$

Learning E_2 *decreases* the likelihood of E_1 .

- $E_1 = \{hh\}$ and $E_2 = \{hh, ht, tt\}$:

$$P(E_1 | E_2) = \frac{1/4}{3/4} = \frac{1}{3} > P(E_1) = \frac{1}{4}.$$

Learning E_2 *increases* the likelihood of E_1 .

- The pairs

$$(\{hh, ht\}, \{hh, th\}), (\{hh, ht\}, \{hh, tt\}), (\{hh, ht\}, \{ht, th\})$$

are all independent: e.g. for the second pair, “first toss gave h ” and “the two tosses agree”, $P(E_1 \cap E_2) = P(\{hh\}) = 1/4 = \frac{1}{2} \cdot \frac{1}{2}$. Knowing that the coins agree tells us nothing about the first coin. (In fact nearly all pairs of two-outcome events in this space are independent.)

Remark (When does conditioning increase a probability?). Write $\Delta \equiv P(E_1 \cap \bar{E}_2)$ and $\bar{\Delta} \equiv P(\bar{E}_1 \cap \bar{E}_2)$ for the probability that is “removed” from E_1 and from its complement when we learn E_2 . Then

$$P(E_1) = \frac{P(E_1)}{P(E_1) + P(\bar{E}_1)}, \quad P(E_1 | E_2) = \frac{P(E_1) - \Delta}{P(E_1) - \Delta + P(\bar{E}_1) - \bar{\Delta}}.$$

The two expressions have the same structure; conditioning simply subtracts the excluded probability. One checks that

$$P(E_1 | E_2) > P(E_1) \iff \frac{\Delta}{\bar{\Delta}} < \frac{P(E_1)}{P(\bar{E}_1)},$$

i.e. E_1 becomes more likely exactly when it loses *proportionally less* probability than its complement.

2.2. The surprisal

Each event specifies the outcome to a larger or lesser degree, i.e. provides more or less information about it. Let us make this quantitative by constructing a function

$$I : \mathcal{F} \rightarrow \mathbb{R}_+$$

that assigns to each event the information it provides about the outcome.

2.2.1. Requirements

Intuitively we should agree that:

1. Events with the same probability provide the same amount of information. Hence I depends on E only through its probability: $I(E) = f(P(E))$ for some unknown $f : [0, 1] \rightarrow \mathbb{R}_+$.
2. More probable events (more outcomes, vaguer answers) provide *less* information: f is **monotonically decreasing**. In particular $I(\Omega) = f(1)$ should be zero — the trivial answer tells us nothing.
3. **Additivity for independent events.** Independent events are complementary answers, so the information of the combined answer $E = E_1 \cap E_2$ should be the sum of the two pieces: $I(E_1 \cap E_2) = I(E_1) + I(E_2)$. Since $P(E_1 \cap E_2) = P(E_1)P(E_2)$, this means

$$f(x_1 x_2) = f(x_1) + f(x_2).$$

2. Defining information

Example 2.3 (Additivity in the two-coin space)

We could first say “the first coin gave h ” ($E_1 = \{hh, ht\}$) and then “the second coin gave h ” ($E_2 = \{hh, th\}$). These events are independent, and together they say exactly “the outcome was hh ” ($E_1 \cap E_2 = \{hh\}$). It is natural to demand that the information in the single answer $\{hh\}$ equal the sum of the information in the two partial answers.

2.2.2. The logarithm is forced

Theorem 2.1 (Characterisation of the surprisal). *A monotonically decreasing function $f : (0, 1] \rightarrow \mathbb{R}_+$ satisfying $f(x_1 x_2) = f(x_1) + f(x_2)$ must be of the form*

$$f(x) = -\alpha \log x, \quad \alpha > 0.$$

Proof. Set $x = e^y$ and $g(y) = f(e^y)$. The functional equation becomes

$$f(e^{y_1} e^{y_2}) = f(e^{y_1 + y_2}) = f(e^{y_1}) + f(e^{y_2}) \implies g(y_1 + y_2) = g(y_1) + g(y_2).$$

This is Cauchy’s functional equation; its only monotone solutions are linear, $g(y) = \alpha' y$. Hence $f(e^y) = \alpha' y$, i.e. $f(x) = \alpha' \log x$. Monotonic decrease requires $\alpha' < 0$; write $\alpha' = -\alpha$ with $\alpha > 0$. $\square$

Definition 2.3 (Surprisal). The **surprisal** (or self-information) of an event E is

$$I(E) = -\log_2 P(E) \quad \text{bits.}$$

For a random variable X taking value x we write $I(x) = -\log_2 p_X(x)$.

The name reflects that low-probability events are more “surprising”. It is worth stressing what the surprisal measures: **the amount of information we obtain when we learn that the outcome of a random process lies in a given subset E of all possible outcomes.**

Remark (Units). The constant α is arbitrary and only fixes the unit. Choosing α is the same as choosing a base $\beta = e^{1/\alpha}$ for the logarithm:

$$-\alpha \log P = -\frac{\log P}{\log \beta} = -\log_\beta P.$$

With $\beta = 2$ we measure information in **bits** (binary digits); with $\beta = e$ in **nats**; with $\beta = 10$ in hartleys. Since $\ln 2 \approx 0.693$, 1 bit ≈ 0.693 nat. From now on $\log \equiv \log_2$ unless stated otherwise.

The bit has a simple interpretation: **it is the information revealed when we learn the outcome of a process with two equiprobable outcomes**, such as the toss of a fair coin.

Example 2.4 (Coin toss)

For a fair coin each outcome carries $\log_2 2 = 1$ bit. For a biased coin with $p(H) = 0.99$, seeing heads carries only $-\log_2 0.99 \approx 0.014$ bits — the outcome was almost certain. Seeing tails carries $-\log_2 0.01 \approx 6.6$ bits — a much more informative event.

Example 2.5 (*Surprisal in the two-coin space*)

$I(\{hh\}) = -\log_2(1/4) = 2$ bits; $I(\{hh, ht\}) = 1$ bit; $I(\{hh, ht, th\}) = -\log_2(3/4) \approx 0.415$ bits; $I(\Omega) = 0$. The more specific the answer, the more bits it carries; full specification of two coins costs two bits, one per coin.

2.2.3. Conditional surprisal: redundant and synergistic answers

What if we specify the outcome through answers that are *not* independent? If we already know that the outcome is in E_1 , the natural information carried by E_2 is

$$I(E_2 | E_1) = -\log P(E_2 | E_1).$$

Since $P(E_1 \cap E_2) = P(E_1)P(E_2 | E_1)$, taking logarithms gives the **chain rule for surprisal**

$$I(E_1 \cap E_2) = I(E_1) + I(E_2 | E_1).$$

In general $I(E_2 | E_1)$ can be larger or smaller than $I(E_2)$. Correspondingly the information provided jointly by two non-independent events can be **less** (redundancy) or **more** (synergy) than the sum of their separate informations.

Example 2.6 (*Redundant answers*)

The event $E = \{hh\}$ can be obtained as the intersection of the non-independent events $E_1 = \{hh, ht, th\}$ (“at least one h ”), $E_2 = \{hh, th\}$ (“heads in the second toss”), $E_3 = \{hh\}$. There is redundancy among these answers:

$$P(E_2 | E_1) = \frac{2}{3} > P(E_2) = \frac{1}{2}, \quad P(E_3 | E_2) = \frac{1}{2} > P(E_3) = \frac{1}{4}.$$

The separate surprisals are

$$I(E_1) = -\log \frac{3}{4} \approx 0.42, \quad I(E_2) = 1, \quad I(E_3) = 2,$$

so naïvely summing gives $3.42 > I(E) = 2$ bits: we have “overcounted” the information. Using conditional surprisals instead,

$$I(E) = I(E_1) + I(E_2 | E_1) + I(E_3 | E_2) = -\log\left(\frac{3}{4} \cdot \frac{2}{3} \cdot \frac{1}{2}\right) = -\log \frac{1}{4} = 2 \text{ bits.} \checkmark$$

Example 2.7 (*Synergistic answers*)

Take $E = \{hh, ht\}$ as the intersection of $E_1 = \{hh, ht, th\}$ and $E_2 = \{hh, ht, tt\}$. Then

$$P(E) = \frac{1}{2} = \frac{8}{16} < P(E_1)P(E_2) = \frac{9}{16},$$

and therefore

$$I(E) = 1 > I(E_1) + I(E_2) = 2 \times 0.415 = 0.83 \text{ bits.}$$

Each answer alone only excludes one outcome, but together they exclude two: the combination is worth more than the parts.

Lemma 2.2 (Chain rule for surprisal). *For any sequence of random variables $X_1, X_2, \dots, X_n$,*

$$I(x_1, x_2, \dots, x_n) = \sum_{i=1}^n I(x_i \mid x_1, \dots, x_{i-1}),$$

where $I(x_i \mid \cdot) = -\log_2 p(x_i \mid \cdot)$ is the conditional surprisal.

This follows from the chain rule of probability $p(x_1, \dots, x_n) = \prod_i p(x_i \mid x_1, \dots, x_{i-1})$ and the additivity of the logarithm.

2.3. Probability trees

Possible answers to “What was the outcome?” can be arranged in a **probability tree**:

- each node is a possible answer, i.e. an event $E \in \mathcal{F}$, carrying probability $P(E)$; the root is Ω ;
- each branching splits an answer E into more detailed, mutually exclusive answers $E_1, \dots, E_k$ with $\bigcup_k E_k = E$; the k -th edge carries the conditional probability $P(E_k \mid E)$;
- at each level, the nodes form a partition of Ω .

Conceptually, **each branching is a sub-question** and each edge an answer to it. The probability of a node is the product of the conditional probabilities along the path from the root; correspondingly, since $I = -\log P$, **the information of a node is the sum of the informations along the edges** leading to it. The same probability space can be split in many different ways, giving different trees.

Example 2.8 (Two trees for two coins)

Tree A (one coin at a time):

```

Omega --1/2--> {hh,ht}  "first coin h?"  --1/2--> {hh}
      |                                     +-1/2--> {ht}
      +---1/2--> {th,tt}  --1/2--> {th}
                        +-1/2--> {tt}

```

Each edge carries 1 bit; each leaf is reached after two edges, i.e. 2 bits, consistent with $I(\{hh\}) = 2$.

Tree B (an unbalanced split): first ask “was it tt ?”

```

Omega --1/4--> {tt}
      +---3/4--> {hh,ht,th}  --2/3--> {hh,ht}  --1/2--> {hh}
                        |                                     +-1/2--> {ht}
                        +-1/3--> {th}

```

Now the leaf $\{hh\}$ is reached through edges carrying $-\log \frac{3}{4} - \log \frac{2}{3} - \log \frac{1}{2} = 2$ bits — the same total, distributed differently along the path. The leaf $\{tt\}$ is reached in one step, also with $-\log \frac{1}{4} = 2$ bits.

2.4. Information for a probability space

The surprisal $I(E)$ measures how much information event E gives about the outcome. We now ask: **how much information is needed, on average, to completely specify the outcome?** This will be the information content of the probability space as a whole.

Example 2.9 (*Urn with three colours*)

An urn contains red, blue and green balls in proportions 50%, 25%, 25%. We draw one ball. To identify its colour we may first ask “is it red?” ($E_1 = \{r\}$ vs $\bar{E}_1 = \{b, g\}$, each answer worth 1 bit). If not red, we ask “is it blue?” — a second bit. Thus red is identified with 1 bit, blue and green with 2 bits. On average

$$\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 2 = 1.5 \text{ bits,}$$

which is the *average number of yes/no questions* needed to identify the outcome.

Example 2.10 (*Urn with five colours*)

Now red, blue, green, yellow, purple in proportions 50%, 12.5%, 12.5%, 12.5%, 12.5%. Ask “red?” (1 bit); if not, “blue or green?” (2nd bit); then “blue?” or “yellow?” (3rd bit). Red costs 1 question, every other colour 3. On average

$$\frac{1}{2} \cdot 1 + 4 \cdot \frac{1}{8} \cdot 3 = 2 \text{ bits.}$$

Note that each question was designed to split the remaining probability *in half*, so every answer was worth exactly one bit.

Formalising the argument: for each outcome $a \in \Omega$ of a discrete space there is an elementary event $\{a\}$ with information $-\log P(\{a\})$. The average information needed to specify the outcome is

$$H(S) = \langle I \rangle = - \sum_{a \in \Omega} P(\{a\}) \log P(\{a\}),$$

a single number for the whole space $S = (\Omega, \mathcal{F}, P)$. Introducing a random variable $X : \Omega \rightarrow \{x_1, \dots, x_N\}$ with $p_X(x_i) = P(\{a\})$ when $X(a) = x_i$ (in the urn, $r \mapsto x_1, b \mapsto x_2, g \mapsto x_3$):

Definition 2.4 (Shannon entropy). The **Shannon entropy** of a discrete random variable X with distribution p_X is

$$H(X) = - \sum_{x \in \mathcal{X}} p_X(x) \log_2 p_X(x) = \mathbb{E}[I(X)].$$

By convention $0 \log 0 = 0$.

The entropy is the *expected* number of bits carried by a random outcome — the average information needed to specify it. This operational reading will be made rigorous by the source-coding theorem in Chapter 3.

Example 2.11 (*Entropy of Bernoulli*)

For $X \sim \text{Bern}(p)$, $H(X) = -p \log_2 p - (1-p) \log_2 (1-p) \equiv h_2(p)$. The **binary entropy function** h_2 is symmetric around $p = 1/2$, where it takes its maximum value $h_2(1/2) = 1$ bit. It vanishes at $p = 0$ or $p = 1$ — no information from a certain outcome. A few values worth remembering: $h_2(0.1) \approx 0.469$, $h_2(0.11) \approx 0.5$, $h_2(0.25) \approx 0.811$.

Example 2.12 (*Four-outcome distribution (Cover and Thomas)*)

Let X take values a, b, c, d with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. Then

$$H(X) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75 \text{ bits.}$$

An efficient questioning strategy is “Is $X = a$?”, then “Is $X = b$?”, then “Is $X = c$?”; the expected number of questions is exactly 1.75. We will see in Chapter 3 that no strategy can use fewer than

$H(X)$ questions on average.

Example 2.13 (*The game of sixty-three (MacKay)*)

An integer x is chosen uniformly in $\{0, 1, \dots, 63\}$. Ask: “is $x \geq 32$?”, “is $x \bmod 32 \geq 16$?”, “is $x \bmod 16 \geq 8$?”, “is $x \bmod 8 \geq 4$?”, “is $x \bmod 4 \geq 2$?”, “is $x \bmod 2 = 1$?”. Each question halves the set of candidates and therefore yields exactly one bit; after six questions x is known, consistent with $H = \log_2 64 = 6$ bits. Writing the answers as 1 = yes, 0 = no gives precisely the binary expansion of x — the bits of information *are* the binary digits.

Example 2.14 (*Uniform distribution maximises entropy*)

For a variable X taking M values, the entropy is maximised when p_X is uniform, in which case $H(X) = \log_2 M$. Non-uniform distributions give strictly less entropy — a preview of the general result proved in Chapter 2.

2.5. An axiomatic derivation of entropy

Had we not derived the surprisal first, we could have obtained Shannon’s formula directly from requirements on H itself, following an argument reported by Bialek. We look for a functional $H[X]$ measuring the information content of the distribution of X .

Step 1: equiprobable outcomes. If $p_X(x_i) = 1/N$ for all i , then H should be a monotonically increasing function of the number of outcomes: $H[X] = g(N)$.

Step 2: composition. Let $N = k^m$. We can draw a tree with m levels and k equiprobable branches at each level, and map $X \mapsto (X_1, \dots, X_m)$ where each X_j is an independent variable with k equally likely outcomes — each outcome of X is identified by answering m independent sub-questions. (Two coins: $k = 2$, $m = 2$.) The total information must be the sum of the parts:

$$g(k^m) = \sum_{j=1}^m H[X_j] = m g(k).$$

Lemma 2.3. *The only monotonic function satisfying $g(k^m) = m g(k)$ for all integers k, m is $g(x) = \alpha \log x$.*

Proof. Fix integers $k, \ell \geq 2$ and $n \geq 1$, and choose m such that $k^m \leq \ell^n < k^{m+1}$, i.e. $m = \lfloor n \log \ell / \log k \rfloor$. (E.g. $k = 4$, $\ell = 3$, $n = 3$: $m = \lfloor 2.37 \rfloor = 2$.) Taking logarithms and dividing by $n \log k$:

$$\frac{m}{n} \leq \frac{\log \ell}{\log k} < \frac{m}{n} + \frac{1}{n} \quad \Rightarrow \quad \left| \frac{\log \ell}{\log k} - \frac{m}{n} \right| \leq \frac{1}{n}.$$

By monotonicity $g(k^m) \leq g(\ell^n) < g(k^{m+1})$, i.e. $m g(k) \leq n g(\ell) < (m+1)g(k)$; dividing by $n g(k)$,

$$\left| \frac{g(\ell)}{g(k)} - \frac{m}{n} \right| \leq \frac{1}{n}.$$

By the triangle inequality, $|g(\ell)/g(k) - \log \ell / \log k| \leq 2/n$ for every n . Letting $n \rightarrow \infty$ gives $g(\ell)/g(k) = \log \ell / \log k$ for all k, ℓ , i.e. $g(x) = \alpha \log x$. $\square$

Thus for N equiprobable outcomes $H[X] = \log N$ (choosing α so that we work in bits).

Step 3: non-uniform probabilities. Suppose the probabilities p_i are rational (any real can be approximated arbitrarily well). Write $p_i = r_i/s_i$ and let $M = \prod_l s_l$; then $p_i = k_i/M$ with $k_i = r_i \prod_{l \neq i} s_l \in \mathbb{N}$ and $\sum_i k_i = M$.

Imagine a variable Y with M equiprobable outcomes arranged into N groups, group i containing k_i outcomes. Let X be the group label (so $P(X = i) = k_i/M = p_i$) and Z_i the position within group i (uniform on k_i values). Identifying Y takes two sub-questions — first the group, then the element — so

$$H[Y] = H[X] + \sum_i p_i H[Z_i],$$

where the second term is averaged because the different branches are reached with different probabilities. Since $H[Y] = \log M$ and $H[Z_i] = \log k_i$,

$$\log M = H[X] + \sum_i p_i \log k_i \implies H[X] = - \sum_i p_i \log \frac{k_i}{M} = - \sum_i p_i \log p_i. \quad \blacksquare$$

Example 2.15 (*Step 3 in action*)

Take $p = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$. With $M = 6$: $k = (3, 2, 1)$. Then $H[Y] = \log 6 \approx 2.585$, and the within-group terms give $\frac{1}{2} \log 3 + \frac{1}{3} \log 2 + \frac{1}{6} \log 1 \approx 0.792 + 0.333 = 1.126$. Hence $H[X] \approx 2.585 - 1.126 = 1.459$ bits, which agrees with $-\frac{1}{2} \log \frac{1}{2} - \frac{1}{3} \log \frac{1}{3} - \frac{1}{6} \log \frac{1}{6} \approx 0.5 + 0.528 + 0.431 = 1.459$. ✓

The composition rule used in Step 3 is the **grouping property** of entropy, which we state in general below.

2.6. Elementary properties

The entropy is well-behaved as a function of the distribution (proofs in Chapter 2).

Proposition 2.4 (Basic properties of entropy). *For any discrete random variable X with distribution on $\mathcal{X}$:*

1. $H(X) \geq 0$, with equality iff X is deterministic.
2. $H(X) \leq \log_2 |\mathcal{X}|$, with equality iff X is uniform.
3. H is a concave function of p_X .
4. **Grouping property.** If two outcomes x_1, x_2 are merged into one with probability $p_1 + p_2$, then

$$H(X) = H(X_{\text{merged}}) + (p_1 + p_2) h_2(p_1/(p_1 + p_2)).$$

Corollary 2.5 (Entropy of a joint variable). *For a pair (X, Y) with joint distribution $p(x, y)$,*

$$H(X, Y) = - \sum_{x, y} p(x, y) \log_2 p(x, y) \leq H(X) + H(Y),$$

with equality iff X, Y are independent (sub-additivity, proved in Chapter 2).

2.7. Exercises

Exercise 2.1

A fair die is rolled. Compute (i) the surprisal of the event “the outcome is 6”, (ii) the entropy of the rolling process, (iii) the surprisal of the event “the outcome is even”.

Solution

- (i) $I(\{6\}) = -\log_2(1/6) = \log_2 6 \approx 2.585$ bits.
- (ii) $H(X) = \log_2 6 \approx 2.585$ bits (uniform on 6 outcomes).
- (iii) $I(\{2, 4, 6\}) = -\log_2(3/6) = 1$ bit — exactly one bit of information (equivalent to a coin toss).

Exercise 2.2

Show that for a Bernoulli(p) variable the entropy is maximised at $p = 1/2$ by direct differentiation. What is the second derivative at the maximum?

Solution

$H'(p) = -\log_2 p + \log_2(1-p) = \log_2((1-p)/p)$. Setting $H' = 0$ gives $p = 1/2$. The second derivative is $H''(p) = -[1/(p \ln 2) + 1/((1-p) \ln 2)]$, which at $p = 1/2$ equals $-4/\ln 2 \approx -5.77$. Negative H'' confirms concavity — a global maximum.

Exercise 2.3

Verify the grouping property for a three-outcome variable with probabilities $(1/2, 1/4, 1/4)$.

Solution

Direct: $H = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 1.5$ bits. Grouping the last two into a merged outcome of probability $1/2$: $H_{\text{merged}} = h_2(1/2) = 1$ bit, plus $1/2 \cdot h_2(1/2) = 0.5$ bits, total 1.5 bits. ✓

Exercise 2.4

In the two-coin space, check that $E_1 = \{hh, ht\}$ (“first coin h ”) and $E_2 = \{ht, th\}$ (“coins differ”) are independent, and compute $I(E_1)$, $I(E_2)$ and $I(E_1 \cap E_2)$.

Solution

$E_1 \cap E_2 = \{ht\}$, $P = 1/4 = P(E_1)P(E_2) = \frac{1}{2} \cdot \frac{1}{2}$: independent. $I(E_1) = I(E_2) = 1$ bit and $I(\{ht\}) = 2 = I(E_1) + I(E_2)$ bits, as additivity requires.

Exercise 2.5

Twelve balls look identical, but one is either heavier or lighter than the others. How much information is needed to identify the odd ball *and* whether it is heavy or light? A balance has three outcomes (left heavy, right heavy, balanced). What is the minimum number of weighings needed in principle?

Solution

There are $12 \times 2 = 24$ equiprobable hypotheses, so $\log_2 24 \approx 4.58$ bits are needed. A weighing has three outcomes and can deliver at most $\log_2 3 \approx 1.58$ bits. Hence at least $\lceil 4.58/1.58 \rceil = 3$ weighings are needed — and three indeed suffice if each weighing is designed so that its three outcomes are as close to equiprobable as possible (MacKay §4.1). The lesson: *good experiments maximise the entropy of their outcome.*

2.8. Callouts

i Note

Shannon's definition is *operational* — it derives its meaning from the coding theorems of Chapter 3. Purely axiomatic derivations (like the one above) show that the formula is *unique*, but do not tell you *why* it is useful.

💡 Tip

When computing entropies by hand, remember the identity $\log_2 n = \log_2 e \cdot \ln n \approx 1.4427 \ln n$: it lets you convert between nats and bits without a calculator.

⚠ Warning

The convention $0 \log 0 = 0$ is genuine, not a hack: $p \log(1/p) \rightarrow 0$ as $p \rightarrow 0^+$. But if you compute $I(x)$ for an event that has empirical probability zero from a finite sample, the value is $+\infty$, and downstream computations will be nonsense. Always regularise.

2.9. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 1 (events, surprisal, probability trees, Bialek's derivation). Cover & Thomas, §2.1 (entropy, Example 2.1.2). MacKay, §2.4 and §4.1 (the weighing problem, the game of sixty-three).

3. Information-theoretic measures

Entropy is only the first in a family of information measures. This chapter first proves the basic properties of the entropy, then introduces the three other quantities we will use constantly — the Kullback–Leibler divergence, the conditional entropy and the mutual information — and develops the identities and inequalities that connect them. Every result here will be used in the coding theorems of Chapters 3–4 and, later, in variational inference (Chapter 10).

3.1. Properties of Shannon entropy

3.1.1. Range

Proposition 3.1 (Range of the entropy). *Let X take values in an alphabet $\mathcal{A}_X$ with $|\mathcal{A}_X| = N$. Then*

$$0 \leq H[X] \leq \log N.$$

The minimum is attained iff one outcome has probability one ($\exists i : p_i = 1$); the maximum is attained iff all outcomes are equiprobable ($p_i = 1/N$ for all i).

Proof. Lower bound. $0 \leq p_i \leq 1$ implies $-p_i \log p_i \geq 0$, with equality only for $p_i \in \{0, 1\}$; so $H = 0$ iff a single outcome has probability one.

Upper bound. Maximise H under the constraint $\sum_i p_i = 1$ with a Lagrange multiplier (natural logs for convenience):

$$\mathcal{L}[\{p_i\}, \lambda] = -\sum_i p_i \ln p_i + \lambda \left(\sum_i p_i - 1 \right).$$

Stationarity gives

$$\frac{\partial \mathcal{L}}{\partial p_i} = -1 - \ln p_i + \lambda = 0 \Rightarrow p_i = e^{\lambda-1} \equiv \mu \quad \forall i, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow \sum_i p_i = 1.$$

All p_i are equal, and normalisation gives $\mu = 1/N$. Because H is concave (below), this stationary point is the global maximum, $H = \log N$. A second proof, via the KL divergence, is given below. $\square$

3.1.2. Concavity

Recall that f is **concave** if $f(\lambda x_1 + (1-\lambda)x_2) \geq \lambda f(x_1) + (1-\lambda)f(x_2)$ for all $\lambda \in [0, 1]$ (the chord lies below the graph), and **convex** if the inequality is reversed (the chord lies above).

Proposition 3.2 (Concavity of entropy). *The Shannon entropy is a concave function of the distribution: for any two distributions p, q on the same alphabet and $\lambda \in [0, 1]$,*

$$H(\lambda p + (1-\lambda)q) \geq \lambda H(p) + (1-\lambda)H(q),$$

with equality (for $0 < \lambda < 1$) iff $p = q$.

3. Information-theoretic measures

The mixture $\lambda p + (1 - \lambda)q$ has a concrete meaning. Let $X_1 \sim p$ and $X_2 \sim q$, and let $Z = X_1$ with probability λ and $Z = X_2$ with probability $1 - \lambda$. Then $p_Z = \lambda p + (1 - \lambda)q$, and concavity states that **mixing two distributions increases entropy**: not knowing *which* source produced the symbol adds uncertainty. The proof is given after we introduce the KL divergence.

Example 3.1 (Mixing deterministic variables)

Let $p = (1, 0)$ and $q = (0, 1)$: both X_1 and X_2 are deterministic, $H(p) = H(q) = 0$. Their mixture is $(\lambda, 1 - \lambda)$, with entropy $h_2(\lambda) > 0$ for every $0 < \lambda < 1$. All the uncertainty of Z comes from the coin that selects the source.

3.2. Kullback–Leibler divergence

The KL divergence (or **relative entropy**) measures how *different* a distribution p is from another distribution q .

Definition 3.1 (Kullback–Leibler divergence). For distributions p, q on a discrete alphabet $\mathcal{X}$ (write $p_i = p(x_i)$, $q_i = q(x_i)$),

$$D_{KL}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

By convention $0 \log(0/q) = 0$ and $p \log(p/0) = +\infty$ for $p > 0$.

Theorem 3.3 (Gibbs' inequality). For any two distributions p, q on the same alphabet, $D_{KL}(p\|q) \geq 0$, with equality iff $p = q$.

The proof (below) uses Jensen's inequality. Because of Gibbs' inequality, D_{KL} can be read as a measure of **discriminability** between two distributions — a kind of “distance” on the space of distributions.

Remark. The KL divergence is *not* a metric: it is not symmetric ($D(p\|q) \neq D(q\|p)$ in general) and it does not satisfy the triangle inequality. It does, however, satisfy $D(p\|q) = 0 \iff p = q$, and it is jointly convex (below).

3.2.1. Interpretation: the cost of a wrong model

Suppose X takes value x_i with probability p_i , but an observer (mistakenly) believes the probabilities are q_i . When the outcome x_i is revealed, the observer assigns it surprisal $-\log q_i$ — they think they have received $-\log q_i$ bits. Since x_i occurs with probability p_i , the average perceived information is the **cross entropy**

$$H(p, q) = - \sum_i p_i \log q_i.$$

The cross entropy is always at least the true entropy, and the excess is exactly the KL divergence:

$$D_{KL}(p\|q) = H(p, q) - H(p) = - \sum_i p_i \log q_i + \sum_i p_i \log p_i = \sum_i p_i (\log p_i - \log q_i) \geq 0.$$

The observer always overestimates the information obtained; the overestimate is $D_{KL}(p\|q)$. In Chapter 3 this becomes concrete: $D(p\|q)$ is the number of *extra bits per symbol* paid by a code designed for q when the source is really p .

Example 3.2 (*Wrongly assuming a fair coin*)

Let $p = (1 - \epsilon, \epsilon)$ with $\epsilon \ll 1$. The entropy is tiny,

$$H(p) = -\epsilon \log \epsilon - (1 - \epsilon) \log(1 - \epsilon) \approx \epsilon (\log(1/\epsilon) + \log e),$$

because the outcome is nearly always x_1 , which carries almost no information ($-\log(1 - \epsilon) \approx \epsilon \log e$). An observer who assumes $q = (\frac{1}{2}, \frac{1}{2})$ will nearly always see x_1 and believe they received one full bit:

$$H(p, q) = -(1 - \epsilon) \log \frac{1}{2} - \epsilon \log \frac{1}{2} = 1 \gg H(p).$$

Example 3.3 (*Wrongly assuming a biased coin*)

Conversely, let $p = (\frac{1}{2}, \frac{1}{2})$, so $H(p) = 1$, but the observer believes $q = (1 - \epsilon, \epsilon)$. Half of the time they see the “rare” outcome x_2 and think they received $-\log \epsilon \gg 1$ bits:

$$H(p, q) = -\frac{1}{2} \log(\epsilon(1 - \epsilon)) \approx \frac{1}{2} \log \frac{1}{\epsilon} \gg 1.$$

Here $D(p\|q) \approx \frac{1}{2} \log(1/\epsilon) - 1$ diverges as $\epsilon \rightarrow 0$, whereas in the previous example the divergence $D(p\|q) = 1 - H(p) \rightarrow 1$ stays bounded: a vivid illustration of the asymmetry of KL. Assigning near-zero probability to something that actually happens is punished very severely.

Example 3.4 (*KL of Bernoullis*)

For $\text{Bern}(a)$ vs $\text{Bern}(b)$,

$$D(a\|b) = a \log \frac{a}{b} + (1 - a) \log \frac{1 - a}{1 - b}.$$

When $b = 1/2$ this reduces to $1 - h_2(a)$, the “distance from uniform” of a biased coin. Numerically, $D(0.5\|0.25) \approx 0.208$ bits but $D(0.25\|0.5) \approx 0.189$ bits.

3.2.2. Entropy as distance from uniform

Let $u_i = 1/N$ be the uniform distribution on the same alphabet. Then

$$H(p) = - \sum_i p_i \log \left(\frac{p_i}{u_i} \cdot u_i \right) = -D_{KL}(p\|u) - \sum_i p_i \log \frac{1}{N} = \log N - D_{KL}(p\|u).$$

Combined with Gibbs’ inequality this gives a one-line proof of $H(p) \leq \log N$, with equality iff $p = u$. **The entropy is the maximal entropy minus how far the distribution is from uniform.**

3.2.3. Proofs

Lemma 3.4 (Jensen’s inequality). *If f is convex and X is a random variable, $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. If f is strictly convex, equality holds iff X is constant (almost surely). For concave f the inequality is reversed.*

Proof. For a discrete X with values x_i and probabilities p_i , peel off one term at a time. Define $p_i^{(1)} = p_i/(1 - p_1)$ for $i \neq 1$ (a normalised distribution). By convexity,

$$f\left(\sum_i p_i x_i\right) = f\left(p_1 x_1 + (1 - p_1) \sum_{i \neq 1} p_i^{(1)} x_i\right) \leq p_1 f(x_1) + (1 - p_1) f\left(\sum_{i \neq 1} p_i^{(1)} x_i\right).$$

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Repeating the step on the last term with $p_i^{(2)} = p_i/(1 - p_1 - p_2)$, and so on, after $N - 1$ steps we arrive at $f(\sum_i p_i x_i) \leq \sum_i p_i f(x_i)$. (Equivalently, a short induction on N .) $\square$

Proof of Gibbs' inequality. Let $Z = q(X)/p(X)$ with $X \sim p$, and apply Jensen to the convex function $-\log$:

$$D_{KL}(p\|q) = \sum_i p_i \left(-\log \frac{q_i}{p_i} \right) = \mathbb{E}[-\log Z] \geq -\log \mathbb{E}[Z] = -\log \sum_i p_i \frac{q_i}{p_i} = -\log \sum_i q_i = 0.$$

Since $-\log$ is strictly convex, equality requires q_i/p_i constant, i.e. $p = q$. $\square$

Lemma 3.5 (Log-sum inequality). *For non-negative a_i and positive b_i ,*

$$\sum_i a_i \log \frac{a_i}{b_i} \geq \left(\sum_i a_i \right) \log \frac{\sum_i a_i}{\sum_i b_i},$$

with equality iff a_i/b_i is constant across i .

Proof. $t \mapsto t \log t$ is strictly convex. Let $A = \sum a_i$, $B = \sum b_i$, weights $w_i = b_i/B$ and points $t_i = a_i/b_i$. Jensen gives $\sum_i w_i t_i \log t_i \geq (\sum_i w_i t_i) \log (\sum_i w_i t_i)$, i.e. $\frac{1}{B} \sum_i a_i \log \frac{a_i}{b_i} \geq \frac{A}{B} \log \frac{A}{B}$. Multiply by B . $\square$

The log-sum inequality is the workhorse behind almost every information-theoretic inequality that follows.

Proposition 3.6 (Joint convexity of KL). *$D(p\|q)$ is convex in the pair (p, q) : for $\lambda \in [0, 1]$,*

$$D(\lambda p_1 + (1 - \lambda)p_2 \parallel \lambda q_1 + (1 - \lambda)q_2) \leq \lambda D(p_1 \parallel q_1) + (1 - \lambda)D(p_2 \parallel q_2).$$

*In words, **mixing decreases discriminability**.*

Proof. For each i , apply the log-sum inequality to the two-term sums $a = (\lambda p_{1i}, (1 - \lambda)p_{2i})$, $b = (\lambda q_{1i}, (1 - \lambda)q_{2i})$:

$$(\lambda p_{1i} + (1 - \lambda)p_{2i}) \log \frac{\lambda p_{1i} + (1 - \lambda)p_{2i}}{\lambda q_{1i} + (1 - \lambda)q_{2i}} \leq \lambda p_{1i} \log \frac{\lambda p_{1i}}{\lambda q_{1i}} + (1 - \lambda)p_{2i} \log \frac{(1 - \lambda)p_{2i}}{(1 - \lambda)q_{2i}}.$$

The factors λ and $1 - \lambda$ cancel inside the logarithms; summing over i gives the claim. $\square$

Proof of concavity of entropy. Using $H(p) = \log N - D(p\|u)$ and convexity of D in its first argument (take $q_1 = q_2 = u$):

$$H(\lambda p + (1 - \lambda)q) = \log N - D(\lambda p + (1 - \lambda)q \parallel u) \geq \log N - \lambda D(p \parallel u) - (1 - \lambda)D(q \parallel u) = \lambda H(p) + (1 - \lambda)H(q).$$

$\square$

3.3. Joint entropy, conditional entropy, mutual information

Let X, Y take values $\{x_i\}, \{y_j\}$, with joint probabilities $p_{ij} \equiv p_{X,Y}(x_i, y_j)$ and marginals $p_i = \sum_j p_{ij}$, $q_j = \sum_i p_{ij}$.

Definition 3.2 (Joint entropy). The **joint entropy** is the entropy of the joint distribution,

$$H[X, Y] = - \sum_{i,j} p_{ij} \log p_{ij}.$$

If we learn that $X = x_i$, then Y has the conditional distribution $p_{Y|X}(y_j | x_i) = p_{ij}/p_i$, with entropy

$$H[Y | x_i] = - \sum_j \frac{p_{ij}}{p_i} \log \frac{p_{ij}}{p_i}.$$

Definition 3.3 (Conditional entropy). The **conditional entropy** of Y given X is the average of $H[Y | x_i]$ over the outcomes of X :

$$H[Y | X] = \sum_i p_i H[Y | x_i] = - \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_i}.$$

It is the average uncertainty remaining in Y once X is known. Since each $H[Y | x_i]$ is a genuine entropy, $H[Y | X] \geq 0$.

Proposition 3.7 (Chain rule for entropy).

$$H[X, Y] = H[X] + H[Y | X] = H[Y] + H[X | Y].$$

More generally, $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$.

Proof. Write $p_{ij} = p_i \cdot (p_{ij}/p_i)$ inside the logarithm:

$$- \sum_{i,j} p_{ij} \log p_{ij} = - \sum_{i,j} p_{ij} \log p_i - \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_i} = - \sum_i p_i \log p_i + H[Y | X].$$

The second form follows by symmetry; the n -variable version by induction. $\square$

The interpretation is simple: **the uncertainty about the pair is the uncertainty about X plus the uncertainty about Y once X is known.** This generalises the chain rule for surprisal of Chapter 1 (take expectations of $I(x, y) = I(x) + I(y | x)$).

Definition 3.4 (Mutual information). The **mutual information** between X and Y is

$$I[X : Y] = H[X] + H[Y] - H[X, Y] = H[Y] - H[Y | X] = H[X] - H[X | Y].$$

Rearranging,

$$H[X] = H[X | Y] + I[X : Y], \quad H[Y] = H[Y | X] + I[X : Y].$$

So $I[X : Y]$ is **the information that X provides about Y , and vice versa**: the information needed to specify X splits into the part supplied by knowing Y plus the residual $H[X | Y]$.

Proposition 3.8 (Mutual information as a KL divergence).

$$I[X : Y] = D_{KL}(p_{ij} \| p_i q_j) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_i q_j} \geq 0,$$

with equality iff $p_{ij} = p_i q_j$, i.e. iff X and Y are independent.

Independent variables carry no information about each other; and mutual information measures *how far* the joint distribution is from the independent one. Similarly, the conditional entropy is itself a KL divergence:

3. Information-theoretic measures

with $u_j = 1/N_Y$,

$$H[Y | X] = - \sum_{ij} p_{ij} \log \frac{p_{ij}}{p_i} = \log N_Y - D_{KL}(p_{ij} \| p_i u_j).$$

Corollary 3.9 (Consequences). 1. **Conditioning reduces entropy (on average):** $H[X | Y] \leq H[X]$, with equality iff $X \perp Y$.
 2. **Sub-additivity:** $H[X, Y] \leq H[X] + H[Y]$, with equality iff $X \perp Y$.
 3. **Independence bound:** $H(X_1, \dots, X_n) \leq \sum_i H(X_i)$.

Example 3.5 (A worked joint distribution (Cover and Thomas Ex. 2.2.1))

Let (X, Y) have the joint distribution

$p(x, y)$	$x = 1$	$x = 2$	$x = 3$	$x = 4$
$y = 1$	1/8	1/16	1/32	1/32
$y = 2$	1/16	1/8	1/32	1/32
$y = 3$	1/16	1/16	1/16	1/16
$y = 4$	1/4	0	0	0

The marginals are $p_X = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ and $p_Y = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, so $H(X) = 7/4$ and $H(Y) = 2$ bits. Conditioning on each row,

$$H(X | Y) = \frac{1}{4} [H(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}) + H(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}) + H(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}) + H(1, 0, 0, 0)] = \frac{1}{4} [\frac{7}{4} + \frac{7}{4} + 2 + 0] = \frac{11}{8}.$$

Then $H(X, Y) = H(Y) + H(X | Y) = 27/8$, $H(Y | X) = 27/8 - 7/4 = 13/8$, and $I(X; Y) = 7/4 - 11/8 = 3/8$ bits. Note $H(X | Y) \neq H(Y | X)$. Note also that observing $y = 3$ *increases* the uncertainty about X ($H(X | Y = 3) = 2 > H(X)$); only the *average* over y is guaranteed to decrease it.

Example 3.6 (Three balls and a balance (MacKay))

Of three balls, two have equal weight and one is either heavier or lighter. Let $X \in \{1^+, 1^-, 2^+, 2^-, 3^+, 3^-\}$, where $n^\pm$ means “ball n is odd and heavier/lighter”, all equiprobable: $H[X] = \log 6 = 1 + \log 3$ bits. Each weighing has three outcomes (g, l, e = left greater, lighter, equal) and yields at most $\log 3$ bits, so at least two weighings are needed.

Weigh ball 1 against 2 (outcome Y_1), then ball 1 against 3 (outcome Y_2). This identifies X :

Y_1	Y_2	X
e	g	3^-
e	l	3^+
g	g	1^+
g	e	2^-
l	l	1^-
l	e	2^+

Y_1 has three equiprobable outcomes, so $H[Y_1] = \log 3$. Given Y_1 , only two values of Y_2 remain possible, each with probability $1/2$ (e.g. $P(Y_2 = e \mid Y_1 = e) = 0$), hence $H[Y_2 \mid Y_1] = 1$ and $H[Y_1, Y_2] = 1 + \log 3 = H[X]$. Since (Y_1, Y_2) determines X , $I[X : Y_1, Y_2] = H[X]$. Individually $I[X : Y_1] = I[X : Y_2] = \log 3$, so

$$I[X : Y_1, Y_2] = 1 + \log 3 < I[X : Y_1] + I[X : Y_2] = 2 \log 3 :$$

the two measurements are not independent and their information about X is partly redundant.

Example 3.7 (Bit conveyed by a noisy channel)

A binary symmetric channel with crossover probability f and uniform input has $H(Y) = 1$ and $H(Y \mid X) = h_2(f)$, so it carries $I(X; Y) = 1 - h_2(f)$ bits per symbol. At $f = 0$ (noise-free) it carries the full 1 bit; at $f = 1/2$ (pure noise) zero. Chapter 4 shows this is the channel's capacity.

3.4. Venn diagram of information

The quantities $H(X), H(Y), H(X, Y), H(X \mid Y), H(Y \mid X), I(X; Y)$ obey the same set-theoretic identities as two overlapping sets: $H(X, Y)$ is the union, $I(X; Y)$ the intersection, $H(X \mid Y)$ and $H(Y \mid X)$ the two set differences. This is the device generations of textbooks use to remember the relations:

$$\begin{array}{ccccccc} & & \text{<----- H(X,Y) ----->} & & & & \\ & \text{<----- H(X) ----->} & & & & & \\ | & \text{H(X|Y)} & | & \text{I(X;Y)} & | & \text{H(Y|X)} & | \\ & & \text{<----- H(Y) ----->} & & & & \end{array}$$

Remark. The Venn-diagram picture *breaks* for three or more random variables because the “interaction information” $I(X; Y; Z) = I(X; Y) - I(X; Y \mid Z)$ can be negative. Example: X, Y independent fair bits and $Z = X \oplus Y$. Then $I(X; Y) = 0$ but $I(X; Y \mid Z) = 1$ — knowing Z , X determines Y — so $I(X; Y; Z) = -1$. Don't over-extend the analogy.

3.5. Conditional mutual information and the data-processing inequality

Definition 3.5 (Conditional mutual information). $I(X; Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z)$ — the information Y provides about X when Z is already known. It satisfies the chain rule

$$I(X; Y, Z) = I(X; Z) + I(X; Y \mid Z).$$

An operational classic: post-processing cannot increase information.

Theorem 3.10 (Data-processing inequality). If $X \rightarrow Y \rightarrow Z$ is a Markov chain (i.e. $Z \perp\!\!\!\perp X \mid Y$), then

$$I(X; Y) \geq I(X; Z).$$

Equality holds iff $I(X; Y \mid Z) = 0$, i.e. Z preserves all the information Y had about X .

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Proof. Expand $I(X; Y, Z)$ in two ways with the chain rule: $I(X; Y, Z) = I(X; Y) + I(X; Z | Y) = I(X; Z) + I(X; Y | Z)$. The Markov property makes $I(X; Z | Y) = 0$, so $I(X; Y) = I(X; Z) + I(X; Y | Z) \geq I(X; Z)$. $\square$

In particular, for any function g , $I(X; g(Y)) \leq I(X; Y)$: **no processing of data can create information about the source that was not already there.** This is the theoretical basis of “sufficient statistics”: $T(Y)$ is sufficient for X exactly when equality holds.

3.6. Concavity of mutual information

Proposition 3.11 (Concavity/convexity of mutual information). *For fixed conditional $p(y | x)$, $I(X; Y)$ is a concave function of $p(x)$. For fixed marginal $p(x)$, $I(X; Y)$ is a convex function of $p(y | x)$.*

The first statement guarantees that maximising $I(X; Y)$ over input distributions — the definition of channel capacity in Chapter 4 — is a concave maximisation problem with no spurious local maxima. The second underlies rate-distortion theory.

3.7. Exercises

Exercise 3.1

Show that $H(X, Y) \leq H(X) + H(Y)$ with equality iff X, Y are independent, using the log-sum inequality directly (not via mutual information).

Solution

Write $H(X) + H(Y) - H(X, Y) = \sum_{x,y} p(x, y) \log[p(x, y)/(p(x)p(y))]$. By the log-sum inequality with $a_{xy} = p(x, y)$ and $b_{xy} = p(x)p(y)$ (both summing to 1),

$$\sum_{x,y} p(x, y) \log \frac{p(x,y)}{p(x)p(y)} \geq 1 \cdot \log \frac{1}{1} = 0.$$

So the difference is ≥ 0 , with equality iff $p(x, y)/[p(x)p(y)]$ is constant — necessarily equal to 1 — iff independence.

Exercise 3.2

Compute $I(X; Y)$ for a binary symmetric channel with error probability f when the input is *not* uniform, $P(X = 1) = \pi$.

Solution

$P(Y = 1) = \pi(1 - f) + (1 - \pi)f \equiv \rho$. $H(Y) = h_2(\rho)$ and $H(Y | X) = h_2(f)$ regardless of π . So $I(X; Y) = h_2(\pi(1 - f) + (1 - \pi)f) - h_2(f)$, which is maximal ($= 1 - h_2(f)$) at $\pi = 1/2$ where $\rho = 1/2$.

Exercise 3.3

Prove the data-processing inequality for the specific case $Z = g(Y)$ with g a deterministic function.

Solution

$X \rightarrow Y \rightarrow g(Y)$ is automatically a Markov chain, since given Y the value of Z is fixed. Then $I(X; Z) = H(X) - H(X | Z) \leq H(X) - H(X | Y, Z)$ (conditioning reduces entropy), and $H(X | Y, Z) = H(X | Y)$ because Z is a function of Y . Hence $I(X; Z) \leq H(X) - H(X | Y) = I(X; Y)$.

Exercise 3.4

Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ and $q = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Compute $H(p)$, $H(p, q)$, $D(p\|q)$ and $D(q\|p)$, and verify $H(p) = \log 3 - D(p\|q)$.

Solution

$H(p) = 1.5$ bits. $H(p, q) = \log 3 \approx 1.585$. $D(p\|q) = H(p, q) - H(p) \approx 0.085$ bits, and indeed $\log 3 - 0.085 = 1.5$ since q is uniform. $D(q\|p) = \frac{1}{3}[\log \frac{1/3}{1/2} + 2 \log \frac{1/3}{1/4}] = \frac{1}{3}[-0.585 + 2(0.415)] \approx 0.082$ bits $\neq D(p\|q)$.

3.8. Callouts

i Note

The three interpretations of mutual information — as a KL divergence, as a reduction of entropy, and as an inclusion–exclusion of joint and marginal entropies — are all useful. Which one you invoke depends on what you want to prove.

💡 Tip

When bounding an information quantity, try (i) Jensen’s inequality applied to $-\log$; (ii) the log-sum inequality; (iii) the data-processing inequality. These three moves cover most of the proofs in Cover & Thomas.

⚠ Warning

The KL divergence $D(p\|q)$ is *asymmetric*. In machine learning the direction matters: minimising $D(p_{\text{data}}\|q_{\theta})$ is maximum likelihood (“mass-covering”), whereas minimising $D(q_{\theta}\|p)$ is what variational inference does (“mode-seeking”, Chapter 10).

3.9. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 2. Cover & Thomas, §2.1–2.8 (Example 2.2.1, log-sum inequality, data processing). MacKay, Chapter 2 and §8.1 (three-ball weighing, Venn diagrams).

4. Source coding: compression

Information theory was born to solve telecommunication problems. A **sender** must transmit a message to a distant **receiver** through a **channel** — any physical means that carries signals. The sender first converts the message into a form suitable for the channel (**encoding**); the receiver reconstructs the message from what arrives (**decoding**). In a digital phone call, for instance, a sound is converted into a string of numbers and back.

A good encoding must satisfy two competing requirements:

- it must allow **faithful reconstruction** of the message (minimal information loss);
- it must be **as short as possible**, so as to maximise the information transmission rate.

Whatever the message (sound, image, text), the encoding converts it into the common currency of communication: a **string of binary digits**. Shannon's motivation for defining entropy was to answer the question: *what is the minimum number of binary digits per symbol needed to encode a source?* This chapter proves that the answer is $H(X)$, in two different settings:

- **lossy (fixed-length) coding**: all codewords have the same length, but we accept that a small fraction of messages are lost — the error probability δ can be made as small as we like;
- **lossless (variable-length, symbol) coding**: every message is reconstructed exactly, but probable messages get short codewords and improbable ones long codewords.

Both rely on the same idea — exploiting the non-uniformity of p — and both give $H(X)$ as the fundamental limit.

4.1. Raw bit content

Let the message x be drawn from a finite set $\mathcal{A}_X = \{x_1, \dots, x_N\}$ with probability $p_X(x)$. A **fixed-length binary code** is a map

$$E : \mathcal{A}_X \rightarrow \mathcal{B}^K, \quad \mathcal{B}^K = \{0, 1\}^K,$$

and decoding is $D = E^{-1}$. For E to be invertible it must be injective; since $|\mathcal{B}^K| = 2^K$, we need $2^K \geq |\mathcal{A}_X|$:

$$K \geq H_0(X) \equiv \log_2 |\mathcal{A}_X|.$$

Definition 4.1 (Raw bit content). $H_0(X) = \log_2 |\mathcal{A}_X|$ is the **raw bit content** of X . By a simple counting argument, no invertible fixed-length code can use fewer than $\lceil H_0 \rceil$ binary digits.

Example 4.1 (A biased coin)

Toss a coin with $P(h) = 0.9$, $P(t) = 0.1$. Then $H_0(X) = 1$ but

$$H(X) = -0.9 \log 0.9 - 0.1 \log 0.1 \approx 0.47 \text{ bits.}$$

To communicate one toss we must still use one binary digit ($E : h \mapsto 0, t \mapsto 1$). The information

content of the outcome is less than half of the raw bit content, which suggests that one digit per toss is wasteful.

The intuition is correct, but the gain can only be realised for **sequences** of messages $x^{(n)} = (x_1, \dots, x_n) \in \mathcal{A}_X^n$, each drawn independently from the same source. There are then two ways to compress:

- **lossy coding:** two different sequences may be mapped to the same codeword (or some sequences are discarded), but this happens with small probability;
- **lossless coding:** use variable-length codewords of lengths ℓ_x . We cannot have $\ell_x < H_0$ for *all* x (counting argument), but we can make the *average* $\langle \ell \rangle = \sum_x p(x)\ell_x$ smaller than H_0 by giving probable x short codewords.

4.2. Lossy coding

Consider sequences $X^{(n)} = (X_1, \dots, X_n)$ of i.i.d. symbols, i.e. a random vector in $\mathcal{A}_X^n$, and a fixed-length code $E^{(n)} : \mathcal{A}_X^n \rightarrow \mathcal{B}^K$. A perfectly faithful code needs $K \geq H_0(X^{(n)}) = nH_0(X)$.

We now relax faithfulness. Choose a subset $S \subset \mathcal{A}_X^n$ and assign codewords only to sequences in S ; sequences in $\bar{S}$ are either discarded ($E(x^{(n)}) = \emptyset$) or mapped onto codewords already in use. The decoder makes an error exactly when $x^{(n)} \in \bar{S}$, so the **error probability** is

$$P_e = P(D(E(X^{(n)})) \neq X^{(n)}) = P(X^{(n)} \in \bar{S}).$$

Example 4.2 (Three tosses of the biased coin)

For $n = 3$ tosses with $P(h) = 0.9$:

outcome	hhh	hht, hth, thh	htt, tht, tth	ttt
probability (each)	0.729	0.081	0.009	0.001

$H(X^{(3)}) = 3H(X) \approx 1.41$ bits while $H_0(X^{(3)}) = 3$. The invertible code

$$hhh \mapsto 000, hht \mapsto 001, hth \mapsto 010, thh \mapsto 100, htt \mapsto 011, tht \mapsto 101, tth \mapsto 110, ttt \mapsto 111$$

uses $K = 3$ and has $P_e = 0$. But the outcomes with two or more tails have total probability $3(0.009) + 0.001 = 0.028$. If we keep only

$$S = \{hhh, hht, hth, thh\}, \quad \tilde{E} : hhh \mapsto 00, hht \mapsto 01, hth \mapsto 10, thh \mapsto 11,$$

and send nothing for the other outcomes, we use only **2 digits** at the price of $P_e = 2.8\%$.

To keep the error below a tolerance δ , S must be **δ -sufficient**: $P(X^{(n)} \in S) \geq 1 - \delta$.

Definition 4.2 (Smallest δ -sufficient subset and essential bit content). The **smallest δ -sufficient subset** is

$$S_\delta(X^{(n)}) = \arg \min_S \{ |S| : S \subseteq \mathcal{A}_X^n, P(X^{(n)} \in S) \geq 1 - \delta \}.$$

It is obtained by sorting the outcomes by decreasing probability and keeping the first k of them, stopping

as soon as their cumulative probability reaches $1 - \delta$. The **essential bit content** is

$$H_\delta(X^{(n)}) = \log_2 |S_\delta(X^{(n)})|,$$

the number of binary digits needed if we tolerate error probability δ .

Example 4.3 (*Essential bit content for three tosses*)

Adding outcomes in order of decreasing probability:

δ range	$ S_\delta $	H_δ (bits)
$0 \leq \delta < 0.001$	8	3.00
$0.001 \leq \delta < 0.010$	7	2.81
$0.010 \leq \delta < 0.019$	6	2.58
$0.019 \leq \delta < 0.028$	5	2.32
$0.028 \leq \delta < 0.109$	4	2.00
$0.109 \leq \delta < 0.190$	3	1.58
$0.190 \leq \delta < 0.271$	2	1.00
$0.271 \leq \delta < 1$	1	0.00

H_δ is a decreasing staircase function of δ ; at $n = 3$ it depends strongly on δ .

4.2.1. What happens as n grows

Plot $H_\delta(X^{(n)})/n$ against δ for increasing n (MacKay, Fig. 4.9; Allegra, Fig. 3.1). As n grows the curves become flatter and flatter and collapse onto the horizontal line $H_\delta/n = H(X)$. This has two consequences:

1. **Achievability.** Fix a small tolerance δ_1 . For large n , $H_{\delta_1}(X^{(n)})/n = H(X) + \beta$ with $\beta \ll 1$: the smallest δ_1 -sufficient subset has about $2^{n(H+\beta)}$ elements, and a code with that many codewords — about $H(X)$ digits per symbol — has error at most δ_1 . For the $p = 0.9$ coin with $n = 1000$, $\delta_1 = 0.05$ gives $\beta \approx 0.05$, i.e. about 0.51 digits per toss instead of 1.
2. **Converse.** Fix a small β . For large n , $H_\delta/n < H(X) - \beta$ only when $\delta \rightarrow 1$: using fewer than $2^{n(H-\beta)}$ codewords means losing almost every message.

One can code with vanishing error using $H(X)$ digits per symbol, but not fewer. The mechanism behind this is typicality.

4.3. Typicality and the AEP

In a long sequence each symbol x_i appears, with high probability, roughly $m_i \approx np_i$ times. For the biased coin, sequences of length n have about $0.9n$ heads and $0.1n$ tails. The probability of such a sequence is

$$P(x^{(n)}) \approx p_1^{np_1} p_2^{np_2} \dots p_N^{np_N} = 2^{n \sum_i p_i \log p_i} = 2^{-nH(X)},$$

so its information content is $-\log P(x^{(n)}) \approx nH(X)$. These are the **typical sequences**.

|

Definition 4.3 (Typical set). For $\beta > 0$, the set of (n, β) -**typical sequences** is

$$T_{n\beta} = \left\{ x^{(n)} \in \mathcal{A}_X^n : \left| -\frac{1}{n} \log_2 P(x^{(n)}) - H(X) \right| < \beta \right\}.$$

Equivalently, $2^{-n(H+\beta)} < P(x^{(n)}) < 2^{-n(H-\beta)}$.

Theorem 4.1 (Asymptotic Equipartition Property (AEP)). For $X_1, X_2, \dots$ i.i.d. $\sim p$:

1. $-\frac{1}{n} \log_2 P(X_1, \dots, X_n) \rightarrow H(X)$ in probability as $n \rightarrow \infty$.
2. $P(X^{(n)} \in T_{n\beta}) \rightarrow 1$ as $n \rightarrow \infty$, for any $\beta > 0$.
3. $|T_{n\beta}| < 2^{n(H+\beta)}$, and $|T_{n\beta}| > (1 - \epsilon)2^{n(H-\beta)}$ for n large enough.

So almost all probability concentrates on roughly 2^{nH} sequences, which are (to exponential accuracy) **equiprobable** — hence *equipartition*. Since there are $2^{nH_0} = |\mathcal{A}_X|^n$ sequences in total, the typical set is an exponentially small fraction $2^{-n(H_0-H)}$ of all sequences, yet it carries almost all the probability.

Example 4.4 (Typicality for 100 tosses)

Take $n = 100$, $p(h) = 0.9$. Typical sequences have about 10 tails; their number is $\binom{100}{10} \approx 1.7 \times 10^{13} \approx 2^{44}$, close to $2^{nH} = 2^{46.9}$ (the agreement improves with n). There are 2^{100} sequences in total. Each typical sequence has probability $\approx 2^{-47}$.

Note that the **single most probable sequence**, $hh \dots h$, with $P = 0.9^{100} \approx 2.7 \times 10^{-5}$, is *not* typical: $-\frac{1}{n} \log P = 0.152 \ll H = 0.469$. It is very likely compared to any *single* typical sequence, but there is only one of it, while typical sequences are astronomically many. The histogram of the number of tails m concentrates around $m = 10$ with width $\sqrt{np(1-p)} = 3$; relative to n , the width shrinks as $1/\sqrt{n}$.

Remark (Physical meaning). The AEP is the information-theoretic analogue of the equivalence of ensembles in statistical mechanics: a system with fixed temperature (canonical) has, with overwhelming probability, an energy close to its mean, so it behaves like a microcanonical ensemble of $\approx e^S$ equiprobable microstates. Here nH plays the role of the entropy S .

4.4. Shannon's source coding theorem

4.4.1. Intuitive argument

1. We have sequences $X^{(n)}$ of i.i.d. messages from the same source.
2. We want a lossy fixed-length code with error at most δ : we must identify $S_\delta(X^{(n)})$; codewords have length $H_\delta = \log |S_\delta|$.
3. For large n , a random sequence is almost surely typical, and each typical sequence has $P \approx 2^{-nH}$. Since their probabilities add up to nearly 1, there are $\approx 2^{nH}$ of them.
4. Hence a δ -sufficient subset made of typical sequences has size $\approx 2^{nH}$, and the *smallest* one can be no larger: codewords of length $\approx nH$ suffice.
5. Codewords cannot be much shorter. With length $n(H - \Delta)$ there are at most $2^{n(H-\Delta)}$ codewords; since nearly all the probability sits on typical sequences of probability $\approx 2^{-nH}$ each, the covered probability is at most $2^{n(H-\Delta)}2^{-nH} = 2^{-n\Delta} \rightarrow 0$. The error tends to 1.

4.4.2. Rigorous statement and proof

Theorem 4.2 (Shannon's source coding theorem). *Let X have entropy $H = H(X)$. For every $\delta \in (0, 1)$ and every $\beta > 0$ there exists n_0 such that for all $n > n_0$,*

$$\left| \frac{H_\delta(X^{(n)})}{n} - H \right| < \beta.$$

In words: for long enough sequences, lossy compression with arbitrarily small error is possible with $n(H + \beta)$ bits, i.e. $2^{n(H+\beta)}$ codewords; compression with $n(H - \beta)$ bits is impossible unless we accept an error probability arbitrarily close to 1. Consequently:

- the information in n i.i.d. realisations of X can be transmitted with $\approx H(X)$ binary digits per realisation, with asymptotically vanishing loss;
- it cannot be compressed further.

The proof has three parts.

Part I — typical sequences are typical.

$$P(X^{(n)} \in T_{n\beta}) \geq 1 - \frac{\sigma^2}{n\beta^2}, \quad \sigma^2 = \text{Var}[-\log p(X_i)].$$

Proof. Recall Chebyshev's inequality $P((Z - \mathbb{E}Z)^2 \geq \alpha) \leq \text{Var}(Z)/\alpha$. For the mean $M_n = \frac{1}{n} \sum_i Z_i$ of i.i.d. variables, $\mathbb{E}M_n = \mathbb{E}Z$ and $\text{Var} M_n = \text{Var} Z/n$, so

$$P((M_n - \mathbb{E}Z)^2 < \alpha) \geq 1 - \frac{\text{Var} Z}{n\alpha}.$$

Let $I_i = -\log p(X_i)$, the information content of the i -th symbol: these are i.i.d. with mean H and variance σ^2 . Because the symbols are independent,

$$\bar{I}^{(n)} \equiv -\frac{1}{n} \log P(X^{(n)}) = -\frac{1}{n} \sum_i \log p(X_i) = \frac{1}{n} \sum_i I_i,$$

a mean of i.i.d. variables. Since $X^{(n)} \in T_{n\beta} \iff (\bar{I}^{(n)} - H)^2 < \beta^2$, Chebyshev with $\alpha = \beta^2$ gives the claim. (This is the weak law of large numbers for $\bar{I}^{(n)}$, i.e. part 1 of the AEP.) $\square$

Part II — achievability: for all δ, β there is n_1 such that $H_\delta(X^{(n)}) < n(H + \beta)$ for all $n > n_1$.

Proof. Take $n_1 > \sigma^2/(\beta^2\delta)$. By Part I, for $n > n_1$, $P(X^{(n)} \in T_{n\beta}) > 1 - \delta$: the typical set is δ -sufficient, so $|S_\delta| \leq |T_{n\beta}|$. Every typical sequence has $P(x^{(n)}) > 2^{-n(H+\beta)}$, hence

$$1 \geq P(T_{n\beta}) > |T_{n\beta}| 2^{-n(H+\beta)} \Rightarrow |T_{n\beta}| < 2^{n(H+\beta)}.$$

Therefore $H_\delta = \log |S_\delta| \leq \log |T_{n\beta}| < n(H + \beta)$. $\square$

Part III — converse: for all δ, β there is n_2 such that $H_\delta(X^{(n)}) > n(H - \beta)$ for all $n > n_2$.

Proof. Suppose, for contradiction, that some set S with $|S| \leq 2^{n(H-\beta)}$ is δ -sufficient. Split it into its typical and atypical parts with respect to the narrower set $T_{n,\beta/2}$:

$$P(X^{(n)} \in S) = P(X^{(n)} \in S \cap T_{n,\beta/2}) + P(X^{(n)} \in S \cap \bar{T}_{n,\beta/2}).$$

Every sequence in $T_{n,\beta/2}$ has $P(x^{(n)}) < 2^{-n(H-\beta/2)}$, so the first term is at most

$$|S| 2^{-n(H-\beta/2)} \leq 2^{n(H-\beta)} 2^{-n(H-\beta/2)} = 2^{-n\beta/2}.$$

4. Source coding: compression

The second term is at most $P(X^{(n)} \notin T_{n,\beta/2}) \leq 4\sigma^2/(n\beta^2)$ by Part I. Hence

$$P(X^{(n)} \in S) \leq 2^{-n\beta/2} + \frac{4\sigma^2}{n\beta^2},$$

which decreases to zero; for n large enough it is below $1-\delta$, contradicting δ -sufficiency. So $|S_\delta| > 2^{n(H-\beta)}$. $\square$

Combining Parts II and III with $n_0 = \max(n_1, n_2)$ proves the theorem. $\blacksquare$

Remark (How large must n be?). Part II requires $n \gtrsim \sigma^2/(\beta^2\delta)$: lossy fixed-length coding is efficient only for very long blocks. For the $p = 0.9$ coin, $\sigma^2 = p(1-p)(\log \frac{p}{1-p})^2 = 0.09 \times (\log_2 9)^2 \approx 0.90$, so $\beta = 0.1$ and $\delta = 0.01$ require $n \approx 10^4$. (Chebyshev is pessimistic; sharper Chernoff bounds give exponential convergence, but the message stands.) This motivates *lossless* variable-length codes, which work symbol by symbol.

Remark (Rate-distortion theory). The lossy coding above is “lossy” only in that a vanishing fraction of messages is lost. In engineering, “lossy compression” (JPEG, MP3) usually means something different: every message is reconstructed *approximately*, with an average distortion D . The optimal trade-off is the **rate-distortion function** $R(D) = \min_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} I(X; \hat{X})$ (Cover & Thomas, Ch. 10). For a Gaussian source with variance σ^2 and squared error, $R(D) = \frac{1}{2} \log_2(\sigma^2/D)$ for $D \leq \sigma^2$. We will not develop this theory further.

4.5. Lossless coding: symbol codes

A **symbol code** maps each symbol to a binary string of variable length:

$$E: \mathcal{A}_X \rightarrow \mathcal{B}^+, \quad \mathcal{B}^+ = \mathcal{B} \cup \mathcal{B}^2 \cup \mathcal{B}^3 \cup \dots,$$

$x_i \mapsto y_i \in \mathcal{B}^{\ell_i}$; y_i is the **codeword** and ℓ_i the **codeword length**. It extends to sequences $x^+ \in \mathcal{A}_X^+ = \mathcal{A}_X \cup \mathcal{A}_X^2 \cup \dots$ by concatenation,

$$E^+(x_{i_1} x_{i_2} \dots x_{i_n}) = E(x_{i_1}) E(x_{i_2}) \dots E(x_{i_n}).$$

Example 4.5

For the coin, $E: h \mapsto 0, t \mapsto 101$ (lengths 1 and 3) gives $E^+(hhh) = 000$, $E^+(ht) = 0101$, etc.

A good symbol code should be

- **uniquely decodable** (invertible on sequences),
- **easy to decode**,
- **short on average**: the expected length $L(E, X) = \sum_i p_i \ell_i$ should be minimal.

4.5.1. Unique decodability

Definition 4.4 (Uniquely decodable code). E is **uniquely decodable** if E^+ is injective: $x^+ \neq x'^+ \Rightarrow E^+(x^+) \neq E^+(x'^+)$.

Distinct codewords are necessary but not sufficient.

Example 4.6 (*Distinct codewords are not enough*)

$h \mapsto 0, t \mapsto 101$ is uniquely decodable: 0101101000101 decodes only as *htthhht* (every 1 starts a 101). But for $\mathcal{A} = \{a, b, c, d\}$ with

$$a \mapsto 0, b \mapsto 1, c \mapsto 00, d \mapsto 11,$$

the string 0011011 can be read as *cdad*, *aadad*, *aabbad* or *aabbabb*. All codewords are distinct, yet the code is not uniquely decodable.

4.5.2. Prefix codes

“Easy to decode” means that each codeword can be recognised as soon as its last bit arrives, without looking ahead. This requires that no codeword be a **prefix** of another (c is a prefix of d if $d = ct$ for some string t ; e.g. 1 and 10 are prefixes of 101).

Definition 4.5 (Prefix code). A code is a **prefix code** (also *instantaneous* or *self-punctuating*) if no codeword is a prefix of another codeword.

Example 4.7 (*Uniquely decodable but not instantaneous*)

$a \mapsto 1, b \mapsto 101$ is uniquely decodable (each 0 signals a b). But on receiving 110111 (= *abaa*), after the first 1 we cannot yet decide between a and the beginning of b ; we must wait for the third digit. The culprit is that 1 is a prefix of 101. The code $a \mapsto 0, b \mapsto 101$ is a prefix code and decodes instantaneously.

Proposition 4.3. *Every prefix code is uniquely decodable. In particular, fixed-length codes with distinct codewords are uniquely decodable.*

Proof. Suppose $E^+(x_1 \dots x_k) = E^+(y_1 \dots y_l)$ with $x \neq y$. Strip the longest common initial run of symbols ($x_i = y_i$ for $i \leq m$); the remaining sequences still have equal encodings, and now $x_{m+1} \neq y_{m+1}$ (the remainders cannot both be empty since $x \neq y$). Both encoded strings begin with $E(x_{m+1})$ and $E(y_{m+1})$ respectively. If these have equal length they are equal, so two symbols share a codeword; if not, the shorter is a prefix of the longer. Either way the code is not a prefix code. $\square$

Prefix codes are binary trees. Take a probability tree in which every branching has exactly two children, $E = E_0 \cup E_1$, and label the edges 0 and 1. Reading the labels from the root to a leaf gives a binary string; assigning these strings to the leaves gives a prefix code, because a prefix of a codeword corresponds to an *internal* node, which is not a leaf. Conversely every prefix code defines such a tree.

Example 4.8 (*A dyadic source*)

$X \in \{0, 1, 2, 3\}$ with $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. The tree “is it 0? / is it 1? / is it 2?” gives the prefix code

$$0 \mapsto 0, \quad 1 \mapsto 10, \quad 2 \mapsto 110, \quad 3 \mapsto 111,$$

with $L = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + 2 \cdot \frac{1}{8} \cdot 3 = 1.75$ bits = $H(X)$. Here $L = H$ because $\ell_i = -\log p_i$ exactly; this is possible only when all p_i are powers of $1/2$.

Example 4.9 (A tree code for three tosses)

For $X^{(3)}$ (three tosses, $p(h) = 0.9$), let M be the number of tails: $p_M = (0.729, 0.243, 0.027, 0.001)$. Ask “ $M = 0?$ ”, then “ $M = 1?$ ”, then “ $M = 2?$ ”. If $M = 1$ or 2, two more binary questions identify which of the three arrangements occurred. Reading edge labels gives

$hhh \mapsto 0$, $hht \mapsto 100$, $thh \mapsto 1010$, $hth \mapsto 1011$, $htt \mapsto 1100$, $tht \mapsto 11010$, $tth \mapsto 11011$, $ttt \mapsto 111$,

with expected length

$$L = 0.729 \cdot 1 + 0.081 \cdot 3 + 2(0.081) \cdot 4 + 0.009 \cdot 4 + 2(0.009) \cdot 5 + 0.001 \cdot 3 = 1.749 \text{ bits},$$

much better than the 3 bits of the fixed-length code and not far from $H(X^{(3)}) = 1.41$.

4.5.3. Kraft's inequality

Short average length comes from giving probable outcomes short codewords. We cannot make *all* codewords short: unique decodability constrains the lengths.

Theorem 4.4 (Kraft–McMillan inequality). *If a binary code is uniquely decodable, its codeword lengths satisfy*

$$\sum_i 2^{-\ell_i} \leq 1.$$

*Conversely, for any lengths satisfying this inequality there exists a **prefix** code with those lengths.*

A code with $\sum_i 2^{-\ell_i} = 1$ is called **complete**. Prefix codes read off a full binary tree are complete: replace every edge probability by $1/2$; the leaf at depth ℓ_i then gets probability $q_i = 2^{-\ell_i}$, and these probabilities must sum to one. For the three-toss tree above, $\sum 2^{-\ell_i} = 2^{-1} + 2 \cdot 2^{-3} + 3 \cdot 2^{-4} + 2 \cdot 2^{-5} = 2^{-5}(16 + 8 + 6 + 2) = 1$.

Proof of necessity (McMillan). Let $S = \sum_i 2^{-\ell_i}$. Then

$$S^n = \sum_{i_1, \dots, i_n} 2^{-(\ell_{i_1} + \dots + \ell_{i_n})},$$

where $\ell_{i_1} + \dots + \ell_{i_n}$ is the length of the encoding of the string $x_{i_1} \dots x_{i_n}$. With $\ell_{\min}, \ell_{\max}$ the shortest and longest codeword lengths, group the terms by total length ℓ :

$$S^n = \sum_{\ell=n\ell_{\min}}^{n\ell_{\max}} 2^{-\ell} |A_\ell|,$$

where A_ℓ is the set of length- n source strings encoded into ℓ bits. Unique decodability forces $|A_\ell| \leq 2^\ell$ (there are only 2^ℓ binary strings of length ℓ). Hence

$$S^n \leq \sum_{\ell=n\ell_{\min}}^{n\ell_{\max}} 1 \leq n\ell_{\max}.$$

If $S > 1$, the left side grows exponentially in n while the right side grows linearly — impossible for large n . So $S \leq 1$. $\square$

Proof of sufficiency. Let A_ℓ now denote the number of required codewords of length ℓ ; the inequality reads $\sum_\ell A_\ell 2^{-\ell} \leq 1$. Assign codewords in order of increasing length, keeping the prefix property at every stage. Suppose the codewords of length up to ℓ have been chosen. Each codeword of length $\ell' \leq \ell$ is a prefix

of $2^{\ell+1-\ell'}$ strings of length $\ell + 1$, which are forbidden. The number of available strings of length $\ell + 1$ is therefore

$$2^{\ell+1} - \sum_{\ell' \leq \ell} A_{\ell'} 2^{\ell+1-\ell'},$$

and we can pick $A_{\ell+1}$ of them iff $\sum_{\ell' \leq \ell+1} A_{\ell'} 2^{-\ell'} \leq 1$ — which follows from Kraft's inequality. Starting from the shortest length (no constraints) and inducting gives a prefix code. $\square$

Remark. Kraft's inequality is *necessary* for unique decodability, not sufficient: mapping every symbol to the single codeword 0 (with $|\mathcal{A}| = 2$, $2^{-1} + 2^{-1} = 1$) satisfies it but is obviously not decodable. The sufficiency statement says that *some* prefix code with those lengths exists. A key consequence: **nothing is gained by using uniquely decodable codes that are not prefix codes** — prefix codes achieve every set of lengths that any uniquely decodable code can.

4.6. Source coding theorem for symbol codes

Theorem 4.5 (Source coding theorem for symbol codes). *For any uniquely decodable code,*

$$L(E, X) \geq H(X),$$

with equality iff $\ell_i = -\log_2 p_i$ for all i . Moreover there exists a prefix code with

$$H(X) \leq L(E, X) < H(X) + 1.$$

Proof. Lower bound. Define the “implicit probabilities” of the code $q_i = 2^{-\ell_i}/z$, with $z = \sum_j 2^{-\ell_j} \leq 1$ by Kraft. Then

$$D_{KL}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} = -H(X) + \sum_i p_i \ell_i + \log z,$$

so

$$L(E, X) = H(X) + D_{KL}(p\|q) - \log z \geq H(X) + D_{KL}(p\|q) \geq H(X).$$

Equality requires (1) $z = 1$, a complete code, and (2) $p = q$, i.e. $p_i = 2^{-\ell_i}$.

Upper bound. Condition (2) can be met exactly only if all p_i are negative powers of 2, but the identity suggests choosing $\ell_i \approx -\log p_i$. Take the **Shannon code lengths** $\ell_i = \lceil -\log_2 p_i \rceil$. They satisfy Kraft,

$$\sum_i 2^{-\lceil -\log p_i \rceil} \leq \sum_i 2^{\log p_i} = \sum_i p_i = 1,$$

so a prefix code with these lengths exists, and

$$L = \sum_i p_i \lceil -\log p_i \rceil < \sum_i p_i (-\log p_i + 1) = H(X) + 1.$$

$\square$

The identity $L = H + D(p\|q) - \log z$ gives a precise meaning to the KL divergence of Chapter 2: **a complete code designed for the distribution q but used on a source p wastes exactly $D(p\|q)$ bits per symbol.**

Example 4.10 (Shannon code for a non-dyadic source)

$p = (0.4, 0.35, 0.2, 0.05)$: $-\log p = (1.32, 1.51, 2.32, 4.32)$, so Shannon lengths are $(2, 2, 3, 5)$ and $L = 0.8 + 0.7 + 0.6 + 0.25 = 2.35$ bits, against $H = 1.74$ bits. The bound $L < H + 1$ holds, but the code is far from optimal — Huffman (below) does better.

Proposition 4.6 (Block coding closes the gap). *Applying the theorem to blocks $X^{(n)}$ of n i.i.d. symbols, whose entropy is nH , gives a code with $nH \leq L_n < nH + 1$, i.e. a rate per symbol*

$$H(X) \leq \frac{L_n}{n} < H(X) + \frac{1}{n}.$$

The overhead of at most one bit is spread over n symbols and becomes negligible.

4.7. Huffman codes

Shannon codes satisfy the bounds but are not optimal. **Huffman's algorithm** constructs an *optimal* prefix code by building the binary tree from the leaves up.

Definition 4.6 (Huffman coding algorithm). Given probabilities $p_1, \dots, p_N$:

1. Create a node for each outcome.
2. Take the two least probable parentless nodes and create a parent node for them, with probability equal to their sum. Label the two edges 0 and 1.
3. Repeat step 2 until a single root remains. The codeword of each leaf is the string of labels from the root to the leaf.

By construction the two least probable outcomes are siblings and get codewords of equal length.

Example 4.11 (Huffman for a skewed source)

$p = (0.4, 0.35, 0.2, 0.05)$, $H \approx 1.74$ bits. Merges: $0.05 + 0.2 = 0.25$; $0.25 + 0.35 = 0.60$; $0.60 + 0.4 = 1$. Codewords 0, 10, 110, 111 with lengths $(1, 2, 3, 3)$ and $L = 0.4 + 0.7 + 0.6 + 0.15 = 1.85$ bits — 0.5 bits better than the Shannon code above, and only 0.11 above H .

Example 4.12 (Huffman for three tosses)

For $X^{(3)}$ with $p(h) = 0.9$, Huffman's algorithm gives

$hhh \mapsto 0$, $hht \mapsto 100$, $hth \mapsto 101$, $thh \mapsto 110$, $htt \mapsto 11110$, $tht \mapsto 11100$, $tth \mapsto 11101$, $ttt \mapsto 11111$,

with

$$L = 0.729 \cdot 1 + 3(0.081) \cdot 3 + 3(0.009) \cdot 5 + 0.001 \cdot 5 = 1.598 \text{ bits},$$

better than the tree code above (1.749) and closer to $H(X^{(3)}) = 1.41$. Per toss: 0.53 bits, versus $H(X) = 0.47$.

Theorem 4.7 (Huffman optimality). *No uniquely decodable code for the same source has smaller expected length than the Huffman code. In particular $H(X) \leq L^{Huff} < H(X) + 1$.*

Proof. Order $p_1 \leq p_2 \leq \dots \leq p_N$. By the Kraft–McMillan theorem we may restrict to prefix codes, and an optimal prefix code may be taken complete (otherwise some codeword could be shortened).

Step 1: in some optimal code, the two least probable symbols are siblings of maximal length. In a complete prefix code the deepest leaves come in sibling pairs. If ℓ_1 were not maximal, some x_j with $p_j \geq p_1$ would sit deeper; swapping x_1 and x_j changes L by $(\ell_1 - \ell_j)(p_j - p_1) \leq 0$, so it does not hurt. Likewise we can place x_2 in the sibling slot of x_1 without increasing L . (If $\ell_1 > \ell_2$ in an optimal code and some x_3 with $p_3 > p_2$ had length ℓ_1 , swapping x_2 and x_3 would change L by $(\ell_1 - \ell_2)(p_2 - p_3) < 0$, contradicting optimality.)

Step 2: induction on N . For $N = 2$ any code uses two codewords of length 1, which is optimal. Assume Huffman is optimal for $N - 1$ symbols. Take an optimal tree for $(p_1, \dots, p_N)$ in which x_1, x_2 are siblings and merge them: this gives a tree for $(p_1 + p_2, p_3, \dots, p_N)$, with

$$L_{\text{opt}}(p_1, \dots, p_N) = L(p_1 + p_2, p_3, \dots, p_N) + p_1 + p_2,$$

since the merged codeword is one bit shorter for both. The merged tree must be optimal for the reduced source (otherwise unmerging a better one would beat the optimum). The Huffman tree for $(p_1, \dots, p_N)$ merges x_1, x_2 first and then *is* the Huffman tree for the reduced source, which is optimal by the induction hypothesis. So both satisfy the same relation with the same reduced optimal length, and $L^{\text{Huff}}(p_1, \dots, p_N) = L_{\text{opt}}(p_1, \dots, p_N)$. $\square$

4.8. Beyond symbol codes

Symbol codes waste up to one bit per symbol, which is a lot when $H(X) \ll 1$ (e.g. the $p = 0.99$ coin, $H = 0.08$ bits, still needs one bit per symbol). Two remedies are used in practice:

- **Arithmetic coding** encodes a whole sequence as a sub-interval of $[0, 1)$ of width $P(x^{(n)})$, which needs about $-\log P(x^{(n)})$ bits: the overhead is $O(1)$ bits *per message* rather than per symbol, and it adapts naturally to probabilities that change from symbol to symbol (e.g. from a predictive model).
- **Universal codes** (Lempel–Ziv, context-tree weighting) do not need to know p at all: they learn the statistics from the data as they go.

Proposition 4.8 (LZ universality). *Lempel–Ziv coding, applied to a stationary ergodic source of entropy rate H_∞ , achieves an asymptotic compression rate of H_∞ bits per symbol without any advance knowledge of the source.*

Remark. gzip, bzip2 and xz are variations on Lempel–Ziv and Huffman/arithmetic coding. The theoretical guarantee that they reach the entropy limit is a consequence of the AEP for ergodic sequences (Chapter 6).

4.9. Exercises

Exercise 4.1

Construct a Huffman code for a source with probabilities $(1/2, 1/4, 1/8, 1/16, 1/16)$ and verify it achieves $H(X)$.

Solution

$H(X) = 0.5 + 0.5 + 0.375 + 0.25 + 0.25 = 1.875$ bits. Huffman merges $1/16 + 1/16 = 1/8$, then $1/8 + 1/8 = 1/4$, then $1/4 + 1/4 = 1/2$, then $1/2 + 1/2$. Codewords 0, 10, 110, 1110, 1111, lengths

4. Source coding: compression

$(1, 2, 3, 4, 4)$, $L = 1/2 + 1/2 + 3/8 + 1/4 + 1/4 = 1.875$ bits. Exactly H — because all probabilities are dyadic.

Exercise 4.2

Show that $\sum_i 2^{-\ell_i} = 1$ for a *complete* prefix code (one where every internal node of the tree has two children).

Solution

Replace every edge probability by $1/2$. A leaf at depth ℓ_i then has probability $2^{-\ell_i}$; since the leaves of a full binary tree partition the root event, these probabilities sum to 1. A tree with a missing child has a “hole” of probability 2^{-d} and gives strict inequality.

Exercise 4.3

Is the code $\{0, 01, 011, 0111\}$ uniquely decodable? Is it a prefix code? Does it satisfy Kraft?

Solution

Not a prefix code (0 is a prefix of all others). It is uniquely decodable: every codeword starts with 0, so each 0 marks the start of a new codeword and the number of following 1s identifies it — but decoding needs one bit of lookahead. Kraft: $1/2 + 1/4 + 1/8 + 1/16 = 15/16 \leq 1$. By Kraft–McMillan, the prefix code $\{0, 10, 110, 1110\}$ (the reversed codewords) has the same lengths.

Exercise 4.4

For the $p = 0.9$ coin, compare the expected length per toss of the Huffman code on blocks of $n = 1, 2, 3$ tosses with $H(X) = 0.469$.

Solution

$n = 1$: 1 bit/toss. $n = 2$: probabilities $(0.81, 0.09, 0.09, 0.01)$; Huffman lengths $(1, 2, 3, 3)$, $L_2 = 0.81 + 0.18 + 0.27 + 0.03 = 1.29$, i.e. 0.645 bits/toss. $n = 3$: $L_3 = 1.598$, i.e. 0.533 bits/toss. The rate approaches H as n grows, consistent with $L_n/n < H + 1/n$.

Exercise 4.5

A code designed for $q = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ (lengths 1, 2, 3, 3) is used on a uniform source $p = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$. Compute the expected length and check it equals $H(p) + D(p\|q)$.

Solution

$L = \frac{1}{4}(1 + 2 + 3 + 3) = 2.25$ bits. $H(p) = 2$ and $D(p\|q) = \frac{1}{4}[\log \frac{1/4}{1/2} + \log 1 + 2 \log \frac{1/4}{1/8}] = \frac{1}{4}[-1 + 0 + 2] = 0.25$. Indeed $2 + 0.25 = 2.25$ (the code is complete, $z = 1$).

4.10. Callouts

Note

The fixed-length source-coding theorem is *asymptotic*: it needs $n \sim \sigma^2/(\beta^2\delta)$. Symbol codes work with $n = 1$ but can waste up to one bit per symbol; coding blocks of n symbols reduces this to $1/n$.

Tip

Useful mental picture: an optimal code assigns $\ell_i \approx -\log_2 p_i$, i.e. **codeword length = surprisal**. Conversely, every complete prefix code implicitly defines a probability model $q_i = 2^{-\ell_i}$.

Important

The most probable sequence is not typical. Typicality is about the *information content* $-\frac{1}{n} \log P$, not about the probability being large.

Warning

Universal codes give the *asymptotic* compression rate. On short inputs they often *expand* the data due to header overhead. Don't benchmark compressors on toy examples.

4.11. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapters 3–4. MacKay, §4.2–4.6 (raw and essential bit content, typicality, source coding theorem), §5.1–5.6 (symbol codes, Kraft, Huffman), Ch. 6 (arithmetic and LZ coding). Cover & Thomas, Ch. 3 (AEP), §5.1–5.8 (Kraft, McMillan, Huffman), §10.3 (Gaussian rate-distortion).

5. Noisy channels and coding

So far the channel was perfect: what the sender wrote, the receiver read. Real signals are corrupted by noise, and the receiver may get a string Y different from the transmitted Z . Shannon's second big theorem answers a question that seems, at first, to have a disappointing answer: *can we communicate reliably over a noisy channel without letting the rate go to zero?* Surprisingly, yes — up to a maximum rate, the channel **capacity** $C = \max_{p(x)} I(X; Y)$. Achievability is proved by the beautiful *random-coding* argument, again built on typicality.

Along the way we meet, for the first time, the theme of Part II: **decoding is an inference problem**.

5.1. Decoding as an inference problem

The general scheme is:

$$X \xrightarrow{\text{encoder } E} Z = E(X) \xrightarrow{\text{channel } p(y|z)} Y \xrightarrow{\text{decoder } D} X' = D(Y).$$

The sender encodes the source X with a deterministic encoder $E: \mathcal{A}_X \rightarrow \mathcal{B}^K$; the encoded message Z passes through a noisy channel, which outputs Y with probability $p(y | z)$; the receiver applies a deterministic decoder D and obtains X' . Because of noise the **error probability** $P_e = P(X' \neq X)$ is in general non-zero. Since E is deterministic and injective, $I(X; Y) = I(Z; Y)$.

Definition 5.1 (Discrete memoryless channel). A **discrete memoryless channel (DMC)** is a triple $(\mathcal{A}_Z, \mathcal{A}_Y, p(y | z))$: input alphabet, output alphabet, and a transition probability. Memoryless means that for a length- n transmission

$$p(y^{(n)} | z^{(n)}) = \prod_{i=1}^n p(y_i | z_i).$$

Example 5.1 (Binary symmetric channel)

A fair coin $X \in \{h, t\}$ is encoded as $Z \in \{0, 1\}$ ($h \mapsto 0, t \mapsto 1$) and sent through a channel that flips the bit with probability f :

$$P(Y = Z) = 1 - f, \quad P(Y \neq Z) = f.$$

Decoding Y directly gives $P_e = f$. With $H(Z) = 1$ and $H(Z | Y) = h_2(f)$,

$$I(X; Y) = I(Z; Y) = 1 - h_2(f).$$

At $f = 0$ the received bit carries all the information; at $f = 1/2$ none.

Example 5.2 (Binary erasure channel)

The BEC with erasure probability ε has $\mathcal{A}_Y = \{0, 1, ?\}$, $p(? | z) = \varepsilon$, $p(z | z) = 1 - \varepsilon$. Each input is either received correctly or visibly lost — *never* silently corrupted. For uniform input, $I(Z; Y) = 1 - \varepsilon$.

5.1.1. Estimators and Bayes risk

Reconstructing X from Y is an instance of a very general problem: a latent variable X and an observed variable Y are related by a joint distribution $p_{XY}(x, y)$, and we want an **estimator** $X' = D(Y)$, a deterministic function of the data. Its quality is measured by a **loss** $\ell(x', x)$; for discrete messages the natural one is the 0–1 loss

$$\ell(x', x) = 1 - \delta_{x', x}.$$

Definition 5.2 (Posterior risk and Bayes risk). The **posterior risk** of D given the observation y is the expected loss under the posterior,

$$r_D(y) = \sum_x p(x | y) \ell(D(y), x) = 1 - p(D(y) | y),$$

i.e. the probability of error when y is received. The **Bayes risk** is its average over y ,

$$r_D = \sum_y p(y) r_D(y) = \sum_{x,y} p(x, y) \ell(D(y), x) = P_e.$$

Proposition 5.1 (MAP decoding is optimal). *The decoder that minimises both the posterior risk and the Bayes risk under 0–1 loss is the **maximum a posteriori (MAP)** decoder*

$$D_{MAP}(y) = \arg \max_x p(x | y) = \arg \max_x p(y | x) p(x).$$

*If all messages are equiprobable it coincides with the **maximum-likelihood** decoder $\arg \max_x p(y | x)$.*

Proof. $r_D(y) = 1 - p(D(y) | y)$ is minimised, for each y separately, by choosing $D(y)$ to maximise $p(\cdot | y)$. Minimising pointwise also minimises the average r_D . $\square$

Other losses give other optimal estimators (squared loss $\rightarrow$ posterior mean, absolute loss $\rightarrow$ posterior median); we return to this in Part II.

5.2. Error-correcting codes

To fight noise the sender adds **redundancy**: codewords become longer, and we trade rate for accuracy. Ideally we would use **non-confusable codewords**, i.e. codewords z_i that the channel maps onto *disjoint* sets of outputs, so that every received y is compatible with exactly one z_i .

Definition 5.3 (Rate). If the code assigns codewords of length L to a set S of messages, the **rate** is

$$R = \frac{\log_2 |S|}{L}$$

— effective bits of information per transmitted binary digit. If $|S| = 2^K$, then $R = K/L$. An injective code needs $2^L \geq |S|$, so $R \leq 1$; adding redundancy (using only a subset of the 2^L strings as codewords) lowers R .

5.2.1. Repetition codes

Example 5.3 (*The repetition code R_3*)

Send each bit three times: $h \mapsto 000$, $t \mapsto 111$. Over a BSC the received block has the following probabilities:

sent \ received	000	001, 010, 100	011, 101, 110	111
000	$(1-f)^3$	$f(1-f)^2$	$f^2(1-f)$	f^3
111	f^3	$f^2(1-f)$	$f(1-f)^2$	$(1-f)^3$

Let $m(y)$ be the number of 1s in y . The **majority decoder** outputs h if $m < 2$ and t if $m \geq 2$. This is the MAP decoder: with equiprobable messages, MAP = ML, and

$$\frac{p(y|h)}{p(y|t)} = \frac{f^m(1-f)^{K-m}}{f^{K-m}(1-f)^m} = \left(\frac{f}{1-f}\right)^{2m-K},$$

which for $f < 1/2$ is > 1 iff $m < K/2$.

The error probability is the probability that two or three bits flip:

$$P_e = 3f^2(1-f) + f^3 = 3f^2 - 2f^3.$$

For $f = 0.1$, $P_e \approx 0.028$, much smaller than f — but the rate has dropped to $R = 1/3$.

For a general repetition code R_K (K odd) with majority decoding,

$$P_e = \sum_{n=\lceil K/2 \rceil}^K \binom{K}{n} f^n (1-f)^{K-n} \approx \binom{K}{\lceil K/2 \rceil} f^{\lceil K/2 \rceil} (1-f)^{\lfloor K/2 \rfloor},$$

dominated by the smallest number of flips that causes an error. Asymptotically $P_e \sim 2^{-KD(1/2||f)}$. The error vanishes as $K \rightarrow \infty$, but so does the rate $R = 1/K$: for $f = 0.1$, reaching $P_e \approx 10^{-15}$ needs $K = 61$, i.e. $R \approx 0.016$. Repetition codes buy reliability at a ruinous price.

5.2.2. The Hamming (7,4) code

Example 5.4 (*Hamming (7,4)*)

Encode blocks of 4 source bits $z_1 z_2 z_3 z_4$ into 7 bits by appending three **parity-check bits** (sums modulo 2, $1 \oplus 1 = 0$):

$$z_5 = z_1 \oplus z_2 \oplus z_3, \quad z_6 = z_2 \oplus z_3 \oplus z_4, \quad z_7 = z_1 \oplus z_2 \oplus z_4.$$

Picture three overlapping circles, each containing four bits whose sum must be even. In matrix form $\mathbf{z} = G \mathbf{z}^{(4)} \pmod{2}$ with the 7×4 generator matrix

$$G = \begin{pmatrix} I_4 \\ P \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}.$$

The 16 codewords differ pairwise in at least 3 positions (minimum distance 3), so any single flip leaves the received word closer to the transmitted codeword than to any other. The rate is $R = 4/7$.

Decoding. Write $\mathbf{y} = \mathbf{z} \oplus \epsilon$ with ϵ the noise vector ($\epsilon_i = 1$ if bit i flipped). For a BSC, $P(\epsilon) = f^{m(\epsilon)}(1-f)^{7-m(\epsilon)}$, which (for $f < 1/2$) decreases with the number of flips $m(\epsilon)$. Hence the MAP decoder picks **the codeword closest to $\mathbf{y}$ in Hamming distance**:

$$\hat{\mathbf{z}} = \mathbf{y} \oplus \arg \min_{\epsilon: \mathbf{y} \oplus \epsilon \in \text{codebook}} m(\epsilon).$$

This is done efficiently with the **parity-check matrix**

$$H = (P \ I_3) = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad HG = 0 \pmod{2}.$$

The **syndrome** $\mathbf{s} = H\mathbf{y} = H(\mathbf{z} \oplus \epsilon) = H\epsilon$ depends only on the noise. A flip of bit i produces the syndrome equal to the i -th column of H ; all seven columns are distinct and non-zero, so the seven single-flip patterns are in one-to-one correspondence with the seven non-zero syndromes:

syn- drome	000	001	010	100	011	101	110	111
unflip	none	z_7	z_6	z_5	z_4	z_1	z_3	z_2

(A flip in a source bit $z_1 \dots z_4$ violates at least two checks; a flip in a parity bit violates exactly one.)

Worked example. Source 1000 $\rightarrow$ codeword 1000101. Suppose bit 3 flips: $\mathbf{y} = 1010101$. Then $s_1 = y_1 \oplus y_2 \oplus y_3 \oplus y_5 = 1$, $s_2 = y_2 \oplus y_3 \oplus y_4 \oplus y_6 = 1$, $s_3 = y_1 \oplus y_2 \oplus y_4 \oplus y_7 = 0$: syndrome 110 $\rightarrow$ unflip z_3 , recovering 1000.

Decoding fails only when two or more bits flip in a block, so the block error is $P_B = 1 - (1-f)^7 - 7f(1-f)^6 \approx \binom{7}{2}f^2 = 21f^2$ for small f (exactly 0.15 at $f = 0.1$). The error is $O(f^2)$, **as for R_3 , but at rate $4/7$ rather than $1/3$.**

5.2.3. How good can a code be?

We want both large R and small P_e . Repetition codes suggest that $P_e \rightarrow 0$ forces $R \rightarrow 0$. Shannon's surprise is that this is false: **P_e can be made arbitrarily small at any rate below a finite threshold, the capacity.**

Definition 5.4 (Channel capacity). The **capacity** of a channel is the maximal mutual information between input and output, over all input distributions:

$$C = \max_{p_Z} I(Z; Y).$$

For a noiseless channel $I(Z; Y) = H(Z)$ and $C = \max H(Z) = \log |\mathcal{A}_Z|$, consistent with the source coding theorem.

Example 5.5 (Capacity of the BSC)

With $p_0 = P(Z = 0)$, the output is Bernoulli with $P(Y = 1) = p_0 f + (1 - p_0)(1 - f)$, and $H(Y | Z) = h_2(f)$ whatever the input. So

$$I(Z; Y) = h_2(p_0 f + (1 - p_0)(1 - f)) - h_2(f),$$

maximised at $p_0 = 1/2$ where the output is uniform:

$$C_{\text{BSC}} = 1 - h_2(f).$$

For $f = 0.1$, $C \approx 0.53$: reliable communication is possible at about half a bit per use — far better than the $R \rightarrow 0$ of repetition codes.

Example 5.6 (Capacity of the BEC)

$H(Y) = h_2(\varepsilon) + (1 - \varepsilon)h_2(p_0)$ and $H(Y | Z) = h_2(\varepsilon)$, so $I = (1 - \varepsilon)h_2(p_0)$ and $C_{\text{BEC}} = 1 - \varepsilon$: each non-erased symbol conveys one bit.

Example 5.7 (The noisy typewriter)

The alphabet is $\{A, B, \dots, Z, -\}$ (27 symbols, arranged in a circle so that $-$ precedes A). Each input is received as itself, its predecessor or its successor, each with probability $1/3$ — an inaccurate typewriter hitting neighbouring keys:

$$P(Y = A | Z = B) = P(Y = B | Z = B) = P(Y = C | Z = B) = \frac{1}{3}, \text{ etc.}$$

Since $H(Y | Z) = \log 3$ for every input, $I(Z; Y) = H(Y) - \log 3$, maximised when Y (hence Z) is uniform: $H(Y) = \log 27 = 3 \log 3$ and

$$C = 3 \log 3 - \log 3 = 2 \log 3 = \log 9 \approx 3.17 \text{ bits.}$$

The capacity is achieved *with zero error* by using only every third letter, $S = \{B, E, H, K, N, Q, T, W, Z\}$: the output sets $\{A, B, C\}, \{D, E, F\}, \dots$ are disjoint, so these 9 codewords are non-confusable, and $R = \log 9$ per use. In this case $n_Z \cdot n_{Y|Z} = 9 \times 3 = 27 = n_Y$, so $H(Z | Y) = 0$.

The noisy typewriter is special: non-confusable codewords exist for a single use. Shannon's insight is that *every* channel looks like a noisy typewriter if we use it n times and n is large.

5.3. The noisy-channel coding theorem

Definition 5.5 (Block code). A (K, n) **block code** is a list of 2^K codewords $S = \{\mathbf{z}_1, \dots, \mathbf{z}_{2^K}\} \subset \mathcal{A}_Z^n$, used to encode 2^K messages; its rate is $R = K/n$. A decoder maps $\mathcal{A}_Y^n \rightarrow S$. The **block error probability** is $P_B = P(D(\mathbf{Y}) \neq \mathbf{Z})$, and the **maximal block error probability** is $P_B^{\max} = \max_{\mathbf{z} \in S} P(D(\mathbf{Y}) \neq \mathbf{z} | \mathbf{Z} = \mathbf{z})$.

The repetition code R_n is a $(1, n)$ block code; the Hamming code is a $(4, 7)$ code.

Theorem 5.2 (Shannon's noisy-channel coding theorem). *For a discrete memoryless channel with capacity $C = \max_{p_Z} I(Z; Y)$:*

1. (**Achievability**) *For every $\delta > 0$ and $\beta > 0$, for n large enough there exists a block code of length n and rate $R \geq C - \beta$, with a decoder whose maximal block error probability is below δ .*
2. (**Converse**) *No sequence of codes with rate $R > C$ has error probability tending to zero.*

5.3.1. The intuition: every channel is a noisy typewriter

1. Use the channel n times. For large n , the input $\mathbf{z}$ is (with high probability) typical, one of about $2^{nH(Z)}$ sequences.
2. The output $\mathbf{y}$ is typical, one of about $2^{nH(Y)}$ sequences.
3. The pair $(\mathbf{z}, \mathbf{y})$ is **jointly typical**, one of about $2^{nH(Z,Y)}$ pairs.
4. Hence, for a given input $\mathbf{z}$, the output is essentially one of $2^{nH(Z,Y)-nH(Z)} = 2^{nH(Y|Z)}$ sequences — a “fan” of possible outputs.
5. If we use 2^{nR} codewords, their fans are (nearly) disjoint — non-confusable — only if they fit in the output space: $2^{nR} 2^{nH(Y|Z)} \leq 2^{nH(Y)}$, i.e.

$$R \leq H(Y) - H(Y|Z) = I(Z;Y) \leq C.$$

6. Conversely, choosing codewords at random among typical inputs and decoding by joint typicality achieves any $R < C$ with vanishing error.

5.3.2. Joint typicality

Definition 5.6 (Jointly typical sequences). Given $p(z, y)$, the jointly typical set $T_{n\beta}[Z, Y]$ is the set of pairs $(\mathbf{z}, \mathbf{y})$ such that

$$\left| -\frac{1}{n} \log p(\mathbf{z}) - H(Z) \right| < \beta, \quad \left| -\frac{1}{n} \log p(\mathbf{y}) - H(Y) \right| < \beta, \quad \left| -\frac{1}{n} \log p(\mathbf{z}, \mathbf{y}) - H(Z, Y) \right| < \beta.$$

Lemma 5.3 (Joint AEP). Let $(\mathbf{Z}, \mathbf{Y})$ be drawn i.i.d. from $p(z, y)$. As $n \rightarrow \infty$:

1. $P((\mathbf{Z}, \mathbf{Y}) \in T_{n\beta}) \rightarrow 1$;
2. $|T_{n\beta}[Z, Y]| \leq 2^{n(H(Z,Y)+\beta)}$;
3. if $\tilde{\mathbf{Z}}$ and $\tilde{\mathbf{Y}}$ are drawn **independently** from the marginals $p(\mathbf{z})$ and $p(\mathbf{y})$, then

$$P((\tilde{\mathbf{Z}}, \tilde{\mathbf{Y}}) \in T_{n\beta}) \leq 2^{-n(I(Z;Y)-3\beta)}.$$

Proof.

- (1) and (2) follow exactly as for the single-variable AEP. For (3), every jointly typical pair has $p(\mathbf{z}) \leq 2^{-n(H(Z)-\beta)}$ and $p(\mathbf{y}) \leq 2^{-n(H(Y)-\beta)}$, so

$$P = \sum_{(\mathbf{z}, \mathbf{y}) \in T_{n\beta}} p(\mathbf{z}) p(\mathbf{y}) \leq 2^{n(H(Z,Y)+\beta)} 2^{-n(H(Z)-\beta)} 2^{-n(H(Y)-\beta)} = 2^{-n(I(Z;Y)-3\beta)}.$$

□

Property 3 is the heart of the theorem: **two independently drawn sequences almost never look jointly typical**; the probability of a “spurious match” is exponentially small in $nI(Z;Y)$.

5.3.3. Random coding and typical-set decoding

Proof of achievability. Fix p_Z achieving the capacity.

Random coding. Draw $2^K = 2^{nR}$ codewords $\mathbf{z}_1, \dots, \mathbf{z}_{2^K}$ independently, each i.i.d. from p_Z .

Typical-set decoding. On receiving $\mathbf{y}$, output the codeword $\mathbf{z}'$ that is jointly typical with $\mathbf{y}$; if there is none, or more than one, declare failure.

The error of a given codebook S , $r(S)$, is hard to compute. **Shannon's trick is to average over all codebooks.** By symmetry the averaged error equals the error when message 1 is sent:

$$\langle r(S) \rangle_S = P(\mathbf{z}' \neq \mathbf{z}_1).$$

When $\mathbf{z}_1$ is sent and $\mathbf{y}$ is received, an error occurs only if

- E_1 : $\mathbf{z}_1$ is not jointly typical with $\mathbf{y}$. Since $(\mathbf{z}_1, \mathbf{y})$ is drawn from the joint distribution, by the joint AEP $P(E_1) < \delta/2$ for large n ;
- E_2 : some other codeword $\mathbf{z}_i$, $i \geq 2$, is jointly typical with $\mathbf{y}$. Each $\mathbf{z}_i$ is independent of $\mathbf{y}$, so by the union bound and property 3,

$$P(E_2) \leq \sum_{i=2}^{2^{nR}} P((\mathbf{z}_i, \mathbf{y}) \in T_{n\beta}) \leq 2^{nR} 2^{-n(I(Z;Y)-3\beta)} = 2^{-n(C-R-3\beta)}.$$

This is smaller than $\delta/2$ for large n provided $R < C - 3\beta$.

Hence $\langle r(S) \rangle_S \leq P(E_1) + P(E_2) < \delta$. **Since the average over codebooks is below δ , at least one codebook S^* has $r(S^*) < \delta$.**

From average to maximal error (expurgation). $r(S^*)$ is an average over messages. At most half the codewords can have error above 2δ ; discard them. The remaining 2^{nR-1} codewords have maximal error $< 2\delta$ and rate $R - 1/n$, which is still $\geq C - \beta'$ for large n . Since β, δ are arbitrary, the theorem follows. $\square$

Remark. The genius of the proof is to show existence by averaging. A random codebook is bad in ways that are hard to analyse, but the *ensemble* has good average behaviour — so at least one codebook is at least as good as the average. In fact almost all random codebooks are good; the hard part is decoding them efficiently (typical-set decoding of a random code takes time exponential in n).

5.3.4. The converse: Fano's inequality

Theorem 5.4 (Fano's inequality). *Let X take values in $\mathcal{X}$ and let $\hat{X} = g(Y)$ be any estimate of X from Y , with $P_e = P(\hat{X} \neq X)$. Then*

$$H(X | Y) \leq H(X | \hat{X}) \leq h_2(P_e) + P_e \log_2(|\mathcal{X}| - 1).$$

Fano's inequality says that small error forces small residual uncertainty; contrapositively, if $H(X | Y)$ is large, every estimator errs often.

Proof sketch of the converse. Send a uniformly chosen message $W \in \{1, \dots, 2^{nR}\}$, so $H(W) = nR$. Then

$$nR = H(W) = H(W | \mathbf{Y}) + I(W; \mathbf{Y}) \leq (1 + P_e nR) + I(\mathbf{Z}; \mathbf{Y}) \leq 1 + P_e nR + nC,$$

using Fano, the data-processing inequality ($W \rightarrow \mathbf{Z} \rightarrow \mathbf{Y}$) and memorylessness ($I(\mathbf{Z}; \mathbf{Y}) \leq \sum_i I(Z_i; Y_i) \leq nC$). Dividing by n : $R \leq C + 1/n + P_e R$, i.e. $P_e \geq 1 - C/R - 1/(nR)$. If $R > C$ the error is bounded away from zero. $\square$

Proposition 5.5 (Random-coding exponent). *For $R < C$ the average error over random codebooks decays exponentially, $\bar{P}_e \leq 2^{-nE_r(R)}$, where $E_r(R) > 0$ is Gallager's random-coding exponent. It quantifies how large n must be for a given target error.*

5.4. Structured codes

The random-coding proof does not construct a practical code; it shows one exists. Practical coding needs structured codes with fast decoders.

- **Repetition codes:** rate $1/K$, error $\sim 2^{-KD(1/2\|f)}$. A baseline.
- **Hamming codes:** rate $4/7$, single-error correcting; the first *algebraic* (linear) codes.
- **Modern codes** — LDPC (Gallager 1962), turbo codes (Berrou et al. 1993), polar codes (Arikan 2009) — approach capacity within fractions of a dB and are decoded in (near-)linear time by *message passing*, i.e. approximate Bayesian inference on a graph. Their invention represents sixty years of work catching up with Shannon.

5.5. Exercises

Exercise 5.1

Compute the capacity of the Z-channel: $\mathcal{A}_Z = \mathcal{A}_Y = \{0, 1\}$, $p(0 | 0) = 1$, $p(0 | 1) = q$, $p(1 | 1) = 1 - q$.

Solution

Let $P(Z = 1) = \pi$. Then $P(Y = 1) = \pi(1 - q)$, $H(Y | Z) = \pi h_2(q)$, and $I = h_2(\pi(1 - q)) - \pi h_2(q)$. Setting $dI/d\pi = 0$ gives the optimum, and the capacity has the closed form $C = \log_2(1 + (1 - q)q^{q/(1-q)})$. For $q = 0.5$: $C = \log_2(1.25) \approx 0.322$ bits per use, at $\pi \approx 0.4$. Compare the BSC with $f = 0.5$, whose capacity is zero: the Z-channel's asymmetry ("0" is never corrupted) keeps it useful.

Exercise 5.2

Prove Fano's inequality using the error indicator $E = \mathbf{1}[\hat{X} \neq X]$.

Solution

Expand $H(E, X | \hat{X})$ in two ways. First, $H(E, X | \hat{X}) = H(X | \hat{X}) + H(E | X, \hat{X}) = H(X | \hat{X})$, since E is a function of $(X, \hat{X})$. Second, $H(E, X | \hat{X}) = H(E | \hat{X}) + H(X | E, \hat{X}) \leq h_2(P_e) + P_e \log_2(|\mathcal{X}| - 1)$: conditioning reduces entropy ($H(E | \hat{X}) \leq H(E) = h_2(P_e)$); if $E = 0$ then $X = \hat{X}$ and there is no uncertainty; if $E = 1$, X is one of the remaining $|\mathcal{X}| - 1$ values. Finally $H(X | Y) \leq H(X | \hat{X})$ by data processing.

Exercise 5.3

For the repetition code R_5 on a BSC with $f = 0.1$, compute P_e and compare with R_3 .

Solution

Error iff ≥ 3 flips: $P_e = \binom{5}{3}f^3(1-f)^2 + \binom{5}{4}f^4(1-f) + f^5 = 10(0.001)(0.81) + 5(0.0001)(0.9) + 10^{-5} \approx 0.0081 + 0.00045 + 0.00001 \approx 0.0086$. R_3 had 0.028 at rate $1/3$; R_5 has 0.0086 at rate $1/5$. Each factor ~ 3 improvement costs a big drop in rate.

Exercise 5.4

In the Hamming (7,4) code, the received word is $\mathbf{y} = 0110110$. Decode it.

Solution

$s_1 = y_1 \oplus y_2 \oplus y_3 \oplus y_5 = 0 \oplus 1 \oplus 1 \oplus 1 = 1$; $s_2 = y_2 \oplus y_3 \oplus y_4 \oplus y_6 = 1 \oplus 1 \oplus 0 \oplus 1 = 1$;
 $s_3 = y_1 \oplus y_2 \oplus y_4 \oplus y_7 = 0 \oplus 1 \oplus 0 \oplus 0 = 1$. Syndrome 111 $\rightarrow$ flip z_2 : $\hat{\mathbf{z}} = 0010110$. Check:
 $z_5 = 0 \oplus 0 \oplus 1 = 1$, $z_6 = 0 \oplus 1 \oplus 0 = 1$, $z_7 = 0 \oplus 0 \oplus 0 = 0 \checkmark$. Decoded message: 0010.

5.6. Callouts

i Note

The noisy-channel coding theorem gives mutual information its operational meaning: $\max I(Z; Y)$ is the maximum rate of reliable communication. Decoding is inference, and the optimal decoder under 0–1 loss is MAP.

💡 Tip

For a symmetric channel (every row of $p(y | z)$ a permutation of every other, and likewise columns) the capacity is achieved by the uniform input and equals $\log |\mathcal{A}_Y| - H(\text{row})$. BSC, BEC-like and noisy typewriter examples all follow from this.

⚠ Warning

Shannon's theorem is an existence result. Constructing capacity-achieving codes with efficient decoders was an open problem for decades.

! Important

The capacity C is an asymptotic (long-block) quantity. At finite block length the achievable rate is lower, roughly $C - \sqrt{V/n} Q^{-1}(\epsilon)$ (Polyanskiy, Poor & Verdú 2010).

5.7. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 5. MacKay, §1.1–1.2 (BSC, repetition and Hamming codes), §9.3–9.7 (capacity, noisy typewriter), Ch. 10 (joint typicality, random coding). Cover & Thomas, §7.1–7.11 (capacity, joint AEP, coding theorem, Fano, Hamming codes). Barbier, §2.4 (estimators and Bayes risk).

6. Continuous random variables

Shannon's theory extends from discrete to continuous variables with minor changes in formalism — probabilities become densities, sums become integrals — but with an important caveat: the “entropy” of a continuous variable is *infinite*, and what survives is a finite part, the **differential entropy**, which can be negative and depends on the choice of coordinates. Mutual information and KL divergence, on the other hand, carry over unchanged. This chapter makes these statements precise and then studies the paradigm of a continuous system, the Gaussian channel.

6.1. Differential entropy

Definition 6.1 (Differential entropy). For a random variable X with probability density $f_X(x)$, the **differential entropy** is

$$h[X] = - \int f_X(x) \log f_X(x) dx,$$

provided the integral exists.

It is tempting to call this “the Shannon entropy of a continuous variable”. That identification is not correct, as the simplest example shows.

6.1.1. Discretisation: what h really measures

Example 6.1 (*Uniform variable measured with finite resolution*)

Let $X \sim \mathcal{U}(-L/2, L/2)$. In practice we can only measure X to some resolution $\epsilon \ll 1$. Take for simplicity $L = 2^{m_1}$, $\epsilon = 2^{-m_2}$. The range splits into $n_c = L/\epsilon = 2^{m_1+m_2}$ cells c_i of width ϵ , each with probability $1/n_c$, so at resolution ϵ the variable is effectively a discrete $X_\epsilon \in \{1, \dots, n_c\}$ with

$$H[X_\epsilon] = \log n_c = \log \frac{L}{\epsilon} = \log L + \log \frac{1}{\epsilon}.$$

The differential entropy is instead

$$h[X] = - \int_{-L/2}^{L/2} \frac{1}{L} \log \frac{1}{L} dx = \log L = \log \text{Vol}(\text{Supp } X).$$

Hence

$$H[X_\epsilon] = h[X] + \log \frac{1}{\epsilon}.$$

As $\epsilon \rightarrow 0$, $H[X_\epsilon] \rightarrow \infty$: specifying a real number to infinite precision requires infinite information.

Theorem 6.1 (Differential entropy as a limit). For a (sufficiently regular) density f_X on $\mathbb{R}^k$ and a

partition into cells of volume ϵ^k ,

$$H[X_\epsilon] = h[X] + \log \frac{1}{\epsilon^k} + O(\epsilon).$$

Proof. Take $k = 1$. Choose a region A containing the support, or, for unbounded support, excluding only tails of probability $O(\epsilon^2)$. Split A into cells c_i of width ϵ with centres x_i , and let X_ϵ be the index of the cell (plus one extra value for $\bar{A}$). For small ϵ ,

$$P(X \in c_i) = \int_{c_i} f_X(x) dx = f_X(x_i) \epsilon + O(\epsilon^2).$$

Then

$$H[X_\epsilon] = - \sum_i f_X(x_i) \epsilon \log(f_X(x_i) \epsilon) + O(\epsilon) = - \sum_i f_X(x_i) \log f_X(x_i) \epsilon - \log \epsilon \sum_i f_X(x_i) \epsilon + O(\epsilon).$$

Both sums are Riemann sums: the first converges to $h[X]$, the second to $\int f_X = 1$. Thus $H[X_\epsilon] = h[X] - \log \epsilon + O(\epsilon)$. In k dimensions, cells have volume ϵ^k and $-\log \epsilon$ becomes $-\log \epsilon^k$. $\square$

So $h[X]$ is the **finite part** of the entropy, after subtracting the divergent, resolution-dependent term $\log(1/\epsilon^k)$. The term $\log(1/\epsilon)$ is what keeps $H[X_\epsilon] \geq 0$ even when $h[X] < 0$.

Remark. $h[X]$ is *not* the average number of bits needed to describe X . It measures entropy *relative to a uniform reference on unit cells*: a negative h means that X is more concentrated than a uniform variable on a unit interval, so at any fixed resolution it is cheaper to describe.

Example 6.2 (Uniform and Gaussian differential entropies)

Uniform on an interval of length L : $h = \log L$, negative if $L < 1$.

Gaussian $X \sim \mathcal{N}(\mu, \sigma^2)$: $\log f(x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2} \log e$, and $\mathbb{E}[(X - \mu)^2] = \sigma^2$, so

$$h[X] = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2} \log e = \frac{1}{2} \log(2\pi e \sigma^2), \quad H[X_\epsilon] \simeq \frac{1}{2} \log \frac{2\pi e \sigma^2}{\epsilon^2}.$$

Independent of μ ; negative if $\sigma^2 < 1/(2\pi e)$.

Multivariate Gaussian $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^k$:

$$h[\mathbf{X}] = \frac{1}{2} \log((2\pi e)^k \det \Sigma), \quad H[\mathbf{X}_\epsilon] \simeq \frac{1}{2} \log((2\pi e)^k \det \Sigma) + \log \frac{1}{\epsilon^k}.$$

The entropy grows with the volume of the “error ellipsoid”, $\sqrt{\det \Sigma}$.

Exponential with mean μ : $h = \log(e\mu)$.

6.2. Joint, conditional, mutual, relative

The other quantities generalise in the same way. Consider $\mathbf{X} \in \mathbb{R}^k$, $\mathbf{Y} \in \mathbb{R}^l$.

Conditional entropy. $H_\epsilon[X | Y] = H_\epsilon[X, Y] - H_\epsilon[Y] = h[X, Y] - h[Y] + \log \frac{1}{\epsilon^k}$, i.e.

$$H_\epsilon[X | Y] = h[X | Y] + \log \frac{1}{\epsilon^k}, \quad h[X | Y] = - \int dx dy f_{XY}(x, y) \log \frac{f_{XY}(x, y)}{f_Y(y)}.$$

Like $h[X]$, it is resolution-dependent and can be negative.

Mutual information. The divergent terms cancel:

$$I_\epsilon[X : Y] = h[X] + \log \frac{1}{\epsilon^k} + h[Y] + \log \frac{1}{\epsilon^l} - h[X, Y] - \log \frac{1}{\epsilon^{k+l}} = \int dx dy f_{XY} \log \frac{f_{XY}(x, y)}{f_X(x)f_Y(y)} \equiv i[X : Y].$$

Mutual information does not depend on the resolution and coincides with its differential expression. It remains a KL divergence, hence ≥ 0 .

KL divergence.

$$D_{KL}(f \| g) = \int dx f(x) \log \frac{f(x)}{g(x)} \geq 0,$$

again resolution-independent (the ϵ 's cancel in the ratio).

Example 6.3 (*Mutual information of jointly Gaussian variables*)

If $(\mathbf{X}, \mathbf{Y})$ is jointly Gaussian with marginal covariances Σ_X, Σ_Y and joint covariance Σ_{XY} , then

$$i[\mathbf{X} : \mathbf{Y}] = h[\mathbf{X}] + h[\mathbf{Y}] - h[\mathbf{X}, \mathbf{Y}] = \frac{1}{2} \log \frac{\det \Sigma_X \det \Sigma_Y}{\det \Sigma_{XY}}.$$

For two scalars with variances a, b and correlation coefficient c ,

$$\Sigma_{XY} = \begin{pmatrix} a & c\sqrt{ab} \\ c\sqrt{ab} & b \end{pmatrix}, \quad \det \Sigma_{XY} = ab(1 - c^2),$$

so

$$i[X : Y] = -\frac{1}{2} \log(1 - c^2).$$

It vanishes for $c = 0$ (for Gaussians, uncorrelated = independent) and diverges as $|c| \rightarrow 1$ (one variable determines the other exactly, and a real number carries infinite information). For $c = 0.5$, $i \approx 0.21$ bits; for $c = 0.9$, $i \approx 1.2$ bits.

6.3. Properties of differential entropy

Proposition 6.2 (Properties of differential entropy). 1. $h[X]$ can be negative.

2. Translation: $h[X + c] = h[X]$. Scaling: $h[aX] = h[X] + \log |a|$; for vectors, $h[A\mathbf{X}] = h[\mathbf{X}] + \log |\det A|$.

3. More generally, for an invertible smooth $\mathbf{Y} = g(\mathbf{X})$, $h[\mathbf{Y}] = h[\mathbf{X}] + \mathbb{E}[\log |\det J_g(\mathbf{X})|]$ — h **depends on the coordinates**.

4. Chain rule: $h[X, Y] = h[X] + h[Y | X]$.

5. Conditioning reduces entropy: $h[Y | X] \leq h[Y]$; sub-additivity $h[X, Y] \leq h[X] + h[Y]$, with equality iff independent.

The scaling rule is just the discretisation formula in disguise: stretching X by a is equivalent to measuring it with resolution ϵ/a .

Proposition 6.3 (Invariance of mutual information). For diffeomorphisms ϕ, ψ acting separately on X and Y ,

$$i[\phi(X) : \psi(Y)] = i[X : Y].$$

The Jacobian terms cancel between $h[X] + h[Y]$ and $h[X, Y]$. This makes i a reparameterisation-invariant measure of dependence.

6.4. Maximum-entropy distributions

Discrete entropies are maximised by the uniform distribution. Continuous ones need a constraint (otherwise h is unbounded), and the maximiser depends on the constraint.

Theorem 6.4 (The Gaussian maximises entropy at fixed variance). *Among all real random variables with variance σ^2 , the Gaussian has the largest differential entropy, $h = \frac{1}{2} \log(2\pi e \sigma^2)$. In $\mathbb{R}^k$, among all vectors with covariance Σ , $\mathcal{N}(\mu, \Sigma)$ is the maximiser.*

Proof. Let f have mean μ and variance σ^2 , and ϕ be the Gaussian with the same moments. Since $\log \phi(x)$ is a quadratic polynomial in x , $\int f \log \phi = \int \phi \log \phi$ (only the first two moments enter). Then

$$0 \leq D(f \parallel \phi) = \int f \log f - \int f \log \phi = -h(f) + h(\phi).$$

□

Corollary 6.5 (Other max-ent distributions). *Support $[a, b]$, no other constraint: uniform, $h = \log(b - a)$.*

- *Support $[0, \infty)$, fixed mean μ : exponential, $h = \log(e\mu)$.*
- *In general, fixing $\mathbb{E}[T_j(X)]$ gives an exponential family $f(x) \propto \exp(\sum_j \lambda_j T_j(x))$ — the continuous analogue of the Boltzmann distribution.*

6.5. The Gaussian channel

The noisy-channel framework extends to continuous inputs and outputs. Let Z be the input and Y the output.

Definition 6.2 (Additive white Gaussian noise (AWGN) channel).

$$Y = Z + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2), \quad f_{Y|Z}(y | z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-z)^2}{2\sigma^2}}.$$

The noise is *additive*, and *white* if independent across uses. The input must be constrained — otherwise a single real number could carry infinite information. The standard constraint is a power (variance) constraint $\text{Var}[Z] \leq v^2$.

Theorem 6.6 (Gaussian channel capacity). *Under $\text{Var}[Z] \leq v^2$, the capacity is achieved by a Gaussian input $Z \sim \mathcal{N}(\xi, v^2)$ and equals*

$$C = \frac{1}{2} \log\left(1 + \frac{v^2}{\sigma^2}\right) = \frac{1}{2} \log(1 + \text{SNR}), \quad \text{SNR} = \frac{v^2}{\sigma^2}.$$

Proof via maximum entropy. $I[Z : Y] = h[Y] - h[Y | Z] = h[Y] - h[\eta]$, since given $Z = z$, Y is just η shifted by z . Since Z and η are independent, $\text{Var}[Y] = \text{Var}[Z] + \sigma^2 \leq v^2 + \sigma^2$, so by the max-ent theorem $h[Y] \leq \frac{1}{2} \log(2\pi e(v^2 + \sigma^2))$, with equality iff Y is Gaussian — which happens iff Z is Gaussian (sum of independent variables is Gaussian only if both are). Hence

$$C = \frac{1}{2} \log(2\pi e(v^2 + \sigma^2)) - \frac{1}{2} \log(2\pi e\sigma^2) = \frac{1}{2} \log\left(1 + \frac{v^2}{\sigma^2}\right).$$

Equivalently, using the Gaussian MI formula with $\Sigma_{ZY} = \begin{pmatrix} v^2 & v^2 \\ v^2 & v^2 + \sigma^2 \end{pmatrix}$: $i = \frac{1}{2} \log \frac{v^2(v^2 + \sigma^2)}{v^2(v^2 + \sigma^2) - v^4}$. $\square$

Proof via the calculus of variations. Maximise $I[Z : Y]$ over f_Z with Lagrange multipliers for normalisation and variance:

$$\mathcal{F}[f_Z] = \int dz f_Z(z) \left[\int dy f(y|z) \log \frac{f(y|z)}{f_Y(y)} - \lambda - \mu z^2 \right], \quad f_Y(y) = \int dz f_Z(z) f(y|z).$$

Setting $\delta \mathcal{F} / \delta f_Z(z^*) = 0$, and noting that the variation of f_Y inside the logarithm contributes $-\int dy f(y|z^*) = -1$ (absorbed into λ), gives

$$\int dy f(y|z^*) \log f(y|z^*) - \int dy f(y|z^*) \log f_Y(y) = \lambda' + \mu z^{*2}.$$

The first term is $-h[\eta]$, a constant. So the Gaussian smoothing of $\log f_Y$ must be a quadratic function of z^* , which forces $\log f_Y(y)$ to be quadratic in y : $f_Y \propto e^{a+cy^2}$, i.e. Y is Gaussian, hence so is Z . $\square$

6.5.1. Decoding a Gaussian channel

With prior $Z \sim \mathcal{N}(0, v^2)$ (take $\xi = 0$) and likelihood $Y | Z \sim \mathcal{N}(Z, \sigma^2)$, Bayes' theorem gives a Gaussian posterior:

$$Z | Y = y \sim \mathcal{N}\left(\frac{v^2}{v^2 + \sigma^2} y, \left(\frac{1}{v^2} + \frac{1}{\sigma^2}\right)^{-1}\right).$$

Precisions add; the posterior mean is a **shrinkage** of the observation towards the prior mean. The MAP and posterior-mean estimators coincide (the posterior is symmetric and unimodal):

$$z' = D(y) = \frac{v^2}{v^2 + \sigma^2} y.$$

The mean squared error of this decoder is the posterior variance,

$$\mathbb{E}[(z' - Z)^2] = \frac{v^2 \sigma^2}{v^2 + \sigma^2} = \sigma^2 \frac{\text{SNR}}{1 + \text{SNR}},$$

which is less than both the prior variance v^2 (ignoring the data) and the noise variance σ^2 (trusting the data blindly).

Remark (Two limits). If $\text{SNR} \rightarrow 0$ the posterior reduces to the prior $\mathcal{N}(0, v^2)$: the data are useless. If $\text{SNR} \rightarrow \infty$ the posterior reduces to the likelihood $\mathcal{N}(y, \sigma^2)$, centred on the observation. Bayesian inference interpolates between “believe the prior” and “believe the data” according to their relative precisions.

6.5.2. Continuous-time signals and bandwidth

A Gaussian channel arises naturally when a time-continuous signal $z(t)$, $t \in [0, T]$, is sent through a medium adding white noise. Encode N numbers z_k as coefficients of N orthonormal functions ϕ_k :

$$z(t) = \sum_{k=1}^N z_k \phi_k(t), \quad z_k = \int_0^T \phi_k(t) z(t) dt,$$

e.g. sines and cosines of frequencies up to W , with $N = 2WT$ (the Nyquist count). A power constraint $\frac{1}{T} \int_0^T z(t)^2 dt \leq P$ becomes $\sum_k z_k^2 \leq TP$, i.e. $\langle z_k^2 \rangle \leq TP/N = P/(2W)$.

The receiver gets $y(t) = z(t) + \eta(t)$, so $y_k = z_k + \eta_k$ with $\eta_k = \int_0^T \phi_k(t) \eta(t) dt$. For white noise, $\mathbb{E}[\eta(t)] = 0$

and $\mathbb{E}[\eta(t)\eta(t')] = \sigma^2\delta(t-t')$:

$$\mathbb{E}[\eta_k] = 0, \quad \mathbb{E}[\eta_k\eta_l] = \sigma^2 \int_0^T \int_0^T dt dt' \phi_k(t)\phi_l(t')\delta(t-t') = \sigma^2 \int_0^T \phi_k\phi_l dt = \sigma^2\delta_{kl}.$$

The noise components are independent $\mathcal{N}(0, \sigma^2)$: **sending the signal is equivalent to $N = 2WT$ uses of a Gaussian channel**. Writing the (two-sided) noise spectral density as $\sigma^2 = N_0/2$, the capacity per second is

$$C = \frac{N}{T} \cdot \frac{1}{2} \log\left(1 + \frac{P/(2W)}{N_0/2}\right) = W \log\left(1 + \frac{P}{N_0W}\right) \text{ bits/s},$$

the **Shannon–Hartley formula**, which governs everything from modems to Wi-Fi.

Example 6.4 (*Telephone line*)

A telephone line with $W = 3$ kHz and $\text{SNR} = 30$ dB ($P/(N_0W) = 1000$) has capacity $C = 3000 \log_2 1001 \approx 30$ kbit/s — the ceiling that analogue modems approached in the 1990s.

6.5.3. Sphere packing: the coding theorem for the Gaussian channel

The noisy-channel coding theorem has a beautiful geometric picture in the Gaussian case. Consider n uses of the channel.

- The noise vector $\eta \in \mathbb{R}^n$ has $\sum_i \eta_i^2 \approx n\sigma^2$ by the law of large numbers: it lies, with overwhelming probability, close to the **sphere of radius $\sqrt{n\sigma^2}$** .
- Given the codeword $\mathbf{z}$, the output $\mathbf{y} = \mathbf{z} + \eta$ lies near a sphere of radius $\sqrt{n\sigma^2}$ centred on $\mathbf{z}$.
- The inputs have power v^2 , so all outputs lie within a sphere of radius $\sqrt{n(v^2 + \sigma^2)}$.

For codewords to be non-confusable, their noise spheres must not overlap. The number of codewords is at most the ratio of volumes:

$$|S| \leq \left(\frac{\sqrt{n(v^2 + \sigma^2)}}{\sqrt{n\sigma^2}} \right)^n = \left(1 + \frac{v^2}{\sigma^2} \right)^{n/2} \Rightarrow \frac{\log |S|}{n} \leq \frac{1}{2} \log\left(1 + \frac{v^2}{\sigma^2} \right) = C.$$

Random coding shows that this packing bound is (asymptotically) achievable.

6.6. Parallel Gaussian channels and water-filling

For K independent Gaussian channels with noise variances σ_i^2 and a total power budget $\sum_i P_i \leq P$:

Theorem 6.7 (Water-filling).

$$C = \max_{\sum P_i \leq P} \sum_{i=1}^K \frac{1}{2} \log\left(1 + \frac{P_i}{\sigma_i^2} \right),$$

achieved by $P_i = \max(0, \mu - \sigma_i^2)$, with the “water level” μ fixed by $\sum_i P_i = P$.

Picture pouring a volume P of water over a vessel whose floor has heights σ_i^2 : noisier channels (higher floor) get less power, and very noisy ones none at all. This is what OFDM, DSL and LTE do at the physical layer, allocating power across frequency sub-carriers.

6.7. Exercises

Exercise 6.1

Compute the differential entropy of a Laplace distribution $f(x) = (2b)^{-1}e^{-|x-\mu|/b}$ and compare it with a Gaussian of the same variance.

Solution

$h = -\int f \log f = \log(2b) + \log e \mathbb{E}|X - \mu|/b = \log(2b) + \log e = \log(2be)$. The variance is $2b^2$; the Gaussian with that variance has $h = \frac{1}{2} \log(2\pi e \cdot 2b^2) = \frac{1}{2} \log(4\pi e b^2)$, while Laplace has $\frac{1}{2} \log(4e^2 b^2)$. The difference is $\frac{1}{2} \log(\pi/e) \approx 0.104$ bits in favour of the Gaussian, as the max-ent theorem requires.

Exercise 6.2

A variable $X \sim \mathcal{N}(0, 1)$ is measured with resolution $\epsilon = 10^{-3}$. How many bits does a measurement carry? How many if the unit of measurement is changed so that $\sigma = 1000$?

Solution

$H[X_\epsilon] \approx \frac{1}{2} \log_2(2\pi e) + \log_2 10^3 \approx 2.05 + 9.97 = 12.0$ bits. Changing units so that $\sigma = 1000$ (and keeping $\epsilon = 10^{-3}$ in the new units) adds $\log_2 1000 \approx 9.97$ bits: h grows by $\log |a|$ and the resolution is effectively finer. Only the ratio σ/ϵ matters.

Exercise 6.3

Two jointly Gaussian variables have correlation $c = 0.8$. Compute $i[X : Y]$. What is $i[X^3 : e^Y]$?

Solution

$i = -\frac{1}{2} \log_2(1 - 0.64) = -\frac{1}{2} \log_2 0.36 \approx 0.737$ bits. Since $x \mapsto x^3$ and $y \mapsto e^y$ are diffeomorphisms of their domains, $i[X^3 : e^Y] = 0.737$ bits as well — even though the transformed variables are no longer Gaussian and their *correlation coefficient* is different.

Exercise 6.4

For a Gaussian channel with SNR = 10 dB, compute the capacity in bits per channel use, and the MSE of the optimal decoder relative to σ^2 .

Solution

SNR = 10, so $C = \frac{1}{2} \log_2 11 \approx 1.73$ bits per use (or $\log_2 11 \approx 3.46$ bits/s/Hz with $N = 2WT$ uses).
MSE = $\sigma^2 \cdot 10/11 \approx 0.91 \sigma^2$.

6.8. Callouts

i Note

Differential entropy is *coordinate-dependent*: $h(g(X)) = h(X) + \mathbb{E} \log |g'(X)|$. Only $i[X : Y]$ and D_{KL} are coordinate-free, because the divergent $\log(1/\epsilon)$ terms cancel in them.

 Tip

When you need the entropy of a continuous variable, first check whether it belongs to a maximum-entropy family (Gaussian, exponential, uniform), which admits closed forms. Otherwise quantise and use the discrete formula, remembering the $\log(1/\epsilon)$ offset.

 Warning

Negative differential entropy is not a mistake: it means the distribution is more concentrated than a uniform on a unit cell. Any finite-resolution discrete entropy is still positive.

 Important

The Gaussian capacity $\frac{1}{2} \log(1 + \text{SNR})$ per use, or $W \log(1 + P/N_0 W)$ per second, is *the* benchmark of digital communication.

6.9. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 6. Cover & Thomas, §8.1–8.6 (differential entropy, discretisation, Gaussian examples), §9.1–9.4 (Gaussian channel, sphere packing, band-limited channels, water-filling). MacKay, Ch. 11 (Gaussian channels).

Part II.

Part II: Inference



7. Markov chains and information in time series

The information framework of the previous chapters treated symbols as i.i.d. Real data are almost never i.i.d.: language has grammar, DNA has motifs, neural spike trains and prices have memory. To measure the information content of a sequence with memory we need **stochastic processes**, and Markov chains are the simplest interesting case. Markov chains are also the engine of Part II: every MCMC algorithm is a Markov chain designed to have a prescribed equilibrium distribution, so the material here — stationarity, ergodicity, detailed balance — is used constantly in Chapters 7–11.

7.1. Stochastic processes

Definition 7.1 (Stochastic process). A **stochastic process** is a collection of random variables $\mathcal{X} = \{X_t\}$ taking values in a common space, indexed by time t . It is *discrete-time* if $t \in \mathbb{N}$. For any finite set of times $t_1 < \dots < t_n$, it is described by the joint distribution $p_{X_{t_1} \dots X_{t_n}}(x_1, \dots, x_n)$.

Definition 7.2 (Stationarity). A process is **stationary** if every joint distribution is invariant under time shifts:

$$p_{X_{t_1} \dots X_{t_n}}(x_1, \dots, x_n) = p_{X_{t_1+\tau} \dots X_{t_n+\tau}}(x_1, \dots, x_n) \quad \forall n, \{t_i\}, \tau.$$

Definition 7.3 (Markov property). A process is **Markov** if the future depends on the past only through the present:

$$p(x_n \mid x_{n-1}, \dots, x_1) = p(x_n \mid x_{n-1}),$$

i.e. X_n is conditionally independent of $X_1, \dots, X_{n-2}$ given X_{n-1} . A discrete-time Markov process is a **Markov chain**.

For a Markov chain, the joint distribution of any path factorises:

$$p(x_1, \dots, x_n) = p(x_1) \prod_{k=2}^n p(x_k \mid x_{k-1}).$$

The chain is **time-invariant** (homogeneous) if $p(x_k \mid x_{k-1})$ does not depend on k .

7.2. Finite-state Markov chains

Let $X_k \in \mathcal{S} = \{1, \dots, S\}$. The distribution at time n is a probability vector $\mathbf{p}(n)$ with components $p_k(n) = P(X_n = k)$, and a time-invariant chain is specified by a single **transition matrix**

$$P_{kl} \equiv P(X_n = k \mid X_{n-1} = l), \quad \sum_k P_{kl} = 1.$$

7. Markov chains and information in time series

(Each *column* sums to one. Many texts, and Chapter 7, use the transposed “row” convention $T(l, k) = P_{kl}$, the probability of going *from* l *to* k .) The evolution is linear:

$$p_k(n) = \sum_l P_{kl} p_l(n-1), \quad \mathbf{p}(n) = P \mathbf{p}(n-1) = P^n \mathbf{p}(0).$$

Definition 7.4 (Stationary distribution). A distribution π is **stationary** (invariant) if

$$\pi_k = \sum_l P_{kl} \pi_l, \quad \pi = P\pi,$$

i.e. π is an eigenvector of P with eigenvalue 1. If the chain starts in π it stays there forever.

Example 7.1 (A two-state chain)

Let $\mathcal{S} = \{0, 1\}$ with $P(0 \rightarrow 1) = a$ and $P(1 \rightarrow 0) = b$:

$$P = \begin{pmatrix} 1-a & b \\ a & 1-b \end{pmatrix}.$$

Solving $\pi = P\pi$: $\pi_0 a = \pi_1 b$, so $\pi = (\frac{b}{a+b}, \frac{a}{a+b})$. The eigenvalues of P are 1 and $\lambda_2 = 1 - a - b$, and

$$\mathbf{p}(n) = \pi + \lambda_2^n (\mathbf{p}(0) - \pi) :$$

the deviation from equilibrium decays geometrically with ratio $|1 - a - b|$. For $a = 0.1$, $b = 0.3$: $\pi = (0.75, 0.25)$ and the memory decays as 0.6^n .

7.2.1. Ergodicity

The stationary distribution is important because, under mild conditions, it is the **limiting distribution**: $\mathbf{p}(n) \rightarrow \pi$ as $n \rightarrow \infty$ from any initial condition. The chain “forgets” where it started and reaches *equilibrium*. Two things can prevent this: the chain can get stuck in a subset of states, or it can cycle.

Definition 7.5 (Irreducibility, aperiodicity, ergodicity).

- The chain is **irreducible** if every state can be reached from every other in a finite number of steps: $\forall k, l \exists n : (P^n)_{kl} > 0$.
- The **period** of state k is the gcd of the lengths of all cycles $k \rightarrow \dots \rightarrow k$; the chain is **aperiodic** if every state has period 1.
- An irreducible, aperiodic chain is called **ergodic**.

Example 7.2 (A reducible chain)

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Starting from 1 or 2, states 3 and 4 are never reached: $(P^n)_{31} = (P^n)_{41} = 0$ for all n . Every $\pi = (\alpha/2, \alpha/2, (1-\alpha)/2, (1-\alpha)/2)$ is stationary — no uniqueness — and the limit depends on the starting point. (This chain *is* aperiodic: cycles $1 \rightarrow 1$, $1 \rightarrow 2 \rightarrow 1$, $1 \rightarrow 2 \rightarrow 2 \rightarrow 1$ of all lengths exist.)

Example 7.3 (A periodic chain)

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

moves deterministically $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$. It is irreducible with stationary $\pi = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, but every return to 1 has length a multiple of 4: started in state 1, $\mathbf{p}(n)$ cycles forever and never converges. (Eigenvalues: the four fourth roots of unity, all with modulus 1.)

Theorem 7.1 (Convergence to equilibrium). *A finite ergodic Markov chain has a unique stationary distribution π , with $\pi_k > 0$ for all k , and $\mathbf{p}(n) \rightarrow \pi$ geometrically fast from any initial distribution.*

Sketch. By Perron–Frobenius, an ergodic P has a power P^m with all entries positive; then the eigenvalue 1 is simple and all other eigenvalues satisfy $|\lambda_i| < 1$. Decomposing $\mathbf{p}(0)$ on the eigenvectors, $\mathbf{p}(n) = \pi + \sum_{i \geq 2} c_i \lambda_i^n \mathbf{v}_i \rightarrow \pi$. The limit must be stationary, and since it is the same for every start it is unique (if the chain starts in π it stays there). $\square$

The rate of convergence is set by the second-largest eigenvalue modulus $|\lambda_2|$, as in the two-state example; the **spectral gap** $1 - |\lambda_2|$ determines the **mixing time** $\tau \sim 1/(1 - |\lambda_2|)$. This is developed in Chapter 7.

Corollary 7.2 (Ergodic theorem for Markov chains). *For an ergodic chain and any $f : \mathcal{S} \rightarrow \mathbb{R}$,*

$$\frac{1}{n} \sum_{t=1}^n f(X_t) \xrightarrow{n \rightarrow \infty} \sum_k \pi_k f(k) = \mathbb{E}_\pi[f] \quad \text{almost surely.}$$

Time averages along one trajectory equal ensemble averages under π . This is the theorem that makes MCMC work.

7.2.2. Detailed balance and time reversibility

Definition 7.6 (Reversibility). A chain is **time-reversible** if at equilibrium every trajectory is as probable as its time reverse:

$$P(X_n = j_1, X_{n+1} = j_2, \dots, X_{n+m} = j_m) = P(X_n = j_m, X_{n+1} = j_{m-1}, \dots, X_{n+m} = j_1).$$

Taking $m = 2$, reversibility requires $P(X_{n+1} = i, X_n = j) = P(X_{n+1} = j, X_n = i)$: **the probability flux between every pair of states is balanced**. This is the **detailed balance** condition:

$$P_{ij} \pi_j = P_{ji} \pi_i \quad \forall i, j.$$

Proposition 7.3 (Detailed balance reversibility and stationarity). *If P satisfies detailed balance with respect to π , then*

1. π is stationary;
2. the chain started from π is time-reversible.

Proof.

- (1) Sum detailed balance over j : $\sum_j P_{ij}\pi_j = \sum_j P_{ji}\pi_i = \pi_i$, using $\sum_j P_{ji} = 1$.
- (2) Apply detailed balance repeatedly, moving π along the path:

$$\begin{aligned} P(j_1 \rightarrow j_2 \rightarrow \cdots \rightarrow j_m) &= P_{j_m j_{m-1}} \cdots P_{j_3 j_2} P_{j_2 j_1} \pi_{j_1} = P_{j_m j_{m-1}} \cdots P_{j_3 j_2} \pi_{j_2} P_{j_1 j_2} \\ &= \cdots = \pi_{j_m} P_{j_{m-1} j_m} \cdots P_{j_2 j_3} P_{j_1 j_2} = P(j_m \rightarrow j_{m-1} \rightarrow \cdots \rightarrow j_1). \end{aligned}$$

□

Detailed balance is *sufficient* but not necessary for stationarity: stationarity only requires *global* balance (total inflow = total outflow at each state), while detailed balance requires it for every pair. Its usefulness is practical — it is a *local* condition that is easy to impose when designing a chain, and it is the basis of the Metropolis algorithm (Chapter 7).

Example 7.4 (*Global but not detailed balance*)

The three-state cycle with $P(1 \rightarrow 2) = P(2 \rightarrow 3) = P(3 \rightarrow 1) = p$ and the reverse moves with probability $q \neq p$ (staying put otherwise) has uniform π by symmetry, but $\pi_1 P(1 \rightarrow 2) = p/3 \neq \pi_2 P(2 \rightarrow 1) = q/3$: there is a net probability current circulating around the cycle. Watching a movie of this chain, one can tell whether it is played forwards or backwards.

7.3. Entropy rate

For a process, the entropy of a single symbol oversells the information: correlations mean each new symbol is partly predictable from the past. The information produced by the process up to time n is $H[X_n, \dots, X_1]$.

Definition 7.7 (Entropy rate). The **entropy rate** of a process $\mathcal{X}$ is

$$H[\mathcal{X}] = \lim_{n \rightarrow \infty} \frac{1}{n} H[X_n, \dots, X_1],$$

and the closely related **conditional entropy rate** (the entropy produced per step) is

$$H'[\mathcal{X}] = \lim_{n \rightarrow \infty} H[X_n | X_{n-1}, \dots, X_1],$$

when the limits exist. For i.i.d. variables, $H[\mathcal{X}] = H'[\mathcal{X}] = H[X]$.

Theorem 7.4. For a stationary process both limits exist and $H[\mathcal{X}] = H'[\mathcal{X}]$.

Proof. H' exists. Conditioning reduces entropy and stationarity shifts indices:

$$H[X_{n+1} | X_n, \dots, X_1] \leq H[X_{n+1} | X_n, \dots, X_2] = H[X_n | X_{n-1}, \dots, X_1].$$

So $a_n = H[X_n | X_{n-1}, \dots, X_1]$ is a non-increasing sequence of non-negative numbers, and has a limit H' .

$H = H'$. By the chain rule, $\frac{1}{n} H[X_n, \dots, X_1] = \frac{1}{n} \sum_{i=1}^n a_i$. By the **Cesàro mean lemma** (if $a_n \rightarrow a$ then $\frac{1}{n} \sum_{i=1}^n a_i \rightarrow a$), this converges to H' . □

Proposition 7.5 (Entropy rate of a Markov chain). For a stationary Markov chain, $H[X_n |$

$X_{n-1}, \dots, X_1] = H[X_n | X_{n-1}]$, so

$$H[\mathcal{X}] = H[X_n | X_{n-1}] = \sum_l \pi_l H[X_n | X_{n-1} = l] = - \sum_l \pi_l \sum_k P_{kl} \log P_{kl}.$$

It is the average, over the stationary distribution, of the entropy of each column of P (the “next-step” distribution from each state).

Theorem 7.6 (AEP for stationary ergodic processes (Shannon–McMillan–Breiman)). *For a stationary ergodic process,*

$$-\frac{1}{n} \log p(X_1, \dots, X_n) \rightarrow H[\mathcal{X}] \quad \text{with probability 1.}$$

So all of Chapter 3 carries over: about $2^{nH[\mathcal{X}]}$ typical sequences carry almost all the probability, and the **entropy rate is the compression limit** for sources with memory — the rate achieved asymptotically by Lempel–Ziv.

Example 7.5 (Entropy rate of the two-state chain)

With $a = 0.1$, $b = 0.3$, $\pi = (0.75, 0.25)$:

$$H[\mathcal{X}] = 0.75 h_2(0.1) + 0.25 h_2(0.3) = 0.75(0.469) + 0.25(0.881) \approx 0.572 \text{ bits/step,}$$

less than the marginal entropy $h_2(0.25) \approx 0.811$ bits. The difference, 0.24 bits, is the information about the next symbol already contained in the current one: $I(X_n; X_{n-1}) = H[X_n] - H[X_n | X_{n-1}]$.

Example 7.6 (Random walk on a graph (Cover and Thomas §4.3))

A walker on an undirected graph with edge weights w_{ij} moves from i to j with probability w_{ij}/w_i , $w_i = \sum_j w_{ij}$. Detailed balance holds with $\pi_i = w_i/(2W)$, $W = \sum_{i < j} w_{ij}$ — check: $\pi_i w_{ij}/w_i = w_{ij}/2W$ is symmetric. For an unweighted graph, $\pi_i = d_i/2E$ (degree over twice the number of edges) and the entropy rate is $H = \sum_i \frac{d_i}{2E} \log d_i$. A walk on a d -regular graph has $H = \log d$.

Example 7.7 (Entropy rate of English)

Shannon’s experiments (1951) estimated the entropy rate of English at about 1–1.5 bits per letter — much less than $\log_2 27 \approx 4.75$ bits for uniform letters, or the ≈ 4.1 bits of the single-letter frequencies — because of the strong dependencies between letters, words and grammar. This is the compression ratio that good text compressors approach.

7.4. Entropy production and irreversibility

In physics, “entropy production” usually refers to a different quantity: the degree to which a stationary process breaks time-reversal symmetry. It is naturally expressed as a KL divergence between forward and backward paths.

Definition 7.8 (Entropy production rate). For a stationary chain (row convention $T(i, j) = P_{ji}$),

$$\sigma = \sum_{i,j} \pi_i T(i, j) \log \frac{\pi_i T(i, j)}{\pi_j T(j, i)} = \lim_{n \rightarrow \infty} \frac{1}{n} D_{KL}(p(x_1, \dots, x_n) \| p(x_n, \dots, x_1)).$$

At stationarity the π factors contribute zero on average, so equivalently $\sigma = \sum_{i,j} \pi_i T(i,j) \log \frac{T(i,j)}{T(j,i)}$.

Theorem 7.7 (Second law for Markov chains). $\sigma \geq 0$, with equality iff detailed balance holds. Non-zero entropy production is the information-theoretic signature of non-equilibrium: it measures **how well one can tell the arrow of time** from a trajectory.

A related classical result (Cover & Thomas §4.4): for any Markov chain, the KL divergence between the distribution at time n and the stationary distribution is non-increasing, $D(\mathbf{p}(n+1) \parallel \pi) \leq D(\mathbf{p}(n) \parallel \pi)$ — the chain approaches equilibrium monotonically in relative entropy (an “H-theorem”).

7.5. Transfer entropy

Mutual information is symmetric and blind to the direction of time. To detect *directed* information flow between two processes X and Y (e.g. two neurons, two markets) one asks: does the past of Y help predict the next value of X *beyond* what the past of X already tells us?

Definition 7.9 (Transfer entropy (Schreiber 2000)). With history lengths k, l ,

$$T_{Y \rightarrow X} = I(X_{t+1}; Y_{t-l+1}^t | X_{t-k+1}^t) = H[X_{t+1} | X_{t-k+1}^t] - H[X_{t+1} | X_{t-k+1}^t, Y_{t-l+1}^t] \geq 0.$$

$T_{Y \rightarrow X}$ is a conditional mutual information, hence non-negative; it is zero iff X_{t+1} is conditionally independent of Y ’s past given X ’s past. In general $T_{Y \rightarrow X} \neq T_{X \rightarrow Y}$. For jointly Gaussian processes, transfer entropy is equivalent (up to a factor of 2) to **Granger causality**.

Example 7.8 (A driven chain)

Let Y_t be i.i.d. fair bits and $X_{t+1} = Y_t$ with probability $1 - \epsilon$ (and flipped otherwise). Then X ’s own past says nothing about X_{t+1} , so $H[X_{t+1} | X_t] = 1$ while $H[X_{t+1} | X_t, Y_t] = h_2(\epsilon)$: $T_{Y \rightarrow X} = 1 - h_2(\epsilon)$. Conversely Y is independent of everything in the past, so $T_{X \rightarrow Y} = 0$. Information flows from Y to X only.

7.6. Estimating information from data

In practice the distributions are unknown and information quantities must be estimated from samples. The naive **plug-in estimator** replaces p by empirical frequencies $\hat{p}_i = n_i/N$:

$$\hat{H}_{\text{plug-in}} = - \sum_i \hat{p}_i \log \hat{p}_i.$$

It is consistent but **biased downward**: by concavity of H and Jensen, $\mathbb{E}[\hat{H}] \leq H$. For K occupied bins and N samples, the leading bias is

$$\mathbb{E}[\hat{H}] - H \approx - \frac{K-1}{2N \ln 2} \text{ bits},$$

which motivates the **Miller–Madow correction** $\hat{H}_{MM} = \hat{H} + (K-1)/(2N \ln 2)$. Mutual information estimated as $\hat{H}_X + \hat{H}_Y - \hat{H}_{XY}$ is correspondingly biased *upward* (the joint has more bins and is underestimated most): independent variables appear to share information. Rules of thumb:

- Make sure $N \gg K$; for joint quantities K grows as the product of alphabet sizes, and for entropy rates exponentially with the history length.

- Compare with a **shuffled surrogate** (permute one variable to destroy dependence) to estimate the bias of $\hat{I}$.
- For continuous variables, avoid binning and use nearest-neighbour estimators (Kozachenko–Leonenko for h , Kraskov–Stögbauer–Grassberger for I), which are far less biased.

7.7. Hidden Markov models

Often the state of interest is not directly observable: a hidden Markov chain S_t emits observations Y_t through an emission distribution $p(y | s)$.

Definition 7.10 (Hidden Markov model). An **HMM** is a pair $(\{S_t\}, \{Y_t\})$ where $\{S_t\}$ is a Markov chain and each Y_t is drawn independently given S_t from $p(y | s)$. Thus $p(s_{1:n}, y_{1:n}) = p(s_1) \prod_{t \geq 1} p(s_t | s_{t-1}) \prod_t p(y_t | s_t)$.

Inferring the hidden states (forward–backward for marginals, Viterbi for the MAP path) and learning the parameters (Baum–Welch, an instance of EM) are classic inference problems. Decoding a convolutional code over a noisy channel is exactly Viterbi decoding of an HMM — another instance of “decoding is inference”.

Example 7.9 (HMM for DNA sequences)

Gene-finding algorithms model DNA as an HMM with hidden states $S_t \in \{\text{exon}, \text{intron}, \text{intergenic}, \dots\}$ and observed nucleotides $Y_t \in \{A, C, G, T\}$. Viterbi decoding assigns the most likely state sequence — a genomic annotation.

7.8. Exercises

Exercise 7.1

For the two-state chain with $P(0 \rightarrow 1) = a$, $P(1 \rightarrow 0) = b$, find the stationary distribution, verify detailed balance, and compute the entropy rate.

Solution

$\pi = (b, a)/(a + b)$. Detailed balance: $\pi_0 a = ab/(a + b) = \pi_1 b$ ✓ (every two-state stationary chain is reversible). Entropy rate

$$H[\mathcal{X}] = \frac{b}{a+b} h_2(a) + \frac{a}{a+b} h_2(b).$$

Exercise 7.2

Compute the entropy production of the three-state cyclic chain with $T(0 \rightarrow 1) = T(1 \rightarrow 2) = T(2 \rightarrow 0) = p$ and backward moves with probability q , $p + q \leq 1$.

Solution

$\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and

$$\sigma = 3 \cdot \frac{1}{3} \left[p \log \frac{p}{q} + q \log \frac{q}{p} \right] = (p - q) \log \frac{p}{q} \geq 0,$$

zero iff $p = q$ (detailed balance).

Exercise 7.3

Show that every stationary distribution of a chain that satisfies detailed balance w.r.t. π on a connected state space equals π , and that a chain with a symmetric transition matrix $P_{ij} = P_{ji}$ has the uniform distribution as a stationary distribution.

Solution

If $P_{ij} = P_{ji}$ then $P_{ij} \cdot \frac{1}{S} = P_{ji} \cdot \frac{1}{S}$: detailed balance holds with uniform π , which is therefore stationary. For uniqueness, an irreducible finite chain has a unique stationary distribution (Perron–Frobenius), so it must be the one satisfying detailed balance. This is the principle of the Metropolis algorithm with symmetric proposals.

Exercise 7.4

Let X_t be the two-state chain with $a = b = \epsilon$. Compute $I(X_t; X_{t+k})$ and show it decays to zero.

Solution

$\pi = (\frac{1}{2}, \frac{1}{2})$ and $\lambda_2 = 1 - 2\epsilon$. The k -step chain is a BSC with flip probability $f_k = \frac{1}{2}(1 - \lambda_2^k)$. With uniform input, $I(X_t; X_{t+k}) = 1 - h_2(\frac{1}{2}(1 - (1 - 2\epsilon)^k))$, which tends to zero as $k \rightarrow \infty$ — faster for larger spectral gap 2ϵ . For large k , $1 - h_2(\frac{1}{2}(1 - \delta)) \approx \frac{2}{\ln 2}\delta^2$, so $I \propto (1 - 2\epsilon)^{2k}$: correlations decay at twice the rate of the deviation from equilibrium.

7.9. Callouts

i Note

Markov chains give an analytical framework for time series that fits neatly into information theory. Real time series are only approximately Markov, but the framework is the standard null model, and for MCMC it is exact by construction.

💡 Tip

When estimating an entropy rate from data, use conditional entropies $H[X_n | X_{n-k}^{n-1}]$ for increasing k and look for a plateau; plug-in estimates of $H(X_1, \dots, X_n)/n$ are badly biased for large n .

⚠ Warning

Detailed balance is sufficient, not necessary, for stationarity. Conversely, a stationary chain without detailed balance is not “wrong” — it is out of equilibrium, with a probability current and positive entropy production.

7.10. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 7. Holmes-Cerfon, *Lecture notes on stochastic processes*, §2 (Markov chains, ergodicity, detailed balance). Cover & Thomas, §4.1–4.5 (entropy rate, random walk on graphs, second law). Schreiber, *Measuring information transfer*, PRL **85**, 461 (2000). Paninski, *Estimation of entropy and mutual information*, Neural Comput. **15**, 1191 (2003).

8. Markov Chain Monte Carlo and Metropolis

Part II of the course turns from communication to inference. The central computational problem is this: we want expectations under a distribution π that we can evaluate only *up to a normalisation constant* — a Bayesian posterior, or a Boltzmann distribution — in high dimension. Direct integration is hopeless, and independent sampling is usually impossible, but *dependent* sampling via a cleverly designed Markov chain works. This chapter introduces the Monte Carlo idea, elementary samplers, the Metropolis algorithm, and the theory of its convergence and autocorrelation.

8.1. Inference requires integrals

In Bayesian inference, data $\mathbf{y}$ are explained by parameters θ and latent variables $\mathbf{z}$, and everything we learn is encoded in the posterior

$$f(\theta, \mathbf{z} \mid \mathbf{y}) = \frac{f(\mathbf{y} \mid \theta, \mathbf{z}) f(\theta, \mathbf{z})}{f(\mathbf{y})}, \quad f(\mathbf{y}) = \int d\theta d\mathbf{z} f(\mathbf{y} \mid \theta, \mathbf{z}) f(\theta, \mathbf{z}).$$

Any posterior summary is an integral of the form

$$\mathbb{E}[g \mid \mathbf{y}] = \int d\theta d\mathbf{z} g(\theta, \mathbf{z}) f(\theta, \mathbf{z} \mid \mathbf{y}).$$

For example, the posterior mean of the parameters uses $g = \theta$; posterior variances, credible intervals ($g = \text{indicator}$), and predictive distributions are all of this form. The normalisation $f(\mathbf{y})$ (the **marginal likelihood** or **evidence**) is itself such an integral, usually intractable.

8.1.1. The curse of dimensionality

The obvious numerical approach is to discretise. In d dimensions, cover the integration region with N cells of side ϵ and use a Riemann sum,

$$\int d\mathbf{x} f(\mathbf{x}) = \sum_i f(\mathbf{x}_i) \epsilon^d + O(\epsilon).$$

Since $N \propto V/\epsilon^d$, the resolution is $\epsilon \propto N^{-1/d}$, so the error scales as $O(N^{-1/d})$: **for fixed accuracy, the number of cells grows exponentially with d** . With 10 points per axis, a 20-dimensional grid already has 10^{20} cells. This is the **curse of dimensionality**: volume grows exponentially with dimension, so any uniform covering of the space becomes sparse.

8.2. The Monte Carlo approach

Monte Carlo methods (Metropolis, Ulam, von Neumann, early 1950s) replace sums over cells by sums over *random samples*.

Example 8.1 (Area of a disc by throwing darts)

Let B be the disc of radius $1/2$ centred at $(1/2, 1/2)$; $\text{Vol}(B) = \pi/4$. If $(X, Y) \sim \mathcal{U}([0, 1]^2)$,

$$\text{Vol}(B) = P((X, Y) \in B) = \mathbb{E}[\chi_B(X, Y)],$$

with χ_B the indicator function. Draw N i.i.d. uniform points and let $M_N = \frac{1}{N} \sum_i \chi_B(X_i, Y_i)$ — the fraction of darts inside the disc. Then

$$\mathbb{E}[M_N] = \text{Vol}(B), \quad \text{Var}[M_N] = \frac{\text{Var}[\chi_B]}{N} = \frac{p(1-p)}{N}, \quad p = \frac{\pi}{4}.$$

The estimate is unbiased, with error $\approx 0.41/\sqrt{N}$. In two dimensions this is no better than the Riemann sum ($N^{-1/2}$ as well), but for the volume of a ball in d dimensions the grid error degrades to $N^{-1/d}$ while the Monte Carlo error **stays** $N^{-1/2}$.

Definition 8.1 (Monte Carlo estimator). Given i.i.d. samples $X_1, \dots, X_N \sim f$ and a function g with finite variance, the **Monte Carlo estimator** of $\mathbb{E}_f[g] = \int d\mathbf{x} f(\mathbf{x})g(\mathbf{x})$ is

$$M_N = \frac{1}{N} \sum_{i=1}^N g(X_i), \quad \mathbb{E}[M_N] = \mathbb{E}_f[g], \quad \text{Var}[M_N] = \frac{\text{Var}_f[g]}{N}.$$

By the law of large numbers $M_N \rightarrow \mathbb{E}_f[g]$ almost surely; by the CLT the error is $\approx \sigma_g/\sqrt{N}$.

The rate $1/\sqrt{N}$ is **dimension-free**. The price is that we must be able to *sample* from f — and that is the hard part.

8.3. Simple ways to sample a distribution

All samplers are built from a single primitive: a generator of uniform random numbers $U \sim \mathcal{U}(0, 1)$.

8.3.1. Inverse-CDF sampling

Proposition 8.1 (Inverse-CDF sampling). Let X have a continuous, invertible CDF F_X . If $U \sim \mathcal{U}(0, 1)$, then $X = F_X^{-1}(U)$ has CDF F_X .

Proof. Let $Z = F_X(X)$. Then $F_Z(z) = P(F_X(X) \leq z) = P(X \leq F_X^{-1}(z)) = F_X(F_X^{-1}(z)) = z$ on $[0, 1]$, so $Z \sim \mathcal{U}(0, 1)$. Reading the argument backwards, $F_X^{-1}(U)$ is distributed as X . $\square$

Example 8.2 (Exponential and Cauchy variables)

Exponential. $f(x) = \lambda e^{-\lambda x}$, $F(x) = 1 - e^{-\lambda x}$, so $X = -\frac{1}{\lambda} \log(1 - U)$ (or $-\frac{1}{\lambda} \log U$, since $1 - U \sim U$).

Cauchy. $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$, so $X = \tan(\pi(U - \frac{1}{2}))$.

The Gaussian has no closed-form inverse CDF; the Box–Muller transform $X = \sqrt{-2 \ln U_1} \cos(2\pi U_2)$ works instead, using a 2-D change of variables.

8.3.2. Rejection sampling

When F^{-1} is unavailable, we can sample uniformly under the graph of the density.

Definition 8.2 (Rejection sampling (uniform envelope)). Let f_X be (essentially) supported on $I = [a, b]$, and let $M \geq \max_x f_X(x)$.

1. Draw $Z_1, Z_2 \sim \mathcal{U}(0, 1)$ and set $Y = a + (b - a)Z_1$ (uniform on I).
2. **Accept** Y if $Z_2 \leq f_X(Y)/M$; otherwise **reject** it and start again.

Values are accepted with probability proportional to $f_X(y)$, so the accepted values are distributed as X .

Proof.

$$P(Y \leq x, \text{accept}) = \frac{1}{b-a} \int_a^x dy \int_0^1 dz_2 \chi\left(z_2 \leq \frac{f_X(y)}{M}\right) = \frac{1}{(b-a)M} \int_a^x f_X(y) dy = \frac{F_X(x)}{(b-a)M},$$

and $P(\text{accept}) = \frac{1}{(b-a)M}$. Dividing, $P(Y \leq x \mid \text{accept}) = F_X(x)$. $\square$

More generally, with any proposal density q such that $f \leq Mq$, draw $Y \sim q$ and accept with probability $f(Y)/(Mq(Y))$; the acceptance probability is $1/M$. The method also works if f is known only up to normalisation (then M just absorbs the constant).

Example 8.3 (*Rejection sampling fails in high dimension*)

Sample $X \sim \mathcal{N}(0, \sigma^2)$ on $[-\kappa\sigma, \kappa\sigma]$ with $M = \max f = (2\pi\sigma^2)^{-1/2}$:

$$P(\text{accept}) = \frac{1}{2\kappa\sigma M} = \frac{\sqrt{2\pi}}{2\kappa}.$$

For $\kappa = 4$ this is 0.31. In d dimensions with $\Sigma = \sigma^2 I$, the box has volume $(2\kappa\sigma)^d$ and $M = (2\pi\sigma^2)^{-d/2}$, so

$$P(\text{accept}) = \left(\frac{\sqrt{2\pi}}{2\kappa}\right)^d,$$

which for $\kappa = 4$ is 9×10^{-6} at $d = 10$ and 8×10^{-11} at $d = 20$. **The acceptance rate decays exponentially with dimension.** The same happens for the darts example: the fraction of the unit cube occupied by the inscribed ball is $\pi/4 \approx 0.79$ in $d = 2$, 0.52 in $d = 3$, but 0.0025 in $d = 10$.

Remark. In high dimension, almost all the volume of a box (or of any simple envelope) lies where the target has negligible mass — the “concentration of measure”. Independent proposals are doomed. The way out is to make *local* moves from a point already in the high-probability region: Markov chains.

8.4. Markov Chain Monte Carlo

The MCMC idea: **construct a Markov chain whose stationary distribution is the target π** , run it until it reaches equilibrium, and use its trajectory as a (correlated) sample. By Chapter 6, an ergodic chain satisfies $p_{X_n} \rightarrow \pi$ for $n \rightarrow \infty$, so after a transient (**burn-in**) the states $X_n, X_{n+1}, \dots$ are distributed according to π . Relabelling the first post-burn-in state as X_1 , the ergodic theorem gives

$$G_n = \frac{1}{n} \sum_{i=1}^n g(X_i) \xrightarrow{n \rightarrow \infty} \mathbb{E}_\pi[g].$$

To compute posterior expectations, we need a chain with stationary distribution $\pi(\theta, \mathbf{z}) = f(\theta, \mathbf{z} \mid \mathbf{y})$. The question is how to build one for an arbitrary π .

8.5. The Metropolis algorithm

8.5.1. Discrete state space

Let $X_n \in \{1, \dots, K\}$ and let π be the target. Choose a **symmetric proposal** matrix T , with T_{ij} the probability of proposing a jump $j \rightarrow i$:

$$T_{ij} = T_{ji}, \quad T_{ii} = 0, \quad \sum_i T_{ij} = 1.$$

A proposed jump $j \rightarrow i$ is **accepted** with probability

$$A_{ij} = \min\left\{1, \frac{\pi_i}{\pi_j}\right\}:$$

uphill moves (towards more probable states) are always accepted, downhill moves with probability $\pi_i/\pi_j < 1$. If rejected, the chain stays at j . The total rejection probability from j is $R_j = 1 - \sum_k A_{kj}T_{kj}$, and the transition matrix is

$$P_{ij} = T_{ij}A_{ij} + R_j \delta_{ij}.$$

Theorem 8.2 (Metropolis satisfies detailed balance). $P_{ij}\pi_j = P_{ji}\pi_i$ for all i, j . Hence π is a stationary distribution of the chain, and if the chain is ergodic it is the limiting distribution.

Proof. For $i = j$ there is nothing to prove. For $i \neq j$,

$$P_{ij}\pi_j = T_{ij} \min\left\{1, \frac{\pi_i}{\pi_j}\right\} \pi_j = T_{ij} \min\{\pi_j, \pi_i\} = T_{ji} \min\left\{\frac{\pi_j}{\pi_i}, 1\right\} \pi_i = P_{ji}\pi_i,$$

using the symmetry of T and of $\min$. Stationarity follows from Chapter 6. $\square$

Example 8.4 (Metropolis on three states)

Target $\pi = (0.5, 0.3, 0.2)$, proposal $T_{ij} = 1/2$ for $i \neq j$. Off-diagonal entries $P_{ij} = \frac{1}{2} \min(1, \pi_i/\pi_j)$:

P_{ij}	from 1	from 2	from 3
to 1	R_1	$\frac{1}{2}$	$\frac{1}{2}$
to 2	$\frac{1}{2} \cdot \frac{3}{5} = 0.3$	R_2	$\frac{1}{2}$
to 3	$\frac{1}{2} \cdot \frac{2}{5} = 0.2$	$\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$	R_3

with $R_1 = 1 - 0.3 - 0.2 = 0.5$, $R_2 = 1 - \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, $R_3 = 1 - \frac{1}{2} - \frac{1}{2} = 0$. Check stationarity row by row: $0.5 \cdot 0.5 + 0.5 \cdot 0.3 + 0.5 \cdot 0.2 = 0.5 \checkmark$; $0.3 \cdot 0.5 + \frac{1}{6} \cdot 0.3 + 0.5 \cdot 0.2 = 0.15 + 0.05 + 0.1 = 0.3 \checkmark$; $0.2 \cdot 0.5 + \frac{1}{3} \cdot 0.3 + 0 = 0.2 \checkmark$. Detailed balance, e.g. $P_{21}\pi_1 = 0.3 \cdot 0.5 = 0.15 = P_{12}\pi_2 = 0.5 \cdot 0.3 \checkmark$.

8.5.2. Continuous state space

The same construction works in $\mathbb{R}^d$. With a symmetric proposal density $T(\mathbf{x}' | \mathbf{x}) = T(\mathbf{x} | \mathbf{x}')$ and acceptance

$$A(\mathbf{x}' | \mathbf{x}) = \min\left\{1, \frac{\pi(\mathbf{x}')}{\pi(\mathbf{x})}\right\},$$

the rejection probability is $R(\mathbf{x}) = \int d\mathbf{x}' T(\mathbf{x}' | \mathbf{x})(1 - A(\mathbf{x}' | \mathbf{x}))$ and the transition kernel is

$$P(\mathbf{x}' | \mathbf{x}) = A(\mathbf{x}' | \mathbf{x}) T(\mathbf{x}' | \mathbf{x}) + \delta(\mathbf{x}' - \mathbf{x}) R(\mathbf{x}).$$

Detailed balance $P(\mathbf{x}' | \mathbf{x})\pi(\mathbf{x}) = P(\mathbf{x} | \mathbf{x}')\pi(\mathbf{x}')$ follows exactly as in the discrete case.

Definition 8.3 (Metropolis algorithm (continuous)). From the current state $\mathbf{x}$:

1. propose $\mathbf{x}' \sim T(\cdot | \mathbf{x})$, e.g. $\mathbf{x}' = \mathbf{x} + \sigma\xi$, $\xi \sim \mathcal{N}(0, I)$;
2. draw $u \sim \mathcal{U}(0, 1)$; if $u < \pi(\mathbf{x}')/\pi(\mathbf{x})$, move to $\mathbf{x}'$; otherwise stay at $\mathbf{x}$ (and **record $\mathbf{x}$ again** as the next sample).

Proposition 8.3 (Metropolis does not need the evidence). If $\pi(\mathbf{x}) = f(\mathbf{x} | \mathbf{y}) = f(\mathbf{x}, \mathbf{y})/f(\mathbf{y})$, then

$$A(\mathbf{x}' | \mathbf{x}) = \min\left\{1, \frac{f(\mathbf{x}', \mathbf{y})}{f(\mathbf{x}, \mathbf{y})}\right\} = \min\left\{1, \frac{f(\mathbf{y} | \mathbf{x}')f(\mathbf{x}')}{f(\mathbf{y} | \mathbf{x})f(\mathbf{x})}\right\}:$$

only the **unnormalised posterior** (likelihood $\times$ prior) is needed. The intractable marginal likelihood $f(\mathbf{y})$ cancels.

This is why Metropolis revolutionised Bayesian statistics (and, earlier, statistical physics, where the unknown constant is the partition function Z).

8.5.3. Metropolis–Hastings

With a non-symmetric proposal $q(\mathbf{x}' | \mathbf{x})$, detailed balance is restored by the Hastings correction:

$$A(\mathbf{x}' | \mathbf{x}) = \min\left\{1, \frac{\pi(\mathbf{x}') q(\mathbf{x} | \mathbf{x}')}{\pi(\mathbf{x}) q(\mathbf{x}' | \mathbf{x})}\right\}.$$

The ratio of proposal densities compensates for the bias of proposing some moves more often than their reverses. Gibbs sampling (Chapter 8) and HMC (Chapter 11) are both special cases.

Example 8.5 (Metropolis for a coin's bias)

We observe $k = 7$ heads in $n = 10$ tosses of a coin with unknown bias θ , and use a uniform prior. The unnormalised posterior is $\pi(\theta) \propto \theta^7(1 - \theta)^3$ on $[0, 1]$ (the exact answer is Beta(8, 4), with mean $8/12 = 0.667$). Metropolis with $\theta' = \theta + 0.1\xi$: accept with probability $\min\{1, \pi(\theta')/\pi(\theta)\}$, with $\pi(\theta') = 0$ (always reject) if $\theta' \notin [0, 1]$. Note that rejected proposals outside the support must still be counted as a repeated sample of θ . A run of 10^5 steps reproduces the Beta mean and variance $(8 \cdot 4 / (12^2 \cdot 13)) \approx 0.0171$ to about two decimal places.

Example 8.6 (Metropolis for a 1-D Gaussian)

Target $\pi(x) \propto e^{-x^2/2}$, proposal $x' = x + \sigma_p \xi$, acceptance $\min\{1, e^{(x^2 - x'^2)/2}\}$. Tiny σ_p : almost every move is accepted but the chain crawls (random walk with tiny steps). Huge σ_p : proposals land in the tails and are almost all rejected; the chain sits still. The optimum is $\sigma_p \approx 2.4$, with acceptance rate $\approx 44\%$.

8.6. Convergence and correlation

The Metropolis chain is guaranteed to converge to π . Two practical issues remain:

- **Convergence time:** how long is the transient before equilibrium?

- **Autocorrelation:** the samples are identically distributed but *not independent*, so they carry less information than an i.i.d. sample of the same size.

Both are controlled by the **spectrum** of the transition matrix. We work with a finite state space; the continuous case is analogous, though the operator theory is more delicate.

8.6.1. Spectrum of a reversible chain

π is an eigenvector of P with eigenvalue 1, but P is not symmetric, so its spectrum is not obviously real. Detailed balance fixes this. Let $\Lambda = \text{diag}(\sqrt{\pi_1}, \dots, \sqrt{\pi_K})$ and

$$V = \Lambda^{-1}P\Lambda, \quad V_{ij} = P_{ij}\sqrt{\pi_j/\pi_i}.$$

By detailed balance, $V_{ij} = P_{ij}\pi_j/\sqrt{\pi_i\pi_j} = P_{ji}\pi_i/\sqrt{\pi_i\pi_j} = V_{ji}$: **V is symmetric**. Hence $V = WDW^\top$ with $D = \text{diag}(\lambda_1, \dots, \lambda_K)$ real and W orthogonal (columns $\mathbf{w}_k$). Since $P = \Lambda V \Lambda^{-1}$ is similar to V , it has the same real eigenvalues, with

- right eigenvectors $\phi_k = \Lambda \mathbf{w}_k$: $P\phi_k = \Lambda V \mathbf{w}_k = \lambda_k \phi_k$;
- left eigenvectors $\psi_k = \Lambda^{-1} \mathbf{w}_k$: $\psi_k^\top P = \mathbf{w}_k^\top V \Lambda^{-1} = \lambda_k \psi_k^\top$;
- biorthonormality $\psi_k^\top \phi_l = \mathbf{w}_k^\top \mathbf{w}_l = \delta_{kl}$.

The powers of P have the spectral decomposition

$$P^n = \Lambda V^n \Lambda^{-1} = \sum_k \lambda_k^n \phi_k \psi_k^\top.$$

Proposition 8.4. *All eigenvalues satisfy $|\lambda_k| \leq 1$. The eigenvalue 1 has left eigenvector $\mathbf{1} = (1, \dots, 1)$ and right eigenvector π ; for an ergodic chain it is simple and all others satisfy $|\lambda_k| < 1$.*

Proof. From $P^n \phi_k = \lambda_k^n \phi_k$, for each component j : $|\lambda_k|^n |(\phi_k)_j| \leq \sum_l (P^n)_{jl} |(\phi_k)_l| \leq \gamma \sum_l (P^n)_{jl}$, with $\gamma = \max_l |(\phi_k)_l|$. Choosing j where $|(\phi_k)_j| = \gamma$ gives $|\lambda_k|^n \leq \sum_l (P^n)_{jl} \leq K$ (entries of P^n are probabilities). A bound uniform in n forces $|\lambda_k| \leq 1$. The left eigenvector $\mathbf{1}$ expresses $\sum_i P_{ij} = 1$; simplicity for ergodic chains is Perron–Frobenius. $\square$

Order the eigenvalues as $1 = \lambda_1 > |\lambda_2| \geq |\lambda_3| \geq \dots$.

8.6.2. Convergence time

Start the chain from $\mathbf{p}^{(0)}$. Since $\phi_1 = \pi$ and $\psi_1 = \mathbf{1}$ (so $\psi_1^\top \mathbf{p}^{(0)} = 1$),

$$\mathbf{p}^{(n)} = P^n \mathbf{p}^{(0)} = \pi + \sum_{k \geq 2} \lambda_k^n \phi_k (\psi_k^\top \mathbf{p}^{(0)}) = \pi + O(e^{-n/\tau_{\text{conv}}}),$$

with

$$\tau_{\text{conv}} = -\frac{1}{\log |\lambda_2|} = -\frac{1}{\log(1 - \delta\lambda)} \simeq \frac{1}{\delta\lambda}, \quad \delta\lambda = 1 - |\lambda_2|.$$

The convergence time is the inverse spectral gap: the smaller the gap, the longer the burn-in.

Example 8.7

In the three-state Metropolis example, $\text{tr } P = 2/3$ and $\det P = 0$, so the eigenvalues are 1, 0 and $-1/3$: $|\lambda_2| = 1/3$ and the chain equilibrates within a few steps (each step shrinks the deviation from π by a

factor 3). For the two-state chain of Chapter 6, $\lambda_2 = 1 - a - b$: with $a = b = 0.01$, $\tau_{\text{conv}} \approx 50$ steps.

8.6.3. The step size of random-walk Metropolis

For a continuous Metropolis chain the transition operator splits as $V = K + M$, with

$$(Kf)(\mathbf{x}) = \int d\mathbf{x}' T(\mathbf{x}' | \mathbf{x}) A(\mathbf{x}' | \mathbf{x}) f(\mathbf{x}'), \quad (Mf)(\mathbf{x}) = R(\mathbf{x}) f(\mathbf{x}).$$

The spectrum of M is the range of the rejection probability, $[0, \max R]$, a continuum; K contributes isolated eigenvalues. Which part sets the gap depends on the proposal width σ :

- **Small steps.** $T(\mathbf{x}' | \mathbf{x}) \rightarrow \delta(\mathbf{x}' - \mathbf{x})$ as $\sigma \rightarrow 0$, so $Kf = f + O(\sigma^2)$: the top eigenvalues of K are $1 - O(\sigma^2)$ and the gap closes as σ^2 . The chain is a slow random walk: covering a distance L takes $\sim (L/\sigma)^2$ steps.
- **Large steps.** Most proposals land in low-probability regions; $R(\mathbf{x}) \rightarrow 1$, the top of the spectrum of M approaches 1, and the chain gets stuck.

In D dimensions, Taylor-expanding $A(\mathbf{x}' | \mathbf{x})$ around $\mathbf{x}' = \mathbf{x}$ inside $R(\mathbf{x}) = 1 - \int d^D \mathbf{x}' A(\mathbf{x}' | \mathbf{x}) \mathcal{N}(\mathbf{x}'; \mathbf{x}, \sigma^2 I)$ gives

$$R(\mathbf{x}) = O(D\sigma^2) :$$

each of the D directions contributes a term of order σ^2 . To keep the rejection rate bounded away from 1 we need $\sigma^2 \lesssim 1/D$. **The optimal step size shrinks as** $\sigma \propto D^{-1/2}$, and the number of steps needed to traverse the distribution grows as $\sim D$.

Proposition 8.5 (Optimal scaling (Roberts, Gelman and Gilks 1997)). *For random-walk Metropolis on a D -dimensional target with i.i.d. components, as $D \rightarrow \infty$ the optimal proposal is $\sigma = 2.38/\sqrt{D}$ (in units of the target's standard deviation), and the corresponding acceptance rate is $\alpha^* \approx 0.234$.*

The rule of thumb “aim for 20–40% acceptance” (about 44% in one dimension) comes from this result and works surprisingly well in practice.

8.6.4. Autocorrelation and effective sample size

After burn-in, $X_n, X_{n+1}, \dots$ are distributed according to π but are **autocorrelated**. For any (not necessarily independent) identically distributed sample,

$$\text{Var}\left[\frac{1}{n} \sum_i g(X_i)\right] = \frac{1}{n^2} \left[\sum_i \text{Var}[g(X_i)] + \sum_i \sum_{j \neq i} \text{Cov}[g(X_i), g(X_j)] \right].$$

- **Independent samples:** covariances vanish and the variance is $\text{Var}[g]/n$, the familiar $1/\sqrt{n}$ error.
- **Perfectly correlated samples:** all covariances equal $\text{Var}[g]$ and the variance is $\text{Var}[g]$ — **no improvement with n at all**: it is as if we had a single sample.

A stationary chain lies in between. Its covariance depends only on the lag, $\text{Cov}[g(X_i), g(X_j)] = c_{|i-j|}$, and typically decays to zero beyond some lag τ , so the covariance matrix is banded around the diagonal.

|

Definition 8.4 (Integrated autocorrelation time).

$$A_g = \frac{1}{\text{Var}[g]} \sum_{j=-\infty}^{\infty} \text{Cov}[g(X_i), g(X_{i+j})] = 1 + 2 \sum_{j=1}^{\infty} \frac{c_j}{c_0}.$$

Summing the banded covariance matrix row by row, rows away from the boundaries contribute $c_0 A_g$ each; boundary rows contribute $O(\tau)$ in total. Hence

$$\text{Var}\left[\frac{1}{n} \sum_i g(X_i)\right] = \frac{\text{Var}[g] A_g}{n} + O\left(\frac{1}{n^2}\right) = \frac{\text{Var}[g]}{n_g^{\text{eff}}} + O\left(\frac{1}{n^2}\right), \quad n_g^{\text{eff}} = \frac{n}{A_g}.$$

A correlated chain of length n is worth n/A_g independent samples — the effective sample size.

Example 8.8 (*AR(1) process*)

Let $X_{t+1} = \rho X_t + \sqrt{1-\rho^2} \xi_t$ with $\xi_t \sim \mathcal{N}(0, 1)$ — a Markov chain with stationary $\mathcal{N}(0, 1)$ and $c_j = \rho^{|j|}$. Then

$$A = 1 + 2 \sum_{j \geq 1} \rho^j = \frac{1+\rho}{1-\rho}.$$

For $\rho = 0.9$, $A = 19$: 10^4 steps are worth only about 530 independent draws. For $\rho = 0.99$, $A = 199$.

8.6.5. Autocorrelation from the spectrum

Standardise $h = (g - \mu_g)/\sigma_g$, so $\mathbb{E}_\pi[h] = 0$, $\text{Var}[h] = 1$. At equilibrium, $P(X_n = i, X_0 = j) = (P^n)_{ij} \pi_j$, so

$$\text{Cov}[h(X_n), h(X_0)] = \sum_{ij} h_i h_j \pi_j (P^n)_{ij} = \sum_k \lambda_k^n \sum_{ij} h_i h_j \pi_j (\phi_k)_i (\psi_k)_j.$$

The $k = 1$ term is $\sum_{ij} h_i h_j \pi_i \pi_j = (\mathbb{E}_\pi h)^2 = 0$. Summing the geometric series $\sum_{n \geq 1} \lambda_k^n = \lambda_k / (1 - \lambda_k)$,

$$A_g = 1 + 2 \sum_{n \geq 1} \text{Cov}[h(X_n), h(X_0)] = \sum_{k \geq 2} \frac{1 + \lambda_k}{1 - \lambda_k} \sum_{ij} h_i h_j \pi_j (\phi_k)_i (\psi_k)_j,$$

where we used $1 = \text{Var}[h] = \sum_{k \geq 2} \sum_{ij} h_i h_j \pi_j (\phi_k)_i (\psi_k)_j$. In terms of the orthonormal eigenvectors of V , the weights are $a_k^2 = (\sum_i \sqrt{\pi_i} h_i (\mathbf{w}_k)_i)^2 \geq 0$ with $\sum_{k \geq 2} a_k^2 = 1$, so

$$A_g = \sum_{k \geq 2} a_k^2 \frac{1 + \lambda_k}{1 - \lambda_k}.$$

The autocorrelation time is a weighted average of $(1 + \lambda_k)/(1 - \lambda_k)$: eigenvalues close to 1 (small gaps) inflate it, and the observable “feels” a slow mode only to the extent that it overlaps with it. The AR(1) result $A = (1 + \rho)/(1 - \rho)$ is the single-mode case. (Negative eigenvalues give $A_g < 1$: antithetic, “better than independent” samples are possible.)

8.7. Practical MCMC

- **Burn-in.** Discard the transient (several τ_{conv}); check that discarding more does not change the estimate.
- **Trace plots** of $g(X_t)$ vs t should look like stationary noise around a constant mean.
- **Effective sample size:** estimate A_g from the empirical autocorrelation (truncating the sum when it becomes noisy) and report n/A_g , not n .

- **Gelman–Rubin** $\hat{R}$: run several chains from over-dispersed starting points and compare within-chain and between-chain variances; $\hat{R} \approx 1$ (in practice < 1.01) indicates convergence.
- **Thinning** (keeping one sample every k) reduces storage but never increases the information in the chain.

8.8. Exercises

Exercise 8.1

Prove that with a symmetric proposal the Metropolis–Hastings acceptance reduces to $\min\{1, \pi(y)/\pi(x)\}$, and verify detailed balance for $x \neq y$.

Solution

With $q(y | x) = q(x | y)$ the q 's cancel in the Hastings ratio. Then $\pi(x)q(y | x) \min\{1, \pi(y)/\pi(x)\} = q(y | x) \min\{\pi(x), \pi(y)\}$, which is symmetric in $x \leftrightarrow y$ because q is symmetric: this is $\pi(y)P(x | y)$.

Exercise 8.2

Using the inverse-CDF method, give an algorithm to sample from $f(x) = 2x$ on $[0, 1]$. Then design a rejection sampler for the same density using the uniform envelope and compute its acceptance rate.

Solution

$F(x) = x^2$, so $X = \sqrt{U}$. For rejection: $M = \max f = 2$, $b - a = 1$, so $P(\text{accept}) = 1/((b - a)M) = 1/2$: draw $Y \sim \mathcal{U}(0, 1)$, accept if $U_2 \leq Y$.

Exercise 8.3

A Metropolis chain has $A_g = 25$ for the observable g . How long must the chain be (after burn-in) to estimate $\mathbb{E}[g]$ to within $0.01\sigma_g$ (one standard error)?

Solution

Standard error $= \sigma_g \sqrt{A_g/n} = 0.01\sigma_g \Rightarrow n = A_g/10^{-4} = 2.5 \times 10^5$ steps, versus 10^4 for i.i.d. sampling.

Exercise 8.4

For the two-state chain with $P(0 \rightarrow 1) = a$, $P(1 \rightarrow 0) = b$ and the observable $g(x) = x$, compute A_g .

Solution

Only one non-trivial eigenvalue, $\lambda_2 = 1 - a - b$, and the centred g lies entirely along its eigenvector, so $A_g = (1 + \lambda_2)/(1 - \lambda_2) = (2 - a - b)/(a + b)$. For $a = b = 0.5$ ($\lambda_2 = 0$, i.i.d.) $A_g = 1$; for $a = b = 0.01$, $A_g = 99$; for $a = b = 0.9$ ($\lambda_2 = -0.8$, alternating) $A_g = 1/9$.

8.9. Callouts

Note

Metropolis is not the fastest MCMC algorithm; it is the most general. It needs only pointwise evaluations of the unnormalised target. Specialised samplers (Gibbs, HMC) exploit more structure and do much better.

Warning

The MCMC estimator is biased by the initial transient. Always discard a burn-in. And the error bar is $\sigma_g \sqrt{A_g/n}$, not $\sigma_g/\sqrt{n}$: forgetting A_g gives wildly overconfident results.

Caution

A chain that “looks converged” may still be missing modes of a multimodal target: trace plots detect within-mode stationarity, not between-mode mixing. The spectral gap of a chain with well-separated modes is exponentially small in the barrier height.

Important

Effective sample size per unit of computation is the right figure of merit for comparing samplers.

8.10. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapters 8–9. Bishop, *Pattern Recognition and Machine Learning*, §1.2 (Bayesian inference), §11.1 (inverse CDF, rejection sampling), §11.2 (Metropolis–Hastings). MacKay, §29.1–29.6. Brooks et al., *Handbook of MCMC*, §1.7–1.12 (MCMC, Metropolis–Hastings, autocorrelation, ESS). Holmes-Cerfon, §3 (spectrum of Markov chains). Roberts, Gelman & Gilks, *Ann. Appl. Probab.* **7**, 110 (1997).

9. Gibbs sampling and latent variables

Metropolis is universal but blunt: it proposes blindly and then corrects by rejection. When the target has structure — conditional distributions we can sample exactly, as in Bayesian models with conjugate priors and latent variables — a specialised algorithm, **Gibbs sampling**, exploits it and never rejects. This chapter derives Gibbs sampling as a special case of Metropolis–Hastings and works out in detail its most important application: inference in mixture models through latent variables, which turns out to be a stochastic cousin of K-means clustering.

9.1. Metropolis–Hastings

Metropolis required a symmetric proposal. With a general proposal $T(\mathbf{x}' | \mathbf{x})$, the kernel

$$P(\mathbf{x}' | \mathbf{x}) = T(\mathbf{x}' | \mathbf{x})A(\mathbf{x}' | \mathbf{x}) + R(\mathbf{x})\delta(\mathbf{x} - \mathbf{x}')$$

satisfies detailed balance with the **Hastings acceptance probability**

$$A(\mathbf{x}' | \mathbf{x}) = \min\left\{1, \frac{\pi(\mathbf{x}')T(\mathbf{x} | \mathbf{x}')}{\pi(\mathbf{x})T(\mathbf{x}' | \mathbf{x})}\right\}.$$

Proof. For $\mathbf{x}' \neq \mathbf{x}$ (the δ term is trivially symmetric),

$$\begin{aligned} P(\mathbf{x}' | \mathbf{x})\pi(\mathbf{x}) &= T(\mathbf{x}' | \mathbf{x})\pi(\mathbf{x}) \min\left\{1, \frac{\pi(\mathbf{x}')T(\mathbf{x} | \mathbf{x}')}{\pi(\mathbf{x})T(\mathbf{x}' | \mathbf{x})}\right\} = \min\{T(\mathbf{x}' | \mathbf{x})\pi(\mathbf{x}), T(\mathbf{x} | \mathbf{x}')\pi(\mathbf{x}')\} \\ &= T(\mathbf{x} | \mathbf{x}')\pi(\mathbf{x}') \min\left\{1, \frac{T(\mathbf{x}' | \mathbf{x})\pi(\mathbf{x})}{T(\mathbf{x} | \mathbf{x}')\pi(\mathbf{x}')}\right\} = P(\mathbf{x} | \mathbf{x}')\pi(\mathbf{x}'). \end{aligned}$$

□

The ratio $T(\mathbf{x} | \mathbf{x}')/T(\mathbf{x}' | \mathbf{x})$ corrects for proposals that favour some directions. The freedom to choose T is what we now exploit.

9.2. The Gibbs sampling algorithm

Write $\mathbf{x} = (x_1, \dots, x_d)$ and, for each k ,

$$\mathbf{x}_{<k} = (x_1, \dots, x_{k-1}), \quad \mathbf{x}_{>k} = (x_{k+1}, \dots, x_d), \quad \mathbf{x}_{-k} = (\mathbf{x}_{<k}, \mathbf{x}_{>k}).$$

Gibbs sampling applies when we can sample each **full conditional** $\pi(x_k | \mathbf{x}_{-k})$.

Definition 9.1 (Gibbs sampler (systematic scan)). One sweep updates the coordinates in turn, always

conditioning on the *most recent* values:

$$\begin{aligned} x'_1 &\sim \pi(x_1 \mid \mathbf{x}_{>1}), \\ x'_2 &\sim \pi(x_2 \mid x'_1, \mathbf{x}_{>2}), \\ &\vdots \\ x'_k &\sim \pi(x_k \mid \mathbf{x}'_{<k}, \mathbf{x}_{>k}), \\ &\vdots \\ x'_d &\sim \pi(x_d \mid \mathbf{x}'_{<d}). \end{aligned}$$

In the *random-scan* variant, each step picks a coordinate k uniformly at random and resamples it.

Each update is a Metropolis–Hastings move with proposal “change only x_k , drawing it from its conditional”:

$$T(x'_k \mid \mathbf{x}) = \pi(x'_k \mid \mathbf{x}'_{<k}, \mathbf{x}_{>k}).$$

Theorem 9.1 (Gibbs moves are always accepted). *With this proposal, the Hastings acceptance probability is exactly 1.*

Proof. Abbreviate $\mathbf{c} = (\mathbf{x}'_{<k}, \mathbf{x}_{>k})$, the coordinates that do not change. The move is $(x_k, \mathbf{c}) \rightarrow (x'_k, \mathbf{c})$, and

$$A = \min\left\{1, \frac{\pi(x'_k, \mathbf{c}) \pi(x_k \mid \mathbf{c})}{\pi(x_k, \mathbf{c}) \pi(x'_k \mid \mathbf{c})}\right\}.$$

Factorising $\pi(x'_k, \mathbf{c}) = \pi(x'_k \mid \mathbf{c})\pi(\mathbf{c})$ and $\pi(x_k, \mathbf{c}) = \pi(x_k \mid \mathbf{c})\pi(\mathbf{c})$,

$$A = \min\left\{1, \frac{\pi(x'_k \mid \mathbf{c})\pi(\mathbf{c})\pi(x_k \mid \mathbf{c})}{\pi(x_k \mid \mathbf{c})\pi(\mathbf{c})\pi(x'_k \mid \mathbf{c})}\right\} = 1.$$

□

So every single-coordinate update leaves π invariant, and so does their composition (a full sweep). Note that a systematic sweep as a whole does *not* satisfy detailed balance (the reversed sweep updates coordinates in the opposite order), but it does leave π stationary, which is what matters; the random-scan version is reversible. Under mild conditions (e.g. $\pi > 0$ everywhere) the chain is ergodic and converges to π .

Remark. “Acceptance 1” does not mean “efficient”. Gibbs moves only along coordinate axes, with step sizes set by the *conditional* widths. When the target is strongly correlated, conditionals are narrow and the chain crawls.

Example 9.1 (*Gibbs on a correlated Gaussian*)

Let $(x_1, x_2) \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$. The full conditionals are

$$x_1 \mid x_2 \sim \mathcal{N}(\rho x_2, 1 - \rho^2), \quad x_2 \mid x_1 \sim \mathcal{N}(\rho x_1, 1 - \rho^2).$$

After one sweep, $x_1^{(t+1)} = \rho x_2^{(t)} + \dots = \rho^2 x_1^{(t)} + \text{noise}$: the sequence $x_1^{(t)}$ is an AR(1) process with coefficient ρ^2 . By Chapter 7 its integrated autocorrelation time is

$$A = \frac{1 + \rho^2}{1 - \rho^2}.$$

For $\rho = 0.5$, $A \approx 1.7$ (excellent); for $\rho = 0.99$, $A \approx 100$; for $\rho = 0.999$, $A \approx 1000$. The chain zig-zags along the narrow ridge with steps of size $\sqrt{1 - \rho^2}$ while it needs to travel distances of order 1. **Reparameterising** (e.g. rotating to $u = x_1 + x_2$, $v = x_1 - x_2$, which are independent) or **blocking** (updating (x_1, x_2) jointly) cures the problem.

9.3. When Gibbs sampling helps: conjugacy

Gibbs is useful when the full conditionals belong to families we can sample directly. The paradigm is **conjugacy**: a prior is conjugate to a likelihood if the posterior is in the same family as the prior.

likelihood	conjugate prior	posterior
Bernoulli/Binomial(θ)	Beta(a, b)	Beta($a + k, b + n - k$)
Categorical/Multinomial($\mathbf{p}$)	Dirichlet(γ)	Dirichlet($\gamma + \mathbf{n}$)
Normal(μ), known σ^2	Normal(μ_0, σ_0^2)	Normal (precisions add)
Normal, known μ , unknown σ^2	Inverse-Gamma(α, β)	Inverse-Gamma($\alpha + n/2, \beta + \frac{1}{2} \sum (x_i - \mu)^2$)
Poisson(λ)	Gamma(α, β)	Gamma($\alpha + \sum x_i, \beta + n$)

In a multi-parameter model, the joint posterior is rarely in a standard family, but if each parameter has a conjugate prior *given the others*, every full conditional is standard and Gibbs applies.

Example 9.2 (*Gibbs for Bayesian linear regression*)

Model: $y_i \mid \beta, \sigma^2 \sim \mathcal{N}(\beta^\top \mathbf{x}_i, \sigma^2)$, priors $\beta \sim \mathcal{N}(\mathbf{0}, \tau^2 I)$ and $\sigma^2 \sim \text{InvGamma}(\alpha, \beta)$. The full conditionals are

- $\beta \mid \sigma^2, \mathbf{y} \sim \mathcal{N}(\hat{\beta}, \Sigma_\beta)$, with $\Sigma_\beta = (\mathbf{X}^\top \mathbf{X} / \sigma^2 + I / \tau^2)^{-1}$ and $\hat{\beta} = \Sigma_\beta \mathbf{X}^\top \mathbf{y} / \sigma^2$;
- $\sigma^2 \mid \beta, \mathbf{y} \sim \text{InvGamma}(\alpha + n/2, \beta + \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2)$.

Alternating the two conjugate draws samples the joint posterior, which is not itself a standard distribution.

9.4. Mixture models and latent variables

9.4.1. The mixture model

Suppose each data point x is drawn from one of K Gaussians with means $\mu = (\mu_1, \dots, \mu_K)$ and a common, known standard deviation σ , the k -th being chosen with probability p_k :

$$f(x \mid \mu, \mathbf{p}) = \sum_{k=1}^K p_k (2\pi\sigma^2)^{-1/2} e^{-\frac{(x-\mu_k)^2}{2\sigma^2}}.$$

For N independent points $\mathbf{x} = \{x_i\}$, the likelihood and the posterior are

$$f(\mathbf{x} \mid \mu, \mathbf{p}) = \prod_{i=1}^N \sum_{k=1}^K p_k (2\pi\sigma^2)^{-1/2} e^{-\frac{(x_i-\mu_k)^2}{2\sigma^2}}, \quad f(\mu, \mathbf{p} \mid \mathbf{x}) = \frac{f(\mathbf{x} \mid \mu, \mathbf{p}) f(\mu) f(\mathbf{p})}{c(\mathbf{x})}.$$

One could run Metropolis on this posterior, but it is unpleasant: the “sum inside the product” makes every likelihood evaluation cost $O(NK)$, and the posterior has no convenient structure.

9.4.2. Introducing latent variables

The trick is to **add** variables. For each point introduce a latent label $Z_i \in \{1, \dots, K\}$ saying which component generated it:

$$Z_i \mid \mathbf{p} \sim \text{Cat}(\mathbf{p}), \quad P(Z_i = k) = p_k; \quad f(x_i \mid \mu, Z_i) = (2\pi\sigma^2)^{-1/2} e^{-\frac{(x_i - \mu_{Z_i})^2}{2\sigma^2}}.$$

Summing over the labels recovers the original model, $f(\mathbf{x} \mid \mu, \mathbf{p}) = \sum_{\mathbf{Z}} f(\mathbf{x}, \mathbf{Z} \mid \mu, \mathbf{p})$. But the **complete-data likelihood** — given the labels — is much simpler:

$$f(\mathbf{x}, \mathbf{Z} \mid \mu, \mathbf{p}) = \prod_i p_{Z_i} (2\pi\sigma^2)^{-1/2} e^{-\frac{(x_i - \mu_{Z_i})^2}{2\sigma^2}} = (2\pi\sigma^2)^{-N/2} \prod_{k=1}^K p_k^{n_k} e^{-\frac{1}{2\sigma^2} \sum_{i: Z_i=k} (x_i - \mu_k)^2},$$

with $n_k = \sum_i \chi(Z_i = k)$ the number of points in component k . The product over points has become a product over components, each depending only on its own points. Two benefits:

1. the likelihood is cheap to evaluate;
2. with **conjugate priors**, every full conditional is a standard distribution, so Gibbs sampling applies.

Remark. Sampling the *augmented* posterior $f(\mu, \mathbf{p}, \mathbf{Z} \mid \mathbf{x})$ and simply ignoring $\mathbf{Z}$ in the output gives samples from the marginal $f(\mu, \mathbf{p} \mid \mathbf{x})$. Making the space *bigger* makes sampling *easier* — a recurring theme (cf. the momentum variables of HMC in Chapter 11).

9.4.3. Priors

- Means: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0^2)$ independently, $f(\mu) = (2\pi\sigma_0^2)^{-K/2} \prod_k e^{-(\mu_k - \mu_0)^2 / 2\sigma_0^2}$. Taking $\mu_0 = 0, \sigma_0 \rightarrow \infty$ gives a flat prior.
- Weights: $\mathbf{p} \sim \text{Dir}(\gamma)$, $f(\mathbf{p}) = B(\gamma)^{-1} \prod_k p_k^{\gamma_k - 1}$ on the simplex. $\gamma_k = 1$ is uniform on the simplex.

The joint posterior is

$$f(\mu, \mathbf{Z}, \mathbf{p} \mid \mathbf{x}) \propto \prod_{k=1}^K \left[e^{-\frac{1}{2\sigma^2} \sum_{i: Z_i=k} (x_i - \mu_k)^2} e^{-\frac{(\mu_k - \mu_0)^2}{2\sigma_0^2}} p_k^{n_k + \gamma_k - 1} \right].$$

9.4.4. Full conditionals

Each conditional is obtained by keeping only the factors that contain the variable of interest.

Means. For μ_k , only the k -th factor matters:

$$f(\mu_k \mid \mathbf{x}, \mathbf{Z}, \mathbf{p}, \mu_{-k}) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i: Z_i=k} (x_i - \mu_k)^2 - \frac{(\mu_k - \mu_0)^2}{2\sigma_0^2} \right].$$

The exponent is quadratic in μ_k ; completing the square,

$$\mu_k \mid \dots \sim \mathcal{N}(\mu'_k, \sigma_k'^2), \quad \mu'_k = \frac{\mu_0/\sigma_0^2 + n_k \bar{X}_k/\sigma^2}{1/\sigma_0^2 + n_k/\sigma^2}, \quad \sigma_k'^2 = \left(\frac{1}{\sigma_0^2} + \frac{n_k}{\sigma^2} \right)^{-1},$$

with $\bar{X}_k = \frac{1}{n_k} \sum_{i: Z_i=k} x_i$ the mean of the points currently assigned to k . **Precisions add, and the mean is a precision-weighted average** of the prior mean and the data mean (exactly as for the Gaussian channel of Chapter 5).

Weights. Only $\prod_k p_k^{n_k + \gamma_k - 1}$ depends on $\mathbf{p}$, so

$$\mathbf{p} \mid \dots \sim \text{Dir}(\gamma'), \quad \gamma' = \gamma + \mathbf{n}.$$

The prior parameters γ_k act as “pseudo-counts” added to the observed counts n_k .

Labels. For Z_i , only the factor involving point i matters; normalising over its K possible values,

$$P(Z_i = k \mid \dots) = \frac{p_k e^{-(x_i - \mu_k)^2 / 2\sigma^2}}{\sum_l p_l e^{-(x_i - \mu_l)^2 / 2\sigma^2}} \equiv p'_{ik}, \quad Z_i \mid \dots \sim \text{Cat}(\mathbf{p}'_i).$$

This is Bayes’ theorem for “which component produced x_i ?”, with prior p_k and likelihood $\mathcal{N}(x_i; \mu_k, \sigma^2)$.

Definition 9.2 (Gibbs sampler for the Gaussian mixture). Initialise $\mathbf{Z}$ (e.g. at random). Repeat:

1. for each k : compute n_k , $\bar{X}_k$ and sample $\mu_k \sim \mathcal{N}(\mu'_k, \sigma_k'^2)$;
2. sample $\mathbf{p} \sim \text{Dir}(\gamma + \mathbf{n})$;
3. for each i : compute p'_{ik} and sample $Z_i \sim \text{Cat}(\mathbf{p}'_i)$.

Example 9.3 (*Four well-separated components*)

Generate data from $K = 4$ components with $\mu_k = \{1, 3, 5, 7\}$, $\sigma = 0.5$ and $p_k = 0.25$. Starting from random labels, the sampled μ_k converge to the true values within a handful of sweeps, and the agreement between true and inferred labels rises just as quickly (Allegra, Fig. 10.1). It saturates at about 97%: with neighbouring means 4σ apart, a point falls past the midpoint towards a neighbour with probability $1 - \Phi(2) \approx 2.3\%$ per side, and such points are genuinely ambiguous — no method can classify them correctly. The posterior spread of each μ_k is $\approx \sigma / \sqrt{n_k}$.

9.4.5. Gibbs sampling and K-means

Soft K-means clustering alternates two steps:

- *assignment*: each point gets a “responsibility” for each cluster, $r_{ik} \propto e^{-(x_i - \mu_k)^2 / 2\sigma^2}$ (normalised over k);
- *update*: each mean moves to the responsibility-weighted average, $\mu_k = \sum_i r_{ik} x_i / \sum_i r_{ik}$.

Now take the Gibbs sampler in the limit $\sigma_0 \rightarrow \infty$ (flat prior on the means) and $\gamma_k = \gamma \rightarrow \infty$ (weights pinned at $p_k = 1/K$). Then:

- μ_k is drawn from a Gaussian centred at $\bar{X}_k = \sum_i \chi(Z_i = k) x_i / \sum_i \chi(Z_i = k)$ with small width $\sigma / \sqrt{n_k}$, so $\mu_k \approx \bar{X}_k$;
- Z_i is drawn with $P(Z_i = k) \propto e^{-(x_i - \mu_k)^2 / 2\sigma^2} = r_{ik}$.

Averaging over the random labels, $\mathbb{E}[\chi(Z_i = k)] = r_{ik}$, so $\mu_k \approx \sum_i r_{ik} x_i / \sum_i r_{ik}$: **the Gibbs sampler is a stochastic version of soft K-means**, and the two agree when clusters are well separated. Conversely, this reveals that K-means *implicitly assumes* a mixture of isotropic Gaussians with equal widths and equal weights — not at all obvious from its formulation. (The deterministic counterpart that maximises the likelihood instead of sampling is the EM algorithm.)

9.4.6. Label switching

The mixture likelihood is invariant under permutations of the component labels, so with symmetric priors the posterior has $K!$ identical modes. A Gibbs chain typically stays in one labelling (moving between them requires relabelling all points at once — an exponentially unlikely sequence of single-site updates). For

estimating label-invariant quantities this is harmless; for component-specific summaries one imposes an ordering ($\mu_1 < \mu_2 < \dots$) or post-processes the samples.

9.5. Modern context

Gibbs sampling remains the method of choice for models with **discrete latent variables** — mixtures, hidden Markov models, topic models such as LDA, Ising-type models — where gradient-based methods (HMC, Chapter 11) do not apply. For continuous parameters with strong correlations, blocking, reparameterisation or Hamiltonian methods are usually preferable.

9.6. Exercises

Exercise 9.1

Derive the full conditional of μ_k in the mixture model by completing the square, and check the limits $n_k = 0$ and $\sigma_0 \rightarrow \infty$.

Solution

The exponent is $-\frac{1}{2}[\mu_k^2(n_k/\sigma^2 + 1/\sigma_0^2) - 2\mu_k(n_k\bar{X}_k/\sigma^2 + \mu_0/\sigma_0^2)] + \text{const}$, i.e. a Gaussian with precision $n_k/\sigma^2 + 1/\sigma_0^2$ and mean (linear coefficient)/(precision) = μ'_k . If $n_k = 0$ (empty cluster) the conditional is the prior $\mathcal{N}(\mu_0, \sigma_0^2)$. If $\sigma_0 \rightarrow \infty$, it is $\mathcal{N}(\bar{X}_k, \sigma^2/n_k)$ — the frequentist sampling distribution of the mean.

Exercise 9.2

Write a Gibbs sampler for the joint posterior of (μ, τ) for data $x_i \sim \mathcal{N}(\mu, 1/\tau)$ with priors $\mu \sim \mathcal{N}(\mu_0, 1/\lambda_0)$ and $\tau \sim \text{Gamma}(a, b)$.

Solution

$\mu \mid \tau, \mathbf{x} \sim \mathcal{N}\left(\frac{\lambda_0\mu_0 + \tau\sum_i x_i}{\lambda_0 + N\tau}, \frac{1}{\lambda_0 + N\tau}\right)$; $\tau \mid \mu, \mathbf{x} \sim \text{Gamma}(a + N/2, b + \frac{1}{2}\sum_i (x_i - \mu)^2)$. Alternate the two draws.

Exercise 9.3

For the bivariate Gaussian with correlation ρ , how many Gibbs sweeps are needed for an effective sample size of 1000 when $\rho = 0.95$? How many if one first rotates to $u = (x_1 + x_2)/\sqrt{2}$, $v = (x_1 - x_2)/\sqrt{2}$?

Solution

$A = (1 + 0.9025)/(1 - 0.9025) \approx 19.5$, so about 19,500 sweeps. After rotation u and v are independent ($\text{Var } u = 1 + \rho$, $\text{Var } v = 1 - \rho$), the conditionals equal the marginals, and every sweep is an independent draw: 1000 sweeps.

Exercise 9.4

For a two-component mixture with means ± 3 and $\sigma = 1$, explain why the Gibbs sampler does not visit the “label-swapped” mode $(\mu_1, \mu_2) \rightarrow (\mu_2, \mu_1)$.

Solution

To swap, all points currently labelled 1 must be relabelled 2 and vice versa. Each single-site update is conditioned on the current means, under which relabelling a point costs a factor $\approx e^{-(6)^2/2}$ in probability; the means only move once many labels have changed. The path between the two modes passes through configurations of exponentially small posterior probability, so the chain essentially never makes the transition. This is harmless for label-invariant summaries; otherwise impose $\mu_1 < \mu_2$.

9.7. Callouts

i Note

Gibbs is *simple and exact* when conjugacy holds. When a conditional is not standard, replace that single update by a Metropolis step targeting the conditional (**Metropolis-within-Gibbs**) → still a valid sampler.

💡 Tip

Blocking correlated variables (updating them jointly) and reparameterising to decorrelate them are the standard cures for slow Gibbs mixing.

⚠ Warning

Gibbs can fail on multimodal targets whose conditionals are unimodal: each coordinate move is local in the joint space.

! Important

The diagnostics of Chapter 7 (burn-in, autocorrelation, ESS, $\hat{R}$) apply without change. Acceptance rate 1 says nothing about how well the chain explores the target.

9.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 10. MacKay, §29.5 (Gibbs sampling), §20–22 (K-means and mixtures). Bishop, §11.3 (Gibbs), §9.1–9.2 (K-means, mixtures of Gaussians). Brooks et al., *Handbook of MCMC*, §1.12.4. Marin, Mengersen & Robert, *Bayesian modelling and inference on mixtures of distributions*, §1.2–1.4.

10. A problem of high-dimensional inference

This chapter works through a real research problem that ties together everything in Chapters 7–8: building a likelihood with latent variables, choosing conjugate priors, discovering that the exact posterior is a combinatorial monster, and sampling it with Metropolis and Gibbs. The problem comes from population genetics: Gaggiotti et al. (2002) studied how new colonies of **grey seals** are founded, and asked *where the founders came from*, using only the genomes of pups born in the new colony.

10.1. The problem

Grey seals live in isolated colonies; occasionally a new colony is founded by migrants from older ones. Assume the founders come from P old colonies, in unknown proportions

$$\mathbf{x} = (x_1, \dots, x_P), \quad x_i \geq 0, \quad \sum_{i=1}^P x_i = 1.$$

$\mathbf{x}$ is the unknown we want to infer: a point in the $(P-1)$ -dimensional simplex.

The data: $N = 150$ pups born in the new colony (whose parents were migrants) were genotyped at $L = 9$ loci, chosen among the most variable ones to maximise discrimination. At each locus an individual carries two variants (**alleles**) of the gene, one inherited from the mother ($n = 1$) and one from the father ($n = 2$), each among M_l possible alleles. Taking $M = \max_l M_l$ (and assigning zero frequency to non-existent alleles), the data are

$$y_{k,l,n} \in \{1, \dots, M\}, \quad k = 1..N \text{ (individual)}, \quad l = 1..L \text{ (locus)}, \quad n = 1, 2 \text{ (parent)}.$$

What links $\mathbf{x}$ to $\mathbf{y}$ is that the **allele frequencies in the old colonies are known** from previous studies:

$$p_m^{(i,l)} = \text{frequency of allele } m \text{ at locus } l \text{ in colony } i, \quad \sum_m p_m^{(i,l)} = 1.$$

Different colonies have different allele frequencies, so a pup's genotype carries (weak) information about where its parents came from; with many pups and loci, this adds up to information about $\mathbf{x}$.

10.2. Basic model

We first study a simplified model in which only the **maternal** alleles are observed: data $y_{k,l} \equiv y_{k,l,1}$.

10.2.1. Likelihood

Introduce a **latent variable** for each pup,

$$Z_k \in \{1, \dots, P\} : \text{the colony the mother of pup } k \text{ came from.}$$

Two modelling assumptions connect the variables:

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- the probability of inheriting allele m at locus l from a mother of colony i is the frequency of that allele in colony i :

$$P(Y_{k,l} = m \mid Z_k = i) = p_m^{(i,l)};$$

- the probability that a mother came from colony i is the proportion of founders from colony i :

$$P(Z_k = i \mid \mathbf{x}) = x_i.$$

Marginalising the latent variable, and using that $Y_{k,l}$ depends on $\mathbf{x}$ only through Z_k (conditional independence),

$$P(Y_{k,l} = m \mid \mathbf{x}) = \sum_{i=1}^P P(Y_{k,l} = m \mid Z_k = i) P(Z_k = i \mid \mathbf{x}) = \sum_{i=1}^P x_i p_m^{(i,l)}.$$

Assuming further that the alleles at different loci are independent given the origin (a simplification — probably not exactly true for real genomes),

$$P(\mathbf{Y}_k = \mathbf{y}_k \mid Z_k = i) = \prod_{l=1}^L p_{y_{k,l}}^{(i,l)} \equiv \gamma_{ik}, \quad P(\mathbf{Y}_k = \mathbf{y}_k \mid \mathbf{x}) = \sum_{i=1}^P x_i \gamma_{ik}.$$

γ_{ik} is the likelihood that pup k 's maternal genotype came from colony i . Assuming independent individuals, the full likelihood is

$$P(\mathbf{Y} = \mathbf{y} \mid \mathbf{x}) = \prod_{k=1}^N \sum_{i=1}^P x_i \gamma_{ik} = \prod_{k=1}^N \sum_{i=1}^P x_i \prod_{l=1}^L p_{y_{k,l}}^{(i,l)}.$$

This is a **mixture model** of exactly the type of Chapter 8: each pup is drawn from one of P “components” (colonies) with weights $\mathbf{x}$, but the component distributions γ_{ik} are *known*.

10.2.2. Prior: the Dirichlet distribution

The likelihood is a polynomial in $\mathbf{x}$, so the natural (conjugate) prior on the simplex is the Dirichlet distribution.

Definition 10.1 (Dirichlet distribution). $\mathbf{x} \sim \text{Dir}(\alpha)$, $\alpha_i > 0$, has density on the simplex $\mathcal{D} = \{\mathbf{x} : x_i \geq 0, \sum_i x_i = 1\}$

$$\text{Dir}(\alpha; \mathbf{x}) = \frac{1}{B(\alpha)} \prod_{i=1}^P x_i^{\alpha_i - 1}, \quad B(\alpha) = \frac{\Gamma(\alpha_1) \cdots \Gamma(\alpha_P)}{\Gamma(\alpha_1 + \cdots + \alpha_P)}.$$

With $\alpha_0 = \sum_i \alpha_i$: $\mathbb{E}[x_i] = \alpha_i / \alpha_0$ and $\text{Var}[x_i] = \alpha_i(\alpha_0 - \alpha_i) / (\alpha_0^2(\alpha_0 + 1))$. For $P = 2$ it reduces to the Beta distribution. $\alpha = (1, \dots, 1)$ is uniform on the simplex; large α_0 concentrates the mass around the mean.

The key integral, used repeatedly below, is just the normalisation:

$$\int_{\mathcal{D}} d\mathbf{x} \prod_i x_i^{\beta_i - 1} = B(\beta).$$

The posterior is

$$P(\mathbf{x} \mid \mathbf{y}) = \frac{1}{P(\mathbf{y})} \prod_{k=1}^N \left(\sum_{i=1}^P x_i \gamma_{ik} \right) \cdot \frac{1}{B(\alpha)} \prod_{p=1}^P x_p^{\alpha_p - 1},$$

with evidence $P(\mathbf{y}) = \int_{\mathcal{D}} d\mathbf{x}$ (numerator), an integral over the simplex.

10.2.3. Metropolis approach

Metropolis only needs the unnormalised posterior. A simple proposal moves $\mathbf{x}$ by a small random step and then projects back onto the simplex:

$$\mathbf{x} \rightarrow \mathbf{x} + \mathbf{u} \rightarrow b_{0,1}(\mathbf{x} + \mathbf{u}) \rightarrow \frac{b_{0,1}(\mathbf{x} + \mathbf{u})}{\sum_i b_{0,1}(x_i + u_i)} = \mathbf{x}', \quad u_i \sim \mathcal{U}(-\epsilon, \epsilon),$$

where $b_{0,1}(x) = \min(\max(x, 0), 1)$ clips each component into $[0, 1]$. The acceptance probability is

$$A(\mathbf{x}' | \mathbf{x}) = \min \left\{ 1, \frac{\prod_k (\sum_i x'_i \gamma_{ik}) \prod_p (x'_p)^{\alpha_p - 1}}{\prod_k (\sum_i x_i \gamma_{ik}) \prod_p x_p^{\alpha_p - 1}} \right\},$$

in which the evidence and $B(\alpha)$ cancel. Each evaluation costs $O(NP)$ operations (after precomputing γ_{ik} , which costs $O(NPL)$ once). In practice one works with log-probabilities to avoid underflow: $\prod_k$ of 150 small numbers is far below double-precision range.

Remark (Is this proposal symmetric?). The clip-and-renormalise map is not exactly symmetric, so strictly a Hastings correction would be needed. Cleaner alternatives: (i) parametrise by $P - 1$ free coordinates and reject proposals that leave the simplex (a symmetric proposal on a bounded domain is fine — out-of-domain proposals have $\pi = 0$ and are simply rejected); (ii) propose $\mathbf{x}' \sim \text{Dir}(c\mathbf{x})$ with a large concentration c and include the Hastings ratio $\text{Dir}(c\mathbf{x}'; \mathbf{x}) / \text{Dir}(c\mathbf{x}; \mathbf{x}')$.

10.2.4. Analytic expression of the posterior

The posterior can, in principle, be written in closed form. Expand the product of sums:

$$\prod_{k=1}^N \left(\sum_{i=1}^P x_i \gamma_{ik} \right) = \sum_{i_1=1}^P \cdots \sum_{i_N=1}^P \prod_{k=1}^N x_{i_k} \gamma_{i_k k} = \sum_{\mathbf{i}} \prod_{k=1}^N x_{i_k} \gamma_{i_k k},$$

where the multi-index $\mathbf{i} = (i_1, \dots, i_N)$ runs over all P^N assignments of pups to colonies — i.e. over all values of the latent variables $\mathbf{Z}$. Let $n_p(\mathbf{i}) = \sum_k \delta_{i_k p}$ count how many pups are assigned to colony p . Then $\prod_k x_{i_k} = \prod_p x_p^{n_p(\mathbf{i})}$ and the numerator becomes

$$\frac{1}{B(\alpha)} \sum_{\mathbf{i}} \left(\prod_k \gamma_{i_k k} \right) \prod_p x_p^{n_p(\mathbf{i}) + \alpha_p - 1} = \sum_{\mathbf{i}} \xi(\mathbf{i}) \text{Dir}(\mathbf{n}(\mathbf{i}) + \alpha; \mathbf{x}), \quad \xi(\mathbf{i}) = \frac{B(\mathbf{n}(\mathbf{i}) + \alpha)}{B(\alpha)} \prod_{k=1}^N \gamma_{i_k k}.$$

Since each Dirichlet integrates to one, the evidence is $P(\mathbf{y}) = \sum_{\mathbf{i}} \xi(\mathbf{i})$, and

Proposition 10.1 (The posterior is a mixture of Dirichlets).

$$P(\mathbf{x} | \mathbf{y}) = \sum_{\mathbf{i}} \zeta(\mathbf{i}) \text{Dir}(\mathbf{n}(\mathbf{i}) + \alpha; \mathbf{x}), \quad \zeta(\mathbf{i}) = \frac{\xi(\mathbf{i})}{\sum_{\mathbf{j}} \xi(\mathbf{j})}.$$

Posterior moments follow from those of the Dirichlet, e.g.

$$\mathbb{E}[x_p | \mathbf{y}] = \sum_{\mathbf{i}} \zeta(\mathbf{i}) \frac{n_p(\mathbf{i}) + \alpha_p}{\sum_q (n_q(\mathbf{i}) + \alpha_q)} = \sum_{\mathbf{i}} \zeta(\mathbf{i}) \frac{n_p(\mathbf{i}) + \alpha_p}{N + \alpha_0}.$$

The interpretation is transparent: for each hypothetical assignment $\mathbf{i}$ of pups to colonies, the posterior on $\mathbf{x}$ would be the conjugate $\text{Dir}(\alpha + \text{counts})$; the true posterior averages these over assignments, weighted by how well each assignment explains the genotypes.

Remark (Why the analytic formula is useless). The sum has P^N terms. With $N = 150$ and $P = 7$ (as in the study), that is $7^{150} \approx 10^{127}$ terms — hopeless. This is a general feature of mixture models: marginalising the latent variables *analytically* is exponential in N . Note, by contrast, that the factorised form $\prod_k \sum_i x_i \gamma_{ik}$ costs only $N \cdot P$ operations, which is why Metropolis on $\mathbf{x}$ is perfectly feasible.

10.2.5. A simple example

To see the structure, consider $P = 2$ colonies, $M = 2$ alleles and two loci, “tail length” and “fin length”, with allele 1 = long and 2 = short. The known frequencies are

$$p^{(1,1)} = p^{(1,2)} = \left(\frac{2}{3}, \frac{1}{3}\right), \quad p^{(2,1)} = p^{(2,2)} = \left(\frac{1}{4}, \frac{3}{4}\right) :$$

in colony 1 two thirds of the individuals have long tails (and fins), in colony 2 only a quarter. We observe two alleles (y_1, y_2) , one per locus, and $\mathbf{x} = (x_1, 1 - x_1)$. Following Allegra’s notes, take the likelihood to be

$$P(y_1, y_2 \mid \mathbf{x}) = \prod_{l=1}^2 \left(\sum_{i=1}^2 x_i p_{y_l}^{(i,l)} \right) = \sum_{i,j} x_i x_j p_{y_1}^{(i,1)} p_{y_2}^{(j,2)},$$

i.e. each of the two alleles comes from an *independently drawn* founder. (This is equivalent to $N = 2$ pups with $L = 1$ locus each. For a single pup with two loci, the basic model would instead give $\sum_i x_i p_{y_1}^{(i,1)} p_{y_2}^{(i,2)}$, linear in $\mathbf{x}$, and the posterior would be a mixture of only two Beta components; see the exercises.) For the four possible observations this gives

(y_1, y_2)	$P(y_1, y_2 \mid \mathbf{x})$
(long, long)	$\frac{4}{9}x_1^2 + \frac{1}{3}x_1x_2 + \frac{1}{16}x_2^2$
(long, short), (short, long)	$\frac{2}{9}x_1^2 + \frac{7}{12}x_1x_2 + \frac{3}{16}x_2^2$
(short, short)	$\frac{1}{9}x_1^2 + \frac{1}{2}x_1x_2 + \frac{9}{16}x_2^2$

With a uniform prior ($\alpha_1 = \alpha_2 = 1$), each monomial $x_1^a x_2^b$ is an unnormalised Beta($a+1, b+1$) with integral $B(a+1, b+1)$: $\int x_1^2 = \frac{1}{3}$, $\int x_1 x_2 = \frac{1}{6}$, $\int x_2^2 = \frac{1}{3}$. For (long, long) the evidence is $\frac{4}{27} + \frac{1}{18} + \frac{1}{48} = \frac{97}{432}$ and

$$P(x_1 \mid \text{long, long}) = \frac{64}{97} \text{Beta}(x_1; 3, 1) + \frac{24}{97} \text{Beta}(x_1; 2, 2) + \frac{9}{97} \text{Beta}(x_1; 1, 3).$$

Similarly

(y_1, y_2)	weight of Beta(3,1)	Beta(2,2)	Beta(1,3)	$\mathbb{E}[x_1 \mid y]$
(long, long)	64/97	24/97	9/97	0.642
(long, short) or (short, long)	32/101	42/101	27/101	0.512
(short, short)	16/133	36/133	81/133	0.378

(using $\mathbb{E}[x_1] = 3/4, 1/2, 1/4$ for the three Beta components). The three components correspond to the latent configurations “both founders from colony 1” ($\mathbf{n} = (2, 0)$), “one from each” ($\mathbf{n} = (1, 1)$, two assignments) and “both from colony 2” ($\mathbf{n} = (0, 2)$), each giving a $\text{Dir}(\mathbf{n} + \alpha) = \text{Beta}$ component. Observing long features shifts belief towards colony 1, short features towards colony 2, and mixed features leave the prior mean $1/2$ almost unchanged. With two alleles the posterior remains very broad: each observation carries little information, and precision comes only from many pups.

10.2.6. Gibbs sampling approach

The smart alternative to marginalising $\mathbf{Z}$ is **not** to marginalise it, and to sample $\mathbf{Z}$ jointly with $\mathbf{x}$, exactly as for the Gaussian mixture of Chapter 8. Keeping the latent variables,

$$P(\mathbf{y} | \mathbf{Z}) = \prod_{k=1}^N \gamma_{Z_k k}, \quad P(\mathbf{Z} | \mathbf{x}) = \prod_{k=1}^N x_{Z_k} = \prod_{i=1}^P x_i^{m_i(\mathbf{Z})}, \quad m_i(\mathbf{Z}) = \sum_k \delta(Z_k = i),$$

so the joint posterior is

$$P(\mathbf{x}, \mathbf{Z} | \mathbf{y}) = \frac{1}{P(\mathbf{y})B(\alpha)} \prod_{k=1}^N \gamma_{Z_k k} \prod_{i=1}^P x_i^{m_i(\mathbf{Z}) + \alpha_i - 1}.$$

The full conditionals are standard:

Proportions. Only the last factor depends on $\mathbf{x}$; normalising over the simplex,

$$\mathbf{x} | \mathbf{y}, \mathbf{Z} \sim \text{Dir}(\alpha + \mathbf{m}(\mathbf{Z})) :$$

prior pseudo-counts plus the number of mothers currently assigned to each colony.

Origins. Only the factors $\gamma_{Z_k k} x_{Z_k}$ depend on Z_k , so

$$P(Z_k = i | \mathbf{y}, \mathbf{x}, \mathbf{Z}_{-k}) = \frac{\gamma_{ik} x_i}{\sum_j \gamma_{jk} x_j} :$$

Bayes' theorem for “which colony did this mother come from?”, with prior $\mathbf{x}$ and likelihood $\gamma_{\cdot k}$. Given $\mathbf{x}$, the Z_k are conditionally independent and can all be updated at once.

Definition 10.2 (Block Gibbs sampler (basic model)). Repeat:

1. sample $\mathbf{x} \sim \text{Dir}(\alpha + \mathbf{m}(\mathbf{Z}))$;
2. for all k , sample $Z_k \sim \text{Cat}(\gamma_{\cdot k} \odot \mathbf{x} / \sum_j \gamma_{jk} x_j)$.

Each sweep costs $O(NP)$ and requires no tuning. As a bonus, the samples of Z_k give the posterior probability that each pup's mother came from each colony — an individual-level “assignment test”.

Remark (Data augmentation). The Gibbs sampler explores the joint space of $(\mathbf{x}, \mathbf{Z})$, of dimension $P-1+N$, which is *larger* than the space of $\mathbf{x}$ alone. Yet it is easier to sample: the P^N -term sum is never computed; instead the chain *visits* the relevant assignments $\mathbf{i}$ with frequency $\propto \xi(\mathbf{i})$. The marginal of $\mathbf{x}$ in the samples is the exact mixture-of-Dirichlets posterior.

10.3. Refined model

The model actually used by Gaggiotti et al. is richer in three ways: both parents contribute alleles, parents tend to mate within their own origin, and the proportions $\mathbf{x}$ are linked to the geography of the old colonies.

10.3.1. Two parents and assortative mating

Let $Z_{k,n}$ be the origin of parent n of pup k . As before,

$$P(Y_{k,l,n} = m | Z_{k,n} = i) = p_m^{(i,l)}, \quad P(\mathbf{Y}_{k,n} = \mathbf{y}_{k,n} | Z_{k,n} = i) = \prod_l p_{y_{k,l,n}}^{(i,l)}.$$

10. A problem of high-dimensional inference

If the two parents' origins were independent, $P(Z_{k,1} = i, Z_{k,2} = j \mid \mathbf{x}) = x_i x_j$, and (assuming the maternal and paternal alleles are independent given the origins)

$$P(\mathbf{Y} = \mathbf{y} \mid \mathbf{x}) = \prod_{k=1}^N \sum_{i,j=1}^P x_i x_j \prod_{l=1}^L p_{y_{k,l},1}^{(i,l)} p_{y_{k,l},2}^{(j,l)}.$$

But founders mate preferentially with individuals of the same origin (**assortative mating**). Model this with a parameter $w \in [0, 1]$: with probability w an individual mates with someone of its own origin, with probability $1 - w$ with a random founder:

$$P(Z_{k,1} = i, Z_{k,2} = j \mid \mathbf{x}, w) = w \delta_{ij} x_i + (1 - w) x_i x_j.$$

Defining $\gamma_{ijk} = \prod_l p_{y_{k,l},1}^{(i,l)} p_{y_{k,l},2}^{(j,l)}$, the likelihood becomes

$$P(\mathbf{Y} = \mathbf{y} \mid \mathbf{x}, w) = \prod_{k=1}^N \left[w \sum_{i=1}^P x_i \gamma_{iik} + (1 - w) \sum_{i,j=1}^P x_i x_j \gamma_{ijk} \right].$$

For $w = 0$ we recover random mating; for $w = 1$ both parents always share an origin.

10.3.2. Geography in the prior

Gaggiotti et al. further assume that the expected contribution of each old colony decays exponentially with its (known) distance δ_i from the new colony and grows with its (known) productivity π_i :

$$\mathbb{E}[x_i] \propto ((1 - S) + S\pi_i) e^{-R\delta_i},$$

with an unknown **distance decay** R and **productivity weight** S . This is built into the Dirichlet prior:

$$\mathbf{x} \mid R, S, \alpha_0 \sim \text{Dir}(\alpha), \quad \alpha_i = \alpha_0 \frac{((1 - S) + S\pi_i) e^{-R\delta_i}}{c}, \quad c = \sum_{i=1}^P ((1 - S) + S\pi_i) e^{-R\delta_i},$$

so that $\mathbb{E}[x_i] = \alpha_i / \alpha_0$ has the desired form and α_0 controls how tightly $\mathbf{x}$ follows it. The **hyperparameters** get uniform priors:

$$\alpha_0 \sim \mathcal{U}(P, 100), \quad w \sim \mathcal{U}(0, 1), \quad S \sim \mathcal{U}(0, 1), \quad R \sim \mathcal{U}(0, 5).$$

This is a **hierarchical model**: $\text{data} \leftarrow (\mathbf{x}, w) \leftarrow (R, S, \alpha_0)$. The posterior is

$$P(\mathbf{x}, w, \alpha_0, R, S \mid \mathbf{y}) \propto P(\mathbf{y} \mid \mathbf{x}, w) \text{Dir}(\alpha(R, S, \alpha_0); \mathbf{x}) P(w) P(R) P(S) P(\alpha_0).$$

Note that $B(\alpha)$ now depends on the hyperparameters and does *not* cancel in Metropolis ratios for R, S, α_0 .

The posterior on R and S is the scientifically interesting output: *do seals found colonies preferentially near their origin, and do productive colonies send more migrants?*

10.3.3. Metropolis approach

Update the parameters one at a time (or in blocks) with bounded random-walk moves, $u \sim \mathcal{U}(-\epsilon, \epsilon)$:

$$\mathbf{x} \rightarrow \frac{b(\mathbf{x} + \mathbf{u})}{\sum_i b(x_i + u_i)}, \quad w \rightarrow b_{0,1}(w + u), \quad S \rightarrow b_{0,1}(S + u), \quad R \rightarrow b_{0,5}(R + u), \quad \alpha_0 \rightarrow b_{P,100}(\alpha_0 + u),$$

each accepted with the Metropolis ratio of the unnormalised posterior. (The remark on symmetry of clipped proposals applies here too.)

10.3.4. Gibbs sampling with latent variables

As in the basic model, we can keep the parental origins $Z_{k,1}, Z_{k,2}$ as latent variables. To deal with the mixture in the mating model, introduce one more latent variable per pup,

$$Q_k \in \{0, 1\} : \quad Q_k = 1 \text{ if the parents mated assortatively,} \quad P(Q_k = 1 \mid w) = w,$$

with

$$P(Z_{k,1}, Z_{k,2} \mid Q_k = 1, \mathbf{x}) = \delta_{Z_{k,1}Z_{k,2}} x_{Z_{k,1}}, \quad P(Z_{k,1}, Z_{k,2} \mid Q_k = 0, \mathbf{x}) = x_{Z_{k,1}} x_{Z_{k,2}}.$$

Summing over Q_k reproduces $w\delta_{ij}x_i + (1-w)x_ix_j$. With a Beta prior $w \sim \text{Beta}(\beta_1, \beta_2)$ (uniform for $\beta_1 = \beta_2 = 1$), the augmented posterior is

$$P(\mathbf{x}, w, \mathbf{Z}, \mathbf{Q} \mid \mathbf{y}) \propto \prod_{k=1}^N \gamma_{Z_{k,1}Z_{k,2}k} \prod_{i=1}^P x_i^{m_i(\mathbf{Z}, \mathbf{Q}) + \alpha_i - 1} w^{n_q + \beta_1 - 1} (1-w)^{N - n_q + \beta_2 - 1} \prod_{k: Q_k=1} \delta_{Z_{k,1}Z_{k,2}},$$

where $n_q = \sum_k \delta(Q_k = 1)$ counts assortative matings and

$$m_i(\mathbf{Z}, \mathbf{Q}) = \sum_{k=1}^N \left[\delta(Q_k = 1) \delta(Z_{k,1} = i) + \delta(Q_k = 0) (\delta(Z_{k,1} = i) + \delta(Z_{k,2} = i)) \right]$$

counts the *independent draws* from $\mathbf{x}$ that landed on colony i (an assortative pair counts once, since only one origin was drawn). The full conditionals:

- **Proportions:** $\mathbf{x} \mid \dots \sim \text{Dir}(\alpha + \mathbf{m}(\mathbf{Z}, \mathbf{Q}))$.
- **Mating parameter:** $w \mid \dots \sim \text{Beta}(\beta_1 + n_q, \beta_2 + N - n_q)$.
- **Parental origins**, jointly for each pup:

$$P(Z_{k,1} = i, Z_{k,2} = j \mid Q_k = 1, \dots) = \frac{\gamma_{iik} x_i \delta_{ij}}{\sum_{i', j'} \gamma_{i'j'k} x_{i'} x_{j'}}, \quad P(Z_{k,1} = i, Z_{k,2} = j \mid Q_k = 0, \dots) = \frac{\gamma_{ijk} x_i x_j}{\sum_{i', j'} \gamma_{i'j'k} x_{i'} x_{j'}}.$$

- **Mating type**, given the origins:

$$P(Q_k = 1 \mid \dots) = \frac{w \delta_{Z_{k,1}Z_{k,2}} x_{Z_{k,1}}}{w \delta_{Z_{k,1}Z_{k,2}} x_{Z_{k,1}} + (1-w) x_{Z_{k,1}} x_{Z_{k,2}}}, \quad P(Q_k = 0 \mid \dots) = 1 - P(Q_k = 1 \mid \dots).$$

(If the parents have different origins, $Q_k = 0$ with certainty.)

Remark (A mixing caveat). Updating Q_k given $\mathbf{Z}_k$ and then $\mathbf{Z}_k$ given Q_k can mix slowly: when $Q_k = 1$ the origins are forced equal, and then Q_k can only switch if $\mathbf{Z}_k$ already has equal origins. Sampling $(Q_k, Z_{k,1}, Z_{k,2})$ **jointly** — from the $P+P^2$ possible configurations, with weights $w\gamma_{iik}x_i$ and $(1-w)\gamma_{ijk}x_ix_j$ — avoids the problem at the same cost.

For the hyperparameters R, S, α_0 the conditionals are not standard (they enter through α and $B(\alpha)$), so we use Metropolis steps for them. The result is a **Metropolis-within-Gibbs** sampler:

Definition 10.3 (Sampler for the refined model). Repeat:

1. sample $\mathbf{x} \sim \text{Dir}(\alpha(R, S, \alpha_0) + \mathbf{m}(\mathbf{Z}, \mathbf{Q}))$;
2. sample $w \sim \text{Beta}(\beta_1 + n_q, \beta_2 + N - n_q)$;
3. for all k , sample $(Z_{k,1}, Z_{k,2})$ from their conditional;
4. for all k , sample Q_k from its conditional;
5. Metropolis updates $S \rightarrow b_{0,1}(S + u)$, $R \rightarrow b_{0,5}(R + u)$, $\alpha_0 \rightarrow b_{P,100}(\alpha_0 + u)$, each targeting

$$\text{Dir}(\alpha(R, S, \alpha_0); \mathbf{x}) P(R)P(S)P(\alpha_0).$$

10.4. Lessons

- **Latent variables make likelihoods tractable.** Writing down the generative story (“each pup has a mother from some colony”) gives the likelihood; keeping the story’s hidden variables gives a sampler.
- **Exact marginalisation is exponential.** The analytic posterior is a mixture over P^N assignments; MCMC replaces the sum by a random walk through the relevant terms.
- **Conjugacy turns modelling choices into algorithms.** Dirichlet priors for proportions and Beta priors for probabilities make every latent-variable conditional a standard distribution.
- **Hierarchies encode scientific hypotheses.** Distance decay and productivity enter as hyperparameters whose posterior answers the scientific question; their non-conjugate updates are handled by Metropolis within Gibbs.

10.5. Exercises

Exercise 10.1

In the simple example, verify the evidence $97/432$ for the observation (long, long) and the mixture weights $64/97, 24/97, 9/97$. Then compute the posterior variance of x_1 .

Solution

$\frac{4}{9} \cdot \frac{1}{3} = \frac{4}{27} = \frac{64}{432}$, $\frac{1}{3} \cdot \frac{1}{6} = \frac{1}{18} = \frac{24}{432}$, $\frac{1}{16} \cdot \frac{1}{3} = \frac{1}{48} = \frac{9}{432}$; total $97/432$, and the weights are the three terms divided by the total. For the variance use $\mathbb{E}[x_1^2] = \sum_c w_c \mathbb{E}[x_1^2]$ with $\mathbb{E}[x^2] = a(a+1)/((a+b)(a+b+1))$: $\text{Beta}(3, 1) \rightarrow 3 \cdot 4/(4 \cdot 5) = 0.6$, $\text{Beta}(2, 2) \rightarrow 2 \cdot 3/(4 \cdot 5) = 0.3$, $\text{Beta}(1, 3) \rightarrow 2/20 = 0.1$. $\mathbb{E}[x_1^2] = (64 \cdot 0.6 + 24 \cdot 0.3 + 9 \cdot 0.1)/97 = 46.5/97 \approx 0.479$; $\text{Var} = 0.479 - 0.642^2 \approx 0.067$ (s.d. ≈ 0.26) — barely narrower than the uniform prior’s 0.289 .

Exercise 10.2

Redo the simple example for a *single* pup whose two loci share the same mother, $P(y_1, y_2 \mid \mathbf{x}) = \sum_i x_i p_{y_1}^{(i,1)} p_{y_2}^{(i,2)}$, for the observation (long, long).

Solution

$P = \frac{4}{9}x_1 + \frac{1}{16}x_2$. With a uniform prior, $\int x_1 = \int x_2 = \frac{1}{2}$, so the evidence is $\frac{2}{9} + \frac{1}{32} = \frac{73}{288}$ and the posterior is $\frac{64}{73}\text{Beta}(2, 1) + \frac{9}{73}\text{Beta}(1, 2)$, with mean $\frac{64}{73} \cdot \frac{2}{3} + \frac{9}{73} \cdot \frac{1}{3} \approx 0.626$. The two components correspond to the two values of the single latent variable Z_1 .

Exercise 10.3

Derive the full conditional $P(Z_k = i \mid \mathbf{y}, \mathbf{x}, \mathbf{Z}_{-k}) \propto \gamma_{ik} x_i$ of the basic model by dividing the joint posterior by its sum over Z_k .

Solution

In $P(\mathbf{x}, \mathbf{Z} \mid \mathbf{y}) \propto \prod_{k'} \gamma_{Z_{k'} k'} x_{Z_{k'}} \cdot \prod_p x_p^{\alpha_p - 1}$, all factors with $k' \neq k$ and the prior are common to numerator and denominator and cancel, leaving $\gamma_{Z_k k} x_{Z_k} / \sum_i \gamma_{ik} x_i$.

Exercise 10.4

Show that summing the augmented model over Q_k reproduces $P(Z_{k,1} = i, Z_{k,2} = j \mid \mathbf{x}, w) = w\delta_{ij}x_i + (1-w)x_ix_j$, and that the likelihood reduces to the basic-model form (with both alleles from the same colony) when $w = 1$.

Solution

$\sum_q P(Q_k = q \mid w)P(i, j \mid q, \mathbf{x}) = w\delta_{ij}x_i + (1-w)x_ix_j$. For $w = 1$, $P(\mathbf{y} \mid \mathbf{x}) = \prod_k \sum_i x_i \gamma_{iik}$ with $\gamma_{iik} = \prod_l p_{y_{k,l,1}}^{(i,l)} p_{y_{k,l,2}}^{(i,l)}$: each pup is a single draw from a colony, carrying $2L$ alleles from it — the basic model with twice as many loci.

Exercise 10.5

Estimate the cost per sweep of the refined Gibbs sampler, and compare with the size of the analytic sum.

Solution

Precompute γ_{ijk} : $O(NP^2L)$ once. Per sweep: the $(Z_{k,1}, Z_{k,2})$ updates cost $O(NP^2)$, the rest $O(N+P)$. With $N = 150$, $P = 7$ this is $\sim 10^4$ operations per sweep, against $\sim (P^2)^N \approx 10^{254}$ terms in the exact marginalisation.

10.6. Callouts

i Note

This chapter is a template for applied Bayesian work: (1) write the generative story, (2) translate it into a likelihood with latent variables, (3) choose (conjugate) priors, encoding scientific hypotheses in hyperparameters, (4) sample with Gibbs where conditionals are standard and Metropolis elsewhere.

💡 Tip

Always test a sampler on **simulated data** with known $\mathbf{x}$, w , R , S before running it on real data: if the posterior does not concentrate around the truth as N grows, there is a bug or an identifiability problem.

⚠ Warning

Products of $N \cdot L$ probabilities underflow double precision. Work with $\log \gamma_{ik}$ and use the log-sum-exp trick when normalising categorical conditionals: $\log \sum_i e^{a_i} = a_{\max} + \log \sum_i e^{a_i - a_{\max}}$.

10.7. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 11. Gaggiotti, Jones, Lee, Amos, Harwood & Nichols, *Patterns of colonization in a metapopulation of grey seals*, *Nature* **416**, 424 (2002). Bishop, §2.2 (Dirichlet distribution). Marin, Mengersen & Robert (2005) on mixtures and data augmentation.

11. Variational Bayes

MCMC is asymptotically exact but can be slow, and it is hard to know when it has converged. **Variational Bayes** (VB) trades exactness for speed: it approximates the posterior by the closest member of a tractable family of distributions, turning inference into *optimisation*. The measure of closeness is — once again — the KL divergence of Chapter 2, and the objective is a lower bound on the log-evidence. This chapter derives the variational principle, the mean-field update equations, works out two examples in full (a Gaussian, and the mixture model of Chapter 8), and ends with the variational autoencoder, where the same bound trains deep generative models.

11.1. The evidence lower bound

Let $\mathbf{y}$ be the data and $\mathbf{z}$ all unknowns (parameters and latent variables). For **any** distribution $q(\mathbf{z})$,

$$\begin{aligned}\log f(\mathbf{y}) &= \int d\mathbf{z} q(\mathbf{z}) \log f(\mathbf{y}) = \int d\mathbf{z} q(\mathbf{z}) \left[\log \frac{f(\mathbf{z}, \mathbf{y})}{q(\mathbf{z})} + \log \frac{q(\mathbf{z}) f(\mathbf{y})}{f(\mathbf{z} | \mathbf{y})} \right] \\ &= \underbrace{\int d\mathbf{z} q(\mathbf{z}) \log \frac{f(\mathbf{z}, \mathbf{y})}{q(\mathbf{z})}}_{\mathcal{L}(q)} + \underbrace{\int d\mathbf{z} q(\mathbf{z}) \log \frac{q(\mathbf{z})}{f(\mathbf{z} | \mathbf{y})}}_{D_{KL}(q \| f(\cdot | \mathbf{y}))},\end{aligned}$$

using $f(\mathbf{z}, \mathbf{y})/f(\mathbf{y}) = f(\mathbf{z} | \mathbf{y})$.

Theorem 11.1 (Evidence lower bound (ELBO)).

$$\log f(\mathbf{y}) = \mathcal{L}(q) + D_{KL}(q(\mathbf{z}) \| f(\mathbf{z} | \mathbf{y})), \quad \mathcal{L}(q) = \mathbb{E}_q[\log f(\mathbf{z}, \mathbf{y})] - \mathbb{E}_q[\log q(\mathbf{z})].$$

Since $D_{KL} \geq 0$, $\mathcal{L}(q) \leq \log f(\mathbf{y})$ for every q , with equality iff q is the exact posterior.

The left-hand side does not depend on q , so **maximising the ELBO over q is the same as minimising $D_{KL}(q \| f(\cdot | \mathbf{y}))$** : we force q to be as close as possible to the posterior. Crucially, $\mathcal{L}$ involves only the *joint* $f(\mathbf{z}, \mathbf{y})$ (likelihood $\times$ prior), never the intractable evidence. The VB recipe is:

choose a family $\mathcal{Q}$ of tractable distributions, and find $q^* = \arg \max_{q \in \mathcal{Q}} \mathcal{L}(q)$.

Two other readings of the ELBO are useful:

- $\mathcal{L}(q) = \mathbb{E}_q[\log f(\mathbf{z}, \mathbf{y})] + H[q]$: an “energy” term rewarding q for sitting where the joint is large, plus an entropy term rewarding it for spreading out — exactly a (negative) **variational free energy** of statistical physics, with $\log f(\mathbf{y})$ playing the role of $-F = \log Z$.
- $\mathcal{L}(q) = \mathbb{E}_q[\log f(\mathbf{y} | \mathbf{z})] - D_{KL}(q(\mathbf{z}) \| f_{\text{prior}}(\mathbf{z}))$: data fit minus a penalty for departing from the prior (used for the VAE below).

Remark (Which KL?). VB minimises $D(q \| p)$, the “reverse” KL. Since $D(q \| p) = \int q \log(q/p)$, it is huge wherever $q > 0$ but $p \approx 0$: q is pushed to **avoid** regions where the posterior is small. The result is **mode-seeking** and typically **under-dispersed**: for a multimodal posterior, q locks onto one mode; for a correlated posterior, q is too narrow. The “forward” KL $D(p \| q)$ would be mass-covering instead, but it

needs expectations under p — which is what we cannot compute.

11.2. The mean-field approximation

The most common choice is to assume that q **factorises** over groups of variables,

$$q(\mathbf{z}) = \prod_i q_i(\mathbf{z}_i),$$

with no assumption on the form of each factor. This is the **mean-field** approximation, named after the physics approximation in which each spin feels only the *average* field of the others.

11.2.1. Derivation of the update

Maximise $\mathcal{L}$ over one factor q_j , keeping the others fixed. Separating the terms that involve q_j ,

$$\begin{aligned} \mathcal{L}(q) &= \int d\mathbf{z} \prod_i q_i(\mathbf{z}_i) \left[\log f(\mathbf{z}, \mathbf{y}) - \sum_i \log q_i(\mathbf{z}_i) \right] \\ &= \int d\mathbf{z}_j q_j(\mathbf{z}_j) \left[\int \prod_{i \neq j} d\mathbf{z}_i q_i(\mathbf{z}_i) \log f(\mathbf{z}, \mathbf{y}) \right] - \int d\mathbf{z}_j q_j \log q_j + \text{const} \\ &= \int d\mathbf{z}_j q_j(\mathbf{z}_j) \log \tilde{f}(\mathbf{z}_j, \mathbf{y}) - \int d\mathbf{z}_j q_j(\mathbf{z}_j) \log q_j(\mathbf{z}_j) + \text{const}, \end{aligned}$$

where “const” does not depend on q_j and we defined the distribution $\tilde{f}$ by

$$\log \tilde{f}(\mathbf{z}_j, \mathbf{y}) = \mathbb{E}_{i \neq j} [\log f(\mathbf{z}, \mathbf{y})] + \text{const},$$

the log-joint **averaged over all other factors**. Then

$$\mathcal{L}(q) = -D_{KL}(q_j \| \tilde{f}) + \text{const},$$

which is maximised by $q_j = \tilde{f}$.

Theorem 11.2 (Mean-field update). *For fixed q_i , $i \neq j$, the optimal factor is*

$$\log q_j^*(\mathbf{z}_j) = \mathbb{E}_{i \neq j} [\log f(\mathbf{z}, \mathbf{y})] + \text{const}, \quad q_j^*(\mathbf{z}_j) = \frac{\exp(\mathbb{E}_{i \neq j} [\log f(\mathbf{z}, \mathbf{y})])}{\int d\mathbf{z}_j \exp(\mathbb{E}_{i \neq j} [\log f(\mathbf{z}, \mathbf{y})])}.$$

Recipe: write the log of the joint over all hidden and observed variables, keep only the terms involving $\mathbf{z}_j$, take the expectation over all *other* factors, and recognise the resulting functional form (normalisation is usually read off at the end).

The equations for the different j are coupled — each right-hand side depends on the other factors — so they are solved **iteratively**: initialise all factors and cycle through them, updating each in turn (**coordinate-ascent variational inference**, CAVI). Each update maximises $\mathcal{L}$ with respect to one factor ($\mathcal{L}$ is concave in each q_j separately), so $\mathcal{L}$ increases monotonically and the iteration converges — to a *local* maximum, which may depend on the initialisation.

Remark (VB vs Gibbs). Compare with the Gibbs sampler of Chapter 8. Gibbs updates $\mathbf{z}_j$ by drawing from $f(\mathbf{z}_j | \mathbf{z}_{-j}, \mathbf{y}) \propto \exp \log f(\mathbf{z}, \mathbf{y})$ evaluated at the *current sample* of the others. Mean-field VB updates the *distribution* $q_j \propto \exp \mathbb{E}_{-j} \log f(\mathbf{z}, \mathbf{y})$ using the *average* over the others. With conjugate models the two have the same functional forms; VB replaces random draws by expectations.

11.2.2. A correlated Gaussian: the variance underestimate

Example 11.1 (Mean field on a bivariate Gaussian (Bishop §10.1.2))

Let the target be $f(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mu, \Lambda^{-1})$ with precision matrix Λ , and $q = q_1(z_1)q_2(z_2)$. Then

$$\log q_1^*(z_1) = \mathbb{E}_{z_2} \left[-\frac{1}{2} \Lambda_{11}(z_1 - \mu_1)^2 - \Lambda_{12}(z_1 - \mu_1)(z_2 - \mu_2) \right] + \text{const},$$

which is quadratic in z_1 , so q_1^* is Gaussian — this was *derived*, not assumed:

$$q_1^* = \mathcal{N}(m_1, \Lambda_{11}^{-1}), \quad m_1 = \mu_1 - \Lambda_{11}^{-1} \Lambda_{12}(m_2 - \mu_2),$$

and symmetrically for q_2 . The self-consistent solution is $m_1 = \mu_1$, $m_2 = \mu_2$: **the means are exact**. But the variances are $1/\Lambda_{11}$, the *conditional* variances, smaller than the true marginal variances $(\Lambda^{-1})_{11}$. For unit variances and correlation ρ , q has variance $1 - \rho^2$ against the true 1: for $\rho = 0.9$, the mean-field standard deviation is 0.44 instead of 1. Mean field “fits inside” the elongated posterior ellipse.

11.3. Example: one Gaussian with unknown mean and precision

Data $x_i \sim \mathcal{N}(\mu, \tau^{-1})$, $i = 1..n$, with τ the precision. The likelihood is

$$f(\mathbf{x} | \mu, \tau) = \left(\frac{\tau}{2\pi} \right)^{n/2} \exp \left[-\frac{\tau}{2} \sum_i (x_i - \mu)^2 \right].$$

The conjugate prior is **normal–gamma**:

$$\tau \sim \text{Gamma}(a_0, b_0) : f(\tau) = \frac{b_0^{a_0}}{\Gamma(a_0)} \tau^{a_0-1} e^{-b_0\tau}, \quad \mu | \tau \sim \mathcal{N}(\mu_0, (\lambda_0\tau)^{-1}) : f(\mu | \tau) = \left(\frac{\lambda_0\tau}{2\pi} \right)^{1/2} e^{-\frac{\lambda_0\tau}{2}(\mu - \mu_0)^2}.$$

The exact posterior is again normal–gamma — but *not* factorised (μ and τ are dependent). We approximate it with $q(\mu, \tau) = q_\mu(\mu)q_\tau(\tau)$.

Factor for μ . Keeping the terms with μ ,

$$\log q_\mu^*(\mu) = \mathbb{E}_\tau \left[-\frac{\tau}{2} \sum_i (x_i - \mu)^2 - \frac{\lambda_0\tau}{2} (\mu - \mu_0)^2 \right] + \text{const} = -\frac{\mathbb{E}[\tau]}{2} \left[\sum_i (x_i - \mu)^2 + \lambda_0(\mu - \mu_0)^2 \right] + \text{const}.$$

Completing the square, $q_\mu^* = \mathcal{N}(\mu', \lambda'^{-1})$ with

$$\mu' = \frac{\lambda_0\mu_0 + \sum_i x_i}{\lambda_0 + n} = \frac{\lambda_0\mu_0 + n\bar{x}}{\lambda_0 + n}, \quad \lambda' = (\lambda_0 + n) \mathbb{E}[\tau].$$

Factor for τ . Keeping the terms with τ (note $n/2$ from the likelihood, $1/2$ from the prior on μ , $a_0 - 1$ from the prior on τ),

$$\log q_\tau^*(\tau) = \left(a_0 - 1 + \frac{n+1}{2} \right) \log \tau - \tau \left(b_0 + \frac{1}{2} \mathbb{E}_\mu \left[\sum_i (x_i - \mu)^2 + \lambda_0(\mu - \mu_0)^2 \right] \right) + \text{const},$$

so $q_\tau^* = \text{Gamma}(a', b')$ with

$$a' = a_0 + \frac{n+1}{2}, \quad b' = b_0 + \frac{1}{2} \mathbb{E}_\mu \left[\sum_i (x_i - \mu)^2 + \lambda_0(\mu - \mu_0)^2 \right].$$

11. Variational Bayes

The required moments under $q_\mu = \mathcal{N}(\mu', \lambda'^{-1})$ are $\mathbb{E}[\mu] = \mu'$ and $\mathbb{E}[\mu^2] = \mu'^2 + \lambda'^{-1}$, giving

$$\mathbb{E}_\mu \left[\sum_i (x_i - \mu)^2 + \lambda_0 (\mu - \mu_0)^2 \right] = \sum_i (x_i - \mu')^2 + \lambda_0 (\mu' - \mu_0)^2 + \frac{n + \lambda_0}{\lambda'},$$

and $\mathbb{E}[\tau] = a'/b'$ under the Gamma.

Iteration. μ' and a' do not change; only λ' and b' are coupled. Initialise $\mathbb{E}[\tau]$, then alternate

$$\lambda' = (\lambda_0 + n) \frac{a'}{b'}, \quad b' = b_0 + \frac{1}{2} \left[\sum_i (x_i - \mu')^2 + \lambda_0 (\mu' - \mu_0)^2 \right] + \frac{1}{2} \frac{n + \lambda_0}{\lambda'}.$$

Example 11.2 (Non-informative limit)

With $a_0 = b_0 = \lambda_0 = 0$: $\mu' = \bar{x}$, $a' = (n + 1)/2$. Substituting $\lambda' = na'/b'$ into the b' equation, $b' = \frac{1}{2}ns^2 + \frac{1}{2}b'/a'$ with $s^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$, whose fixed point is $b' = \frac{ns^2 a'}{2a' - 1} = \frac{(n+1)s^2}{2}$. Then $\mathbb{E}[\tau] = a'/b' = 1/s^2$ and $\text{Var}_q[\mu] = 1/\lambda' = s^2/n$: the familiar maximum-likelihood variance estimate and the standard error of the mean. The iteration converges in a handful of steps from any starting $\mathbb{E}[\tau]$, since the map $b' \mapsto \frac{1}{2}ns^2 + b'/(2a')$ is a contraction with factor $1/(n + 1)$.

11.4. Example: mixture distributions

Return to the Gaussian mixture of Chapter 8 (known σ , K components) with latent labels $\mathbf{Z}$, weights $\mathbf{p} \sim \text{Dir}(\gamma)$ and means $\mu_k \sim \mathcal{N}(\mu_{0k}, \sigma_{0k}^2)$. The joint is

$$f(\mathbf{x}, \mathbf{Z}, \mathbf{p}, \mu) = f(\mathbf{x} | \mathbf{Z}, \mu) f(\mathbf{Z} | \mathbf{p}) f(\mathbf{p}) f(\mu).$$

Assume only that the labels decouple from the parameters,

$$q(\mathbf{Z}, \mathbf{p}, \mu) = q(\mathbf{Z}) q(\mathbf{p}, \mu),$$

and write $r_{ik} \equiv q(Z_i = k) = \mathbb{E}_q[\chi(Z_i = k)]$ (the **responsibilities**).

Labels. Since $\log f(\mathbf{Z} | \mathbf{p}) = \sum_{i,k} \chi(Z_i = k) \log p_k$ and $\log f(\mathbf{x} | \mathbf{Z}, \mu) = -\frac{1}{2\sigma^2} \sum_{i,k} \chi(Z_i = k) (x_i - \mu_k)^2 + \text{const}$,

$$\log q^*(\mathbf{Z}) = \sum_{i,k} \chi(Z_i = k) \left[\mathbb{E}[\log p_k] - \frac{1}{2\sigma^2} \mathbb{E}[(x_i - \mu_k)^2] \right] + \text{const}.$$

This factorises over points: $q^*(\mathbf{Z}) = \prod_i \prod_k r_{ik}^{\chi(Z_i=k)}$ with

$$r_{ik} = \frac{\rho_{ik}}{\sum_{k'} \rho_{ik'}}, \quad \rho_{ik} = \exp \left(\mathbb{E}[\log p_k] - \frac{1}{2\sigma^2} \mathbb{E}[(x_i - \mu_k)^2] \right).$$

Parameters. Taking the expectation over $\mathbf{Z}$ (only $\mathbb{E}[\chi(Z_i = k)] = r_{ik}$ enters, by linearity),

$$\log q^*(\mathbf{p}, \mu) = \underbrace{\sum_k \left(\gamma_k - 1 + \sum_i r_{ik} \right) \log p_k}_{\text{depends on } \mathbf{p}} - \underbrace{\frac{1}{2\sigma^2} \sum_{i,k} r_{ik} (x_i - \mu_k)^2 - \sum_k \frac{(\mu_k - \mu_{0k})^2}{2\sigma_{0k}^2}}_{\text{depends on } \mu} + \text{const}.$$

The right-hand side is a sum of a term in $\mathbf{p}$ and a term in μ , so $q^*(\mathbf{p}, \mu) = q^*(\mathbf{p})q^*(\mu)$ **factorises automatically** — a result of the calculation, not an assumption. Reading off the forms, with $N_k = \sum_i r_{ik}$ the expected number of points in cluster k :

$$q^*(\mathbf{p}) = \text{Dir}(\gamma'), \quad \gamma'_k = \gamma_k + N_k; \quad q^*(\mu_k) = \mathcal{N}(\mu'_k, \sigma_k'^2), \quad \mu'_k = \frac{\mu_{0k}/\sigma_{0k}^2 + \sum_i r_{ik} x_i / \sigma^2}{1/\sigma_{0k}^2 + N_k / \sigma^2}, \quad \sigma_k'^2 = \left(\frac{1}{\sigma_{0k}^2} + \frac{N_k}{\sigma^2} \right)^{-1}.$$

These are exactly the Gibbs conditionals of Chapter 8 with **hard counts replaced by soft counts** $N_k = \sum_i r_{ik}$.

Moments needed for r_{ik} .

$$\mathbb{E}[(x_i - \mu_k)^2] = (x_i - \mu'_k)^2 + \sigma_k'^2, \quad \mathbb{E}[\log p_k] = \psi(\gamma'_k) - \psi\left(\sum_j \gamma'_j\right),$$

where $\psi = \Gamma'/\Gamma$ is the digamma function (the second identity is a standard property of the Dirichlet). Hence

$$\rho_{ik} = \exp\left[\psi(\gamma'_k) - \psi\left(\sum_j \gamma'_j\right) - \frac{(x_i - \mu'_k)^2}{2\sigma_k'^2} - \frac{\sigma_k'^2}{2\sigma^2}\right].$$

Definition 11.1 (VB for the Gaussian mixture). Iterate until $\mathcal{L}$ stops increasing:

1. **(E-like step)** with $\mu'_k, \sigma'_k, \gamma'_k$ fixed, compute ρ_{ik} and the responsibilities $r_{ik} = \rho_{ik} / \sum_{k'} \rho_{ik'}$;
2. **(M-like step)** with r_{ik} fixed, compute $N_k = \sum_i r_{ik}$ and $\sum_i r_{ik} x_i$, then update $\gamma'_k, \mu'_k, \sigma'_k$.

This has the structure of the EM algorithm and of soft K-means: the responsibilities are a softmax of “log-prior weight minus squared distance”, corrected by the posterior uncertainty of the parameters ($\sigma_k'^2$ and the digamma terms). A practical bonus: with a small Dirichlet concentration $\gamma_k < 1$, the VB solution drives the weights of superfluous components to zero, effectively *selecting* the number of clusters — something maximum-likelihood EM cannot do.

11.5. The variational autoencoder

Mean field is the most common choice, but VB only requires that q belong to a tractable family. Let the family be **parametrised** by θ : $q = q(\mathbf{z} \mid \theta)$. Writing the joint as likelihood $\times$ prior,

$$\mathcal{L}(q) = \int d\mathbf{z} q(\mathbf{z}) \log \frac{f(\mathbf{y} \mid \mathbf{z}) f_{\text{prior}}(\mathbf{z})}{q(\mathbf{z})} = -D_{KL}(q(\mathbf{z}) \parallel f_{\text{prior}}(\mathbf{z})) + \mathbb{E}_{\mathbf{z} \sim q}[\log f(\mathbf{y} \mid \mathbf{z})].$$

Two contributions compete:

- the **prior term** $-D_{KL}(q \parallel f_{\text{prior}})$ keeps q from departing too far from the prior;
- the **data term** $\mathbb{E}_q \log f(\mathbf{y} \mid \mathbf{z})$ pushes q to concentrate on values of $\mathbf{z}$ that explain the data well.

The best approximation is $q^* = \arg \max_{\theta} \mathcal{L}(q(\cdot \mid \theta))$, and $\log f(\mathbf{y}) \geq \mathcal{L}(q^*)$.

11.5.1. Learning the likelihood too

Now suppose the data and the prior are given, but the likelihood is only known to belong to a family $f(\mathbf{y} \mid \mathbf{z}, \phi)$. A natural way to choose ϕ is to maximise the evidence $\log f(\mathbf{y} \mid \phi)$ (*empirical Bayes* / maximum marginal likelihood). The evidence is intractable, but the ELBO bounds it, so we maximise the bound jointly:

$$\log f(\mathbf{y}) \geq \max_{\theta, \phi} \left[-D_{KL}(q(\mathbf{z} \mid \theta) \parallel f_{\text{prior}}(\mathbf{z})) + \mathbb{E}_{\mathbf{z} \sim q} \log f(\mathbf{y} \mid \mathbf{z}, \phi) \right].$$

Given only the prior and the data, this finds a self-consistent **generative model** of the data within a class of models, together with an approximate posterior.

11.5.2. Encoder and decoder

Take one latent vector $\mathbf{z}_i \in \mathbb{R}^d$ per data point $\mathbf{y}_i \in \mathbb{R}^D$, with $d < D$: a low-dimensional representation of high-dimensional data. A simple and typical choice:

- **prior:** $f_{\text{prior}}(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I)$;
- **decoder (likelihood):** $\mathbf{y} \mid \mathbf{z} \sim \mathcal{N}(\mathbf{d}_\phi(\mathbf{z}), cI)$ — the data point is Gaussian around a nonlinear function of its latent code;
- **encoder (variational posterior):** $\mathbf{z} \mid \mathbf{y} \sim q = \mathcal{N}(\mathbf{e}_{1,\theta}(\mathbf{y}), \text{diag } \mathbf{e}_{2,\theta}(\mathbf{y}))$ — mean and variances are functions of the data. Using one shared function of $\mathbf{y}$ instead of separate parameters for each $q(\mathbf{z}_i)$ is called **amortised inference**.

(Unless $\mathbf{d}_\phi$ is linear, the true posterior is not Gaussian; the encoder gives the best Gaussian approximation.) The KL between Gaussians is available in closed form,

$$D_{KL}(\mathcal{N}(\mathbf{m}, \Sigma) \parallel \mathcal{N}(\mathbf{0}, I)) = \frac{1}{2} \left(\mathbf{m}^\top \mathbf{m} + \text{Tr } \Sigma - d - \log \det \Sigma \right),$$

and the data term is a reconstruction error, so (per data point, up to constants)

$$-\mathcal{L} = \underbrace{\frac{1}{2} \left(\|\mathbf{e}_1(\mathbf{y})\|^2 + \sum_j e_{2j}(\mathbf{y}) - d - \sum_j \log e_{2j}(\mathbf{y}) \right)}_{\text{regulariser}} + \underbrace{\mathbb{E}_{\mathbf{z} \sim q} \frac{\|\mathbf{y} - \mathbf{d}_\phi(\mathbf{z})\|^2}{2c}}_{\text{reconstruction}}.$$

q acts as a **probabilistic encoder** (each $\mathbf{y}$ maps to a *distribution* over codes centred at $\mathbf{e}_1(\mathbf{y})$), and $f(\mathbf{y} \mid \mathbf{z})$ as a **probabilistic decoder**. Implementing $\mathbf{e}_1, \mathbf{e}_2, \mathbf{d}$ as deep neural networks, θ and ϕ are their weights and biases, and the loss is minimised by stochastic gradient descent with back-propagation. Without the regulariser this would be an ordinary autoencoder; the KL term forces the codes to fill the prior $\mathcal{N}(\mathbf{0}, I)$, so that *sampling* $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$ and decoding produces new, realistic data.

11.5.3. The reparametrisation trick

The reconstruction term is an expectation over $\mathbf{z} \sim q(\cdot \mid \theta)$, estimated with samples $\mathbf{z}_k(\theta)$:

$$\mathbb{E}_{\mathbf{z} \sim q} \frac{\|\mathbf{y} - \mathbf{d}_\phi(\mathbf{z})\|^2}{2c} \approx \frac{1}{M} \sum_{k=1}^M \frac{\|\mathbf{y} - \mathbf{d}_\phi(\mathbf{z}_k)\|^2}{2c}.$$

But a random draw is not a differentiable function of θ , so back-propagation cannot pass through it. The fix is to move the randomness *outside* the parameters:

$$\mathbf{z} = \mathbf{e}_{1,\theta}(\mathbf{y}) + \sqrt{\mathbf{e}_{2,\theta}(\mathbf{y})} \odot \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, I).$$

Sample $\boldsymbol{\eta}_k$ once, independently of θ ; then

$$\mathbb{E}_{\mathbf{z} \sim q} \frac{\|\mathbf{y} - \mathbf{d}_\phi(\mathbf{z})\|^2}{2c} \approx \frac{1}{M} \sum_k \frac{\|\mathbf{y} - \mathbf{d}_\phi(\mathbf{e}_1(\mathbf{y}) + \sqrt{\mathbf{e}_2(\mathbf{y})} \odot \boldsymbol{\eta}_k)\|^2}{2c},$$

a deterministic, differentiable function of (θ, ϕ) to which standard back-propagation applies. In practice even $M = 1$ sample per data point per step works, because the noise averages out over mini-batches.

11.6. Exercises

Exercise 11.1

Derive the decomposition $\mathcal{L} = \mathbb{E}_q[\log f(\mathbf{y} | \mathbf{z})] - D(q \| f_{\text{prior}})$.

Solution

$$\mathcal{L} = \mathbb{E}_q[\log f(\mathbf{y}, \mathbf{z}) - \log q(\mathbf{z})] = \mathbb{E}_q[\log f(\mathbf{y} | \mathbf{z}) + \log f_{\text{prior}}(\mathbf{z}) - \log q(\mathbf{z})] = \mathbb{E}_q[\log f(\mathbf{y} | \mathbf{z})] - D(q \| f_{\text{prior}}).$$

Exercise 11.2

Show that mean-field VB *underestimates* posterior variances for a correlated Gaussian target with covariance Σ .

Solution

From the bivariate example, the optimal factor variance is $1/\Lambda_{jj} = 1/(\Sigma^{-1})_{jj}$. By the Schur complement, $1/(\Sigma^{-1})_{11} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \leq \Sigma_{11}$, with equality iff the coordinates are uncorrelated. Mean field reports the *conditional* variance, which is always smaller than the marginal one.

Exercise 11.3

Compute $D_{KL}(\mathcal{N}(m, s^2) \| \mathcal{N}(0, 1))$ in one dimension and check the VAE formula.

Solution

$$D = \mathbb{E}_q[\log q - \log p] = \mathbb{E}_q\left[-\frac{1}{2}\log(2\pi s^2) - \frac{(z-m)^2}{2s^2} + \frac{1}{2}\log 2\pi + \frac{z^2}{2}\right] = -\frac{1}{2}\log s^2 - \frac{1}{2} + \frac{1}{2}(m^2 + s^2),$$

i.e. $\frac{1}{2}(m^2 + s^2 - 1 - \log s^2)$, matching the formula with $d = 1$. It vanishes iff $m = 0, s = 1$.

Exercise 11.4

In the one-Gaussian example with $a_0 = b_0 = \lambda_0 = 0$, $n = 10$ and $s^2 = 4$, iterate the CAVI equations starting from $\mathbb{E}[\tau] = 1$ and verify the fixed point.

Solution

$a' = 5.5$, fixed point $b' = (n+1)s^2/2 = 22$. Start: $\lambda' = 10$, $b' = \frac{1}{2}(10 \cdot 4) + \frac{1}{2} \cdot 10/10 = 20.5$; $\mathbb{E}\tau = 0.268$, $\lambda' = 2.68$, $b' = 20 + 1.86 = 21.86$; $\mathbb{E}\tau = 0.2516$, $\lambda' = 2.516$, $b' = 20 + 1.987 = 21.987$; ... converging to $b' = 22$, $\mathbb{E}[\tau] = 0.25 = 1/s^2$, $\text{Var}_q \mu = 0.4 = s^2/n$.

11.7. Callouts

i Note

VB is biased by construction: the true posterior is generally not in $\mathcal{Q}$. Use it when speed matters and the posterior *mean* is what you need; use MCMC/HMC when calibrated uncertainty matters.

 Tip

Monitor $\mathcal{L}$ during CAVI: it must increase at every step. A decrease always means a bug in the update equations. Run from several initialisations and keep the solution with the largest $\mathcal{L}$.

 Warning

Mean-field VB underestimates posterior variance and can lock onto a single mode. Richer families (full-rank Gaussians, normalising flows) reduce, but do not remove, this bias.

11.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 12. Bishop, §10.1.1–10.1.3 (variational principle, factorised distributions, bivariate Gaussian, univariate Gaussian), §10.2 (variational mixture of Gaussians). Blei, Kucukelbir & McAuliffe, *Variational inference: a review for statisticians*, J. Am. Stat. Assoc. **112**, 859 (2017). Kingma & Welling, *Auto-encoding variational Bayes*, ICLR (2014).

12. Hamiltonian Monte Carlo

We now have several ways to obtain posterior estimates. Gibbs sampling and variational Bayes are very efficient, but both need strong analytical control: closed-form conditionals, typically requiring simple likelihoods and conjugate priors. When these are unavailable we can always fall back on Metropolis–Hastings, but in high dimension its random-walk proposals must be tiny to be accepted (Chapter 7), and the chain diffuses slowly. **Hamiltonian Monte Carlo** (HMC) overcomes this by borrowing classical mechanics: it uses the gradient of the log-posterior to build *distant* proposals that are nevertheless accepted with probability close to 1. Like Metropolis it needs the unnormalised posterior (likelihood $\times$ prior), and in addition **its derivatives with respect to the parameters**.

12.1. Hamiltonian dynamics

Consider coordinates $\mathbf{x} = (x_1, \dots, x_n)$ and conjugate momenta $\mathbf{p} = (p_1, \dots, p_n)$, with Hamiltonian

$$H(\mathbf{x}, \mathbf{p}) = \sum_i \frac{p_i^2}{2m_i} + U(\mathbf{x}),$$

the sum of kinetic and potential energy. Hamilton's equations

$$\dot{x}_i = \frac{\partial H}{\partial p_i} = \frac{p_i}{m_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial x_i} = -\frac{\partial U}{\partial x_i}$$

define a **flow** $\phi_t : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$, $(\mathbf{x}_0, \mathbf{p}_0) \mapsto (\mathbf{x}_t, \mathbf{p}_t)$. Three properties of this flow make it useful for Monte Carlo.

12.1.1. Energy conservation

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial x_i} \dot{x}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) = \sum_i \left(\frac{\partial H}{\partial x_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial x_i} \right) = 0.$$

12.1.2. Volume preservation (Liouville's theorem)

Proposition 12.1 (Volume preservation). *For every region $A \subset \mathbb{R}^{2n}$, $\text{Vol}(\phi_t(A)) = \text{Vol}(A)$; equivalently, the Jacobian determinant of the flow is $|J_{\phi_t}| = 1$ everywhere.*

Proof. Changing variables $(\mathbf{x}', \mathbf{p}') = \phi_t(\mathbf{x}, \mathbf{p})$, $\int_{\phi_t(A)} d\mathbf{x}' d\mathbf{p}' = \int_A |J_{\phi_t}| d\mathbf{x} d\mathbf{p}$, so volume preservation for all A is equivalent to $|J_{\phi_t}| = 1$. For an infinitesimal time step,

$$\phi_{\delta t}(\mathbf{x}, \mathbf{p}) = \begin{pmatrix} \mathbf{x} \\ \mathbf{p} \end{pmatrix} + \delta t \begin{pmatrix} \partial_{\mathbf{p}} H \\ -\partial_{\mathbf{x}} H \end{pmatrix} + O(\delta t^2), \quad J_{\phi_{\delta t}} = \begin{pmatrix} 1 + \delta t \partial_{\mathbf{x}} \partial_{\mathbf{p}} H & \delta t \partial_{\mathbf{p}}^2 H \\ -\delta t \partial_{\mathbf{x}}^2 H & 1 - \delta t \partial_{\mathbf{p}} \partial_{\mathbf{x}} H \end{pmatrix} + O(\delta t^2).$$

The determinant is $1 + \delta t (\partial_{\mathbf{x}} \partial_{\mathbf{p}} H - \partial_{\mathbf{p}} \partial_{\mathbf{x}} H) + O(\delta t^2) = 1 + O(\delta t^2)$: the trace of the first-order term vanishes because mixed partial derivatives commute. For finite $t = n \delta t$, ϕ_t is the n -fold composition, so $|J_{\phi_t}| = (1 + O(\delta t^2))^n = 1 + O(n \delta t^2) = 1 + O(t^2/n) \rightarrow 1$ as $n \rightarrow \infty$. $\square$

12.1.3. Time reversibility

The flow is invertible, and its inverse is obtained by running time backwards: $\phi_t^{-1} = \phi_{-t}$, i.e. integrating $\dot{x}_i = -p_i/m_i$, $\dot{p}_i = +\partial_{x_i}U$. These are the original equations with $\mathbf{p} \rightarrow -\mathbf{p}$. Introducing the **momentum flip** $N_p(\mathbf{x}, \mathbf{p}) = (\mathbf{x}, -\mathbf{p})$,

$$\phi_t^{-1} = \phi_{-t} = N_p \circ \phi_t \circ N_p.$$

To run a trajectory backwards: flip the momentum, run forwards, flip again. For every forward trajectory $(\mathbf{x}, \mathbf{p}) \rightarrow (\mathbf{x}', \mathbf{p}')$ there is a backward one $(\mathbf{x}', -\mathbf{p}') \rightarrow (\mathbf{x}, -\mathbf{p})$.

12.2. Numerical integration of Hamilton's equations

In practice the flow must be approximated numerically, and *how* matters: we need an integrator that retains volume preservation and reversibility.

12.2.1. Euler's method

$$p_i(t + \epsilon) = p_i(t) - \epsilon \partial_{x_i} U(\mathbf{x}(t)), \quad x_i(t + \epsilon) = x_i(t) + \epsilon p_i(t)/m_i.$$

Its Jacobian determinant is $\det \begin{pmatrix} 1 & \epsilon/m \\ -\epsilon \partial_x^2 U & 1 \end{pmatrix} = 1 + \epsilon^2 \partial_x^2 U/m \neq 1$: **Euler does not preserve volume.**

For the harmonic oscillator ($m = 1$, $U = x^2/2$) each step multiplies the phase-space area by $1 + \epsilon^2$ and the radius by $\sqrt{1 + \epsilon^2}$: the exact circular orbit becomes an outward spiral. With $\epsilon = 0.3$, twenty steps already inflate the radius by $(1.09)^{10} \approx 2.4$. Smaller steps ($\epsilon = 0.01$) delay the divergence, but over many steps (thousands) the trajectory still spirals away.

12.2.2. Modified (symplectic) Euler

Update the momentum first, then use the *new* momentum to update the position:

$$p_i(t + \epsilon) = p_i(t) - \epsilon \partial_{x_i} U(\mathbf{x}(t)), \quad x_i(t + \epsilon) = x_i(t) + \epsilon p_i(t + \epsilon)/m_i.$$

This is the composition of a map that changes only momenta and a map that changes only positions, with Jacobians

$$J^{(p)} = \begin{pmatrix} 1 & 0 \\ -\epsilon \partial_{\mathbf{x}}^2 U & 1 \end{pmatrix}, \quad J^{(x)} = \begin{pmatrix} 1 & \epsilon \text{diag}(m^{-1}) \\ 0 & 1 \end{pmatrix},$$

both triangular with unit diagonal, hence determinant 1. **The modified Euler method preserves volume exactly**, and trajectories no longer diverge. But it is not reversible: flipping the momentum, running it, and flipping again does not retrace the trajectory, because the order “momentum then position” is not symmetric in time.

12.2.3. Leapfrog

The fix is to make the scheme symmetric: half a step in momentum, a full step in position, another half step in momentum.

|

Definition 12.1 (Leapfrog step).

$$\begin{aligned} p_i(t + \frac{\epsilon}{2}) &= p_i(t) - \frac{\epsilon}{2} \partial_{x_i} U(\mathbf{x}(t)), \\ x_i(t + \epsilon) &= x_i(t) + \epsilon p_i(t + \frac{\epsilon}{2}) / m_i, \\ p_i(t + \epsilon) &= p_i(t + \frac{\epsilon}{2}) - \frac{\epsilon}{2} \partial_{x_i} U(\mathbf{x}(t + \epsilon)). \end{aligned}$$

Proposition 12.2 (Properties of leapfrog). *The leapfrog map $\phi_\epsilon^{\text{lf}}$ is*

1. **volume-preserving**: it is a composition of three shear maps (momentum, position, momentum), each with unit Jacobian determinant;
2. **time-reversible**: $(\phi_\epsilon^{\text{lf}})^{-1} = \phi_{-\epsilon}^{\text{lf}} = N_p \circ \phi_\epsilon^{\text{lf}} \circ N_p$;
3. **second-order accurate**: the energy error is $O(\epsilon^2)$ and, for stable step sizes, **does not drift** over long trajectories.

Proof of reversibility. Apply $\phi_{-\epsilon}^{\text{lf}}$ to the endpoint $(\mathbf{x}(t + \epsilon), \mathbf{p}(t + \epsilon))$:

$$\begin{aligned} p'_i &= p_i(t + \epsilon) + \frac{\epsilon}{2} \partial_{x_i} U(\mathbf{x}(t + \epsilon)) = p_i(t + \frac{\epsilon}{2}), \\ x'_i &= x_i(t + \epsilon) - \epsilon p_i(t + \frac{\epsilon}{2}) / m_i = x_i(t), \\ p''_i &= p_i(t + \frac{\epsilon}{2}) + \frac{\epsilon}{2} \partial_{x_i} U(\mathbf{x}(t)) = p_i(t), \end{aligned}$$

recovering the starting point exactly (each line undoes the corresponding line of the forward step, in reverse order). And running $\phi_{-\epsilon}$ is the same as flipping $\mathbf{p}$, running $\phi_{+\epsilon}$, and flipping again. $\square$

Example 12.1 (Stability of leapfrog for a harmonic oscillator)

Take $m = 1$, $U(x) = x^2/(2\sigma^2)$ (a Gaussian target of width σ). One leapfrog step is the linear map

$$\begin{pmatrix} x(t + \epsilon) \\ p(t + \epsilon) \end{pmatrix} = \begin{pmatrix} 1 - \frac{\epsilon^2}{2\sigma^2} & \epsilon \\ -\frac{\epsilon}{\sigma^2} + \frac{\epsilon^3}{4\sigma^4} & 1 - \frac{\epsilon^2}{2\sigma^2} \end{pmatrix} \begin{pmatrix} x(t) \\ p(t) \end{pmatrix},$$

with determinant $(1 - \frac{\epsilon^2}{2\sigma^2})^2 + \frac{\epsilon^2}{\sigma^2} - \frac{\epsilon^4}{4\sigma^4} = 1$ (volume preservation). With $a = 1 - \epsilon^2/(2\sigma^2)$ the eigenvalues are $\lambda_{\pm} = a \pm \sqrt{a^2 - 1}$, with $\lambda_+ \lambda_- = 1$.

- If $\epsilon < 2\sigma$, then $|a| < 1$, the eigenvalues are complex conjugates with $|\lambda_{\pm}| = 1$, and the trajectory is a stable (slightly distorted) ellipse.
- If $\epsilon > 2\sigma$, the eigenvalues are real with one of modulus > 1 : the trajectory diverges exponentially.

Rule of thumb: the step size must be smaller than the narrowest width of the target, $\epsilon \lesssim \sigma_{\min}$.

12.3. The HMC algorithm

12.3.1. The augmented distribution

To sample $\pi(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^n$, add n auxiliary momenta, independent of $\mathbf{x}$ and of each other, $p_i \sim \mathcal{N}(0, m_i)$:

$$\Pi(\mathbf{x}, \mathbf{p}) = \pi(\mathbf{x}) \cdot \prod_i (2\pi m_i)^{-1/2} e^{-p_i^2/2m_i}.$$

If we can sample Π , the $\mathbf{x}$ -components of the samples are distributed according to π (just discard $\mathbf{p}$). Define the **potential** $U(\mathbf{x}) = -\log \pi(\mathbf{x})$; then

$$\Pi(\mathbf{x}, \mathbf{p}) = \frac{1}{Z} e^{-H(\mathbf{x}, \mathbf{p})}, \quad H(\mathbf{x}, \mathbf{p}) = U(\mathbf{x}) + \sum_i \frac{p_i^2}{2m_i},$$

a Boltzmann distribution at unit temperature, with the momentum variances playing the role of masses. Below we take $m_i = 1$, so $H = U(\mathbf{x}) + \frac{1}{2}|\mathbf{p}|^2$.

For Bayesian inference, $\pi(\mathbf{x}) = f(\mathbf{y} | \mathbf{x}) f_{\text{prior}}(\mathbf{x}) / f(\mathbf{y})$; since the evidence does not depend on $\mathbf{x}$ it is absorbed into Z , and

$$U(\mathbf{x}) = -\log f(\mathbf{y} | \mathbf{x}) - \log f_{\text{prior}}(\mathbf{x}).$$

12.3.2. The proposal

Each HMC iteration is a composition of two Metropolis–Hastings moves on Π :

1. **Momentum refresh:** draw $\mathbf{p}_r \sim \mathcal{N}(0, I)$, independently of the current $\mathbf{p}$.
2. **Hamiltonian move:** evolve $(\mathbf{x}, \mathbf{p}_r)$ under the Hamiltonian flow for a time t and flip the momentum: $(\mathbf{x}', \mathbf{p}') = N_p \circ \phi_t(\mathbf{x}, \mathbf{p}_r)$.

Proposition 12.3. *With the exact Hamiltonian flow, both moves are accepted with probability 1.*

Proof. *Move 1* is a Gibbs update: it samples $\mathbf{p}$ from its exact conditional $\Pi(\mathbf{p} | \mathbf{x}) = \mathcal{N}(0, I)$. Explicitly, the proposal is $T(\mathbf{x}', \mathbf{p}_r | \mathbf{x}, \mathbf{p}) = \delta(\mathbf{x}' - \mathbf{x})(2\pi)^{-n/2} e^{-|\mathbf{p}_r|^2/2}$ and the Hastings ratio is

$$\frac{\Pi(\mathbf{x}, \mathbf{p}_r) T(\mathbf{x}, \mathbf{p} | \mathbf{x}, \mathbf{p}_r)}{\Pi(\mathbf{x}, \mathbf{p}) T(\mathbf{x}, \mathbf{p}_r | \mathbf{x}, \mathbf{p})} = \frac{\pi(\mathbf{x}) e^{-|\mathbf{p}_r|^2/2} \cdot e^{-|\mathbf{p}|^2/2}}{\pi(\mathbf{x}) e^{-|\mathbf{p}|^2/2} \cdot e^{-|\mathbf{p}_r|^2/2}} = 1.$$

Move 2 is deterministic: $T(\mathbf{x}', \mathbf{p}' | \mathbf{x}, \mathbf{p}_r) = \delta((\mathbf{x}', -\mathbf{p}') - \phi_t(\mathbf{x}, \mathbf{p}_r))$. Energy conservation gives $\Pi(\mathbf{x}', \mathbf{p}') = \Pi(\mathbf{x}, \mathbf{p}_r)$ (the flip does not change $|\mathbf{p}|^2$), so the acceptance probability reduces to $\min\{1, T(\mathbf{x}, \mathbf{p}_r | \mathbf{x}', \mathbf{p}') / T(\mathbf{x}', \mathbf{p}' | \mathbf{x}, \mathbf{p}_r)\}$. For the reverse move,

$$T(\mathbf{x}, \mathbf{p}_r | \mathbf{x}', \mathbf{p}') = \delta((\mathbf{x}, -\mathbf{p}_r) - \phi_t(\mathbf{x}', \mathbf{p}')) = \delta((\mathbf{x}', \mathbf{p}') - \phi_t^{-1}(\mathbf{x}, -\mathbf{p}_r)) |J_{\phi_t}|^{-1},$$

and $|J_{\phi_t}| = 1$ by volume preservation. By reversibility $\phi_t^{-1} \circ N_p = N_p \circ \phi_t$, so $\phi_t^{-1}(\mathbf{x}, -\mathbf{p}_r) = N_p \phi_t(\mathbf{x}, \mathbf{p}_r)$ and the delta function equals $\delta((\mathbf{x}', -\mathbf{p}') - \phi_t(\mathbf{x}, \mathbf{p}_r)) = T(\mathbf{x}', \mathbf{p}' | \mathbf{x}, \mathbf{p}_r)$. The ratio is 1. $\square$

The proof shows exactly which properties are needed: **volume preservation** (no Jacobian factor), **reversibility with the momentum flip** (the reverse proposal exists and has the same density), and **energy conservation** (Π is unchanged). The momentum flip is required for the proof; in practice it can be skipped, because the momentum is immediately resampled at the next iteration.

12.3.3. With a numerical integrator

The exact flow is replaced by $L = t/\epsilon$ leapfrog steps. Leapfrog is exactly volume-preserving and reversible, so the proposal densities still cancel; but energy is conserved only approximately, $\Pi(\mathbf{x}', \mathbf{p}') \approx \Pi(\mathbf{x}, \mathbf{p}_r)$. The Metropolis correction takes care of this:

Definition 12.2 (Hamiltonian Monte Carlo). From the current state $\mathbf{x}$:

1. draw $\mathbf{p}_r \sim \mathcal{N}(0, I)$;
2. apply L leapfrog steps of size ϵ to $(\mathbf{x}, \mathbf{p}_r)$, obtaining $(\mathbf{x}', \mathbf{p}')$ (and flip $\mathbf{p}'$);
3. accept $\mathbf{x}'$ with probability

$$\min\left\{1, \frac{\Pi(\mathbf{x}', \mathbf{p}')}{\Pi(\mathbf{x}, \mathbf{p}_r)}\right\} = \min\{1, e^{H(\mathbf{x}, \mathbf{p}_r) - H(\mathbf{x}', \mathbf{p}')}\};$$

otherwise keep $\mathbf{x}$.

The acceptance probability depends only on the integration error, not on the distance travelled.

With a stable step size the energy error stays small along arbitrarily long trajectories, so HMC can make large moves with high acceptance. The price is one gradient evaluation $\nabla U(\mathbf{x}) = -\nabla \log f(\mathbf{y} | \mathbf{x}) - \nabla \log f_{\text{prior}}(\mathbf{x})$ per leapfrog step.

Remark (HMC as gradient descent with momentum). The update $\mathbf{p} \leftarrow \mathbf{p} - \epsilon \nabla U$, $\mathbf{x} \leftarrow \mathbf{x} + \epsilon \mathbf{p}$ is “heavy-ball” gradient descent on $U = -\log \pi$: the direction of motion is not just the current gradient but has inertia. HMC balances **exploration** (the random momentum draw, which sets the direction and energy of each trajectory) and **exploitation** (the deterministic flow, which follows the geometry of the posterior). Unlike gradient descent, energy conservation prevents the trajectory from simply sliding to the mode: it converts potential into kinetic energy and back, travelling along the level sets of the posterior.

12.4. Advantages of HMC

The key advantage of HMC is that it **replaces the random walk of Metropolis by directed motion**. A random walk with step ϵ moves a distance $\sim \epsilon\sqrt{k}$ in k steps; a Hamiltonian trajectory moves a distance $\sim \epsilon k$.

Example 12.2 (A strongly correlated 2-D Gaussian)

Sample $\mathcal{N}(\mathbf{0}, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 1 & 0.98 \\ 0.98 & 1 \end{pmatrix}, \quad \text{eigenvalues } 1.98 \text{ and } 0.02.$$

The standard deviations along the principal axes are $\sigma_1 = \sqrt{1.98} \approx 1.41$ and $\sigma_2 = \sqrt{0.02} \approx 0.14$. Both samplers must use steps of size $\approx \sigma_2$ to stay stable/accepted: HMC with $\epsilon = \sigma_2$ and $L = 20$ leapfrog steps per iteration; Metropolis with Gaussian jumps of size σ_2 .

To reach an effectively independent point we must cross the long axis, a distance $\approx \sigma_1 \approx 10\sigma_2$. HMC’s momentum along the long direction is stable, so about **10 leapfrog steps** suffice. Metropolis cannot move consistently in one direction: after k steps it has moved $\sim \sigma_2\sqrt{k}$, so it needs ~ 100 **steps**.

With the same budget — 20 HMC iterations (400 leapfrog steps) versus 400 Metropolis steps — the estimated covariances are (Allegra, Fig. 13.2)

$$\hat{\Sigma}_{\text{HMC}} = \begin{pmatrix} 0.96 & 0.90 \\ 0.90 & 0.88 \end{pmatrix}, \quad \hat{\Sigma}_{\text{Met}} = \begin{pmatrix} 0.26 & 0.26 \\ 0.26 & 0.30 \end{pmatrix}.$$

HMC is already close to the truth; Metropolis has explored only a small part of the long axis and badly underestimates the variances.

Example 12.3 (A 100-dimensional Gaussian)

Take $\Sigma = \text{diag}(0.01^2, 0.02^2, \dots, 1.00^2)$: 100 independent coordinates with widths from 0.01 to 1. Both step sizes must be of the order of the smallest width: HMC uses $\epsilon \sim \mathcal{U}[0.0104, 0.0156]$ with $L = 150$ leapfrog steps and 1000 iterations; Metropolis uses jumps $\sim \mathcal{U}[0.0176, 0.0264]$ and 150,000 steps — the same number of gradient/likelihood evaluations. (Randomising ϵ avoids accidental periodicities in which a trajectory returns exactly to its start.) HMC estimates the means and standard deviations of *all* coordinates well, including the widest ones; Metropolis gets the narrow coordinates right but grossly underestimates the spread of the wide ones, which it has barely explored.

Proposition 12.4 (Dimension scaling). *For targets with d roughly independent components, keeping the acceptance rate fixed requires a step size $\epsilon \propto d^{-1/2}$ for random-walk Metropolis but only $\epsilon \propto d^{-1/4}$ for HMC. Reaching an effectively independent point (moving a distance of order the target's width) therefore takes $O(d)$ Metropolis steps but only $O(d^{1/4})$ leapfrog steps. If each density or gradient evaluation costs $O(d)$, the total cost per independent sample is $O(d^2)$ for Metropolis versus $O(d^{5/4})$ for HMC (Neal 2011, §5.4.4). The optimal HMC acceptance rate in this limit is about 0.65.*

12.5. Practical considerations

- **Step size.** Too large: leapfrog becomes unstable and acceptance collapses (“divergent transitions”). Too small: wasted gradient evaluations. Modern implementations tune ϵ during warm-up to reach a target acceptance (≈ 0.65 – 0.8).
- **Trajectory length.** Too short: HMC degenerates into a random walk. Too long: trajectories loop back (for a Gaussian, after half a period). The **No-U-Turn Sampler** (NUTS, Hoffman & Gelman 2014) chooses L adaptively, stopping when the trajectory starts to turn back; it underlies Stan and PyMC.
- **Mass matrix.** Using $\mathbf{p} \sim \mathcal{N}(0, M)$ with $M \approx \Sigma_\pi^{-1}$ (estimated during warm-up) makes all directions equally “stiff”: it is equivalent to rescaling the target to unit covariance, which removes the $\sigma_{\max}/\sigma_{\min}$ penalty seen in the examples.
- **Limitations.** HMC requires a differentiable log-density on a continuous space: discrete parameters need Gibbs steps (or marginalisation). Multimodality remains hard. Each step costs a gradient, which can be expensive (automatic differentiation helps).

12.6. Exercises

Exercise 12.1

Show that the modified Euler map $p \leftarrow p - \epsilon x$, $x \leftarrow x + \epsilon p$ for the harmonic oscillator has unit Jacobian, and find the condition on ϵ for stability.

Solution

The map is $\begin{pmatrix} x' \\ p' \end{pmatrix} = \begin{pmatrix} 1 - \epsilon^2 & \epsilon \\ -\epsilon & 1 \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}$, determinant $1 - \epsilon^2 + \epsilon^2 = 1$. Trace $2 - \epsilon^2$: eigenvalues are complex with unit modulus iff $|2 - \epsilon^2| < 2$, i.e. $\epsilon < 2$ — the same stability limit as leapfrog with $\sigma = 1$.

Exercise 12.2

For a Gaussian target $\mathcal{N}(\mathbf{0}, \Sigma)$ show that with mass matrix $M = \Sigma^{-1}$ the Hamiltonian dynamics reduce to independent oscillators *all with the same frequency*.

Solution

$H = \frac{1}{2}\mathbf{x}^\top \Sigma^{-1}\mathbf{x} + \frac{1}{2}\mathbf{p}^\top M^{-1}\mathbf{p} = \frac{1}{2}\mathbf{x}^\top \Sigma^{-1}\mathbf{x} + \frac{1}{2}\mathbf{p}^\top \Sigma \mathbf{p}$. Then $\dot{\mathbf{x}} = \Sigma \mathbf{p}$, $\dot{\mathbf{p}} = -\Sigma^{-1}\mathbf{x}$, so $\ddot{\mathbf{x}} = -\mathbf{x}$: every direction oscillates with unit angular frequency. With $M = I$ instead, the mode along eigenvector k has frequency $1/\sigma_k$, and a single step size must resolve the fastest mode while the trajectory must be long enough for the slowest: the ratio $\sigma_{\max}/\sigma_{\min}$ is the cost of a poor mass matrix.

Exercise 12.3

Write the HMC potential U and its gradient for the posterior of μ in the model $x_i \sim \mathcal{N}(\mu, 1)$, $\mu \sim \mathcal{N}(0, \tau^2)$.

Solution

$U(\mu) = \frac{1}{2} \sum_i (x_i - \mu)^2 + \mu^2/(2\tau^2) + \text{const}$, $U'(\mu) = -\sum_i (x_i - \mu) + \mu/\tau^2 = (n + \tau^{-2})\mu - n\bar{x}$. The target is Gaussian with $\sigma^2 = (n + \tau^{-2})^{-1}$, so stable leapfrog needs $\epsilon < 2\sigma$.

Exercise 12.4

Explain why the momentum flip is needed in the proof of detailed balance but can be omitted in practice.

Solution

Without the flip, the reverse of the proposal $(\mathbf{x}, \mathbf{p}_r) \rightarrow \phi_t(\mathbf{x}, \mathbf{p}_r)$ would require running the flow *backwards*, which is not the same proposal mechanism, so $T(\text{reverse})$ would vanish and detailed balance would fail. With the flip, the proposal is an involution: applying it twice returns the start. In practice the momentum is discarded and redrawn from the symmetric $\mathcal{N}(0, I)$ at the next iteration, so its sign never matters for the $\mathbf{x}$ -chain.

12.7. Callouts

i Note

HMC is the default sampler for smooth posteriors over continuous parameters (Stan, PyMC, NumPyro). Use Gibbs for discrete latent variables and random-walk Metropolis only as a baseline or when gradients are unavailable.

💡 Tip

Poor HMC mixing (low ESS, divergences) usually signals bad geometry. Reparameterise to make the posterior more isotropic, or adapt the mass matrix.

! Important

Leapfrog energy errors oscillate but do not drift for stable step sizes. This is why long trajectories remain acceptable — an integrator with energy drift (like Euler) would make acceptance decay with trajectory length.

12.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 13. Neal, *MCMC using Hamiltonian dynamics*, Ch. 5 of Brooks et al., *Handbook of MCMC* (2011): §5.2.1–5.2.3 (Hamiltonian dynamics, properties, leapfrog), §5.3.1–5.3.3 (HMC algorithm, advantages), §5.4.4 (scaling). MacKay, §30.1. Hoffman & Gelman, *The No-U-Turn sampler*, JMLR **15**, 1593 (2014). Betancourt, *A conceptual introduction to HMC*, arXiv:1701.02434.

13. Approximate Bayesian computation

Every method so far — Metropolis, Gibbs, variational Bayes, HMC — requires the **likelihood** $f(\mathbf{y} \mid \mathbf{x})$ to be evaluable (and HMC, its gradient). In many scientific problems it is not: the likelihood may have no closed form, or be too expensive to compute. Often, however, the model is easy to **simulate**: given parameters $\mathbf{x}$ we can generate synthetic data $\mathbf{y} \sim f(\cdot \mid \mathbf{x})$. **Approximate Bayesian computation** (ABC) performs Bayesian inference using simulations alone, by replacing the likelihood with a comparison between simulated and observed data.

13.1. Intractable likelihoods

Typical situations:

- **Latent dynamics.** The observed data are a noisy function of a hidden stochastic process; the likelihood is a sum (or integral) over all hidden trajectories.
- **Implicit models.** The model *is* a simulator — an agent-based model in ecology or epidemiology, a coalescent process in population genetics, a stochastic simulation of a physical system — with no density written down anywhere.
- **Intractable normalisation.** The likelihood is known up to a parameter-dependent constant (e.g. a partition function), so Metropolis ratios cannot be computed.

Example 13.1 (*A hidden Markov model with measurement noise*)

A binary hidden state $z_t \in \{0, 1\}$ evolves as a Markov chain that switches with probability x at each step:

$$P(z_{t+1} \mid z_t) = \begin{pmatrix} 1-x & x \\ x & 1-x \end{pmatrix}.$$

We cannot see z_t ; we measure y_t , which equals z_t with probability γ and is flipped with probability $1 - \gamma$:

$$p_{Y|Z}(0 \mid 0) = p_{Y|Z}(1 \mid 1) = \gamma, \quad p_{Y|Z}(1 \mid 0) = p_{Y|Z}(0 \mid 1) = 1 - \gamma.$$

We want to infer the switching rate x from an observed sequence $y_1, \dots, y_T$.

The likelihood is a sum over all 2^T hidden paths,

$$f(\mathbf{y} \mid x) = \sum_{z_1, \dots, z_T} P(z_1) \prod_{t=1}^{T-1} P(z_{t+1} \mid z_t) \prod_{t=1}^T p_{Y|Z}(y_t \mid z_t),$$

because the observations at different times are correlated through the hidden chain. (For this small model the sum can be organised recursively — the forward algorithm of HMMs — but for richer hidden dynamics no such shortcut exists.) **Simulating** data, on the other hand, is trivial: draw z_1 , flip it with probability x at each step, and corrupt each z_t with probability $1 - \gamma$.

13.2. Exact rejection ABC for discrete data

Suppose the data take values in a discrete set $\mathcal{D}$. Consider the following scheme, repeated to produce samples $x_1, \dots, x_N$:

- Definition 13.1** (Exact-matching ABC). 1. Draw $x' \sim f_{\text{prior}}(x)$.
 2. Simulate $\mathbf{y}' \sim f(\cdot | x')$.
 3. If $\mathbf{y}' = \mathbf{y}$ (the observed data), accept: $x_i = x'$. Otherwise go back to step 1.

Proposition 13.1. *The accepted values are an i.i.d. sample from the exact posterior $f(x | \mathbf{y})$.*

Proof. The probability density of proposing x and accepting it is

$$\sum_{\mathbf{y}' \in \mathcal{D}} f_{\text{prior}}(x) f(\mathbf{y}' | x) \chi(\mathbf{y}' = \mathbf{y}) = f_{\text{prior}}(x) f(\mathbf{y} | x) \propto f(x | \mathbf{y}).$$

The likelihood is never evaluated: it is **sampled**, through the frequency with which x reproduces the data. The overall acceptance probability is $\int f_{\text{prior}}(x) f(\mathbf{y} | x) dx = f(\mathbf{y})$, the evidence. $\square$

Example 13.2 (Coin bias by exact matching)

Observed: $k = 7$ heads in $n = 10$ tosses. Prior $\theta \sim \mathcal{U}(0, 1)$. ABC: draw θ' , toss a θ' -coin ten times, accept if the result has exactly 7 heads. (Matching the *number of heads* rather than the exact sequence is legitimate because k is a sufficient statistic — see below.) The accepted θ 's are distributed as $\text{Beta}(8, 4)$, the exact posterior. The acceptance rate is $f(k = 7) = \int_0^1 \binom{10}{7} \theta^7 (1 - \theta)^3 d\theta = 1/11$. With the full sequence it would be $1/(11 \binom{10}{7}) = 1/1320$ — same posterior, 120 times slower.

Exact matching becomes hopeless as the data grow: the acceptance rate is the evidence, which decays exponentially with the size of the data set. For **continuous** data it is exactly zero, since $P(\mathbf{y}' = \mathbf{y}) = 0$.

13.3. The ABC algorithm

Two approximations make ABC work in general: compare **summary statistics** instead of raw data, and accept **approximate** matches.

Definition 13.2 (ABC rejection algorithm). Choose a summary statistic $\eta : \mathcal{D} \rightarrow \mathbb{R}^s$, a distance (e.g. Euclidean) and a tolerance $\epsilon > 0$. To generate each sample:

1. draw $x' \sim f_{\text{prior}}(x)$;
2. simulate $\mathbf{y}' \sim f(\cdot | x')$;
3. if $\|\eta(\mathbf{y}') - \eta(\mathbf{y})\| < \epsilon$ accept $x_i = x'$; otherwise go back to step 1.

The accepted pairs $(x, \mathbf{y}')$ are distributed as

$$f_{\epsilon}(x, \mathbf{y}' | \mathbf{y}) = \frac{f_{\text{prior}}(x) f(\mathbf{y}' | x) \chi(\|\eta(\mathbf{y}') - \eta(\mathbf{y})\| < \epsilon)}{\int dx d\mathbf{y}' f_{\text{prior}}(x) f(\mathbf{y}' | x) \chi(\|\eta(\mathbf{y}') - \eta(\mathbf{y})\| < \epsilon)},$$

so the samples of x come from the **ABC posterior**

$$f_{\epsilon}(x | \mathbf{y}) = \int d\mathbf{y}' f_{\epsilon}(x, \mathbf{y}' | \mathbf{y}) \propto f_{\text{prior}}(x) P(\|\eta(\mathbf{Y}') - \eta(\mathbf{y})\| < \epsilon | x).$$

The exact likelihood $f(\mathbf{y} | x)$ has been replaced by the probability that a simulation at x lands within ϵ of the observed summaries. The basic intuition is that

a representative (enough) summary statistic η together with a small (enough) tolerance ϵ produces a good (enough) approximation, $f_\epsilon(x | \mathbf{y}) \approx f(x | \mathbf{y})$.

Two sources of error must be kept in mind:

1. **Tolerance.** As $\epsilon \rightarrow 0$, $f_\epsilon(x | \mathbf{y}) \rightarrow f(x | \eta(\mathbf{y}))$, the posterior given the summaries. For $\epsilon > 0$ it is broader: ABC with tolerance ϵ is exact inference for a model in which the observed summaries are corrupted by uniform noise of size ϵ .
2. **Insufficiency.** $f(x | \eta(\mathbf{y}))$ equals $f(x | \mathbf{y})$ only if η is **sufficient**; otherwise information is lost, and the ABC posterior is wider (or even biased).

13.3.1. Summary statistics and sufficiency

Definition 13.3 (Sufficient statistic). $\eta(\mathbf{y})$ is **sufficient** for x if $f(\mathbf{y} | x)$ depends on $\mathbf{y}$ only through $\eta(\mathbf{y})$ (more precisely, $f(\mathbf{y} | x) = g(\eta(\mathbf{y}), x)h(\mathbf{y})$ — Fisher–Neyman factorisation). Equivalently, $f(x | \mathbf{y}) = f(x | \eta(\mathbf{y}))$ for every prior; in information-theoretic terms, $I(X; \mathbf{Y}) = I(X; \eta(\mathbf{Y}))$ — equality in the data-processing inequality of Chapter 2.

For i.i.d. Bernoulli data the number of successes is sufficient; for i.i.d. Gaussian data with unknown mean and variance, $(\sum y_i, \sum y_i^2)$ is. For complex simulators sufficient statistics of low dimension rarely exist, and η is chosen by insight — moments, quantiles, autocorrelations, counts of characteristic events — or learned by regression or neural networks.

There is a trade-off: adding statistics reduces information loss but increases the dimension s , and the acceptance rate of $\|\eta' - \eta\| < \epsilon$ decays like ϵ^s — the curse of dimensionality strikes again.

Example 13.3 (Summaries for the noisy HMM)

For the hidden Markov model above, natural summaries of the observed sequence are

- the fraction of ones, $\bar{y}$ (uninformative here: by symmetry it is $\approx 1/2$ for any x);
- the number of switches $s(\mathbf{y}) = \sum_t \chi(y_{t+1} \neq y_t)$, which increases with x ;
- the lag-1 autocorrelation of y .

Since the hidden state has lag-1 autocorrelation $1 - 2x$ and each noisy observation multiplies it by $(2\gamma - 1)$, the expected switching frequency of y is $\mathbb{E}[s]/(T - 1) = \frac{1}{2}(1 - (1 - 2x)(2\gamma - 1)^2)$. ABC with $\eta = s(\mathbf{y})/(T - 1)$ therefore concentrates around x values that reproduce the observed switching rate. If $\gamma = 1/2$ the data carry no information about x and the ABC posterior correctly reproduces the prior.

13.4. Beyond rejection: ABC-MCMC and ABC-SMC

Plain rejection wastes most simulations on parameters drawn from the prior, far from the posterior. Two standard improvements:

Definition 13.4 (ABC-MCMC (Marjoram et al. 2003)). From the current x , propose $x' \sim q(\cdot | x)$, simulate $\mathbf{y}' \sim f(\cdot | x')$, and accept with probability

$$\min\left\{1, \frac{f_{\text{prior}}(x') q(x | x')}{f_{\text{prior}}(x) q(x' | x)}\right\} \cdot \chi(\|\eta(\mathbf{y}') - \eta(\mathbf{y})\| < \epsilon).$$

This is Metropolis–Hastings on the augmented target $f_\epsilon(x, \mathbf{y}' \mid \mathbf{y})$, using the simulator as part of the proposal; the intractable likelihoods cancel. Its stationary distribution is $f_\epsilon(x \mid \mathbf{y})$.

Definition 13.5 (ABC-SMC). Run a population of particles through a decreasing sequence of tolerances $\epsilon_1 > \epsilon_2 > \dots > \epsilon_T$. At each stage, resample particles from the previous population, perturb them, keep those whose simulations fall within the new tolerance, and reweight to correct for the perturbation. The population is gradually guided from the prior to the region of high posterior mass.

Remark (Simulation-based inference). Modern **neural** methods train a network on simulated pairs $(x, \mathbf{y})$ to approximate the posterior $f(x \mid \mathbf{y})$ (neural posterior estimation), the likelihood, or the likelihood ratio directly. They can be seen as ABC with learned summary statistics and a learned (variational, cf. Chapter 10) posterior, and are far more simulation-efficient in high dimension.

13.5. Applications

Example 13.4 (Population genetics)

Coalescent models generate gene genealogies whose likelihood involves an intractable sum over genealogies, but which are easy to simulate. ABC was developed in this field (Tavaré et al. 1997; Pritchard et al. 1999; Beaumont et al. 2002) to infer population sizes, migration rates and split times from DNA data — the same kind of question as the grey-seal problem of Chapter 9, but with demographic history included.

Example 13.5 (Epidemics and ecology)

Stochastic compartmental and agent-based models of epidemics and of population dynamics are straightforward to simulate but have intractable likelihoods, especially with partial observation. ABC-SMC and neural simulation-based inference are routinely used to fit them to incidence time series.

13.6. Exercises

Exercise 13.1

Show that ABC rejection with $\epsilon = 0$ has acceptance probability zero for continuous data, and that for discrete data its acceptance probability equals the evidence $f(\mathbf{y})$.

Solution

For continuous $\mathbf{y}$, $P(\mathbf{y}' = \mathbf{y} \mid x) = 0$ for every x . For discrete data, $P(\text{accept}) = \int dx f_{\text{prior}}(x) P(\mathbf{y}' = \mathbf{y} \mid x) = \int dx f_{\text{prior}}(x) f(\mathbf{y} \mid x) = f(\mathbf{y})$. (This even gives a — very inefficient — Monte Carlo estimator of the evidence.)

Exercise 13.2

For $\mathbf{y} \mid \theta \sim \mathcal{N}(\theta \mathbf{1}, I_n)$ with prior $\theta \sim \mathcal{N}(0, \tau^2)$, show that ABC with $\eta = \bar{y}$ recovers the exact posterior as $\epsilon \rightarrow 0$, and compute the ABC posterior for finite ϵ when the uniform window is replaced by a Gaussian kernel of width ϵ .

Solution

$\bar{y}$ is sufficient ($f(\mathbf{y} \mid \theta) \propto e^{-n(\bar{y}-\theta)^2/2} \times (\text{terms without } \theta)$), so $f(\theta \mid \bar{y}) = f(\theta \mid \mathbf{y})$ and the $\epsilon \rightarrow 0$ limit is exact. With a Gaussian kernel, the ABC “likelihood” is $P(\bar{y}' \approx \bar{y} \mid \theta) \propto \mathcal{N}(\bar{y}; \theta, 1/n + \epsilon^2)$: the ABC posterior is the exact posterior for a model with noise variance inflated from $1/n$ to $1/n + \epsilon^2$, i.e. $\mathcal{N}(\frac{\bar{y}/(1/n+\epsilon^2)}{1/\tau^2+1/(1/n+\epsilon^2)}, (1/\tau^2 + 1/(1/n + \epsilon^2))^{-1})$. The tolerance should be small compared to the statistical uncertainty $1/\sqrt{n}$.

Exercise 13.3

In the noisy HMM, why is $\bar{y}$ a useless summary statistic for x ? Suggest a statistic and explain how it depends on x and γ .

Solution

The hidden chain is symmetric, so its stationary distribution is $(1/2, 1/2)$ for every x , and the symmetric noise preserves this: $\mathbb{E}[\bar{y}] = 1/2$ regardless of x . The switching frequency of $\mathbf{y}$ depends on x : $\mathbb{E}[s]/(T-1) = \frac{1}{2}[1 - (1-2x)(2\gamma-1)^2]$. It increases with x (for $x < 1/2$) and its sensitivity to x is proportional to $(2\gamma-1)^2$, vanishing for completely noisy measurements.

13.7. Callouts

Note

ABC turns a simulator into an inference engine. It is the natural tool whenever the model is *defined* by a simulation rather than by a density.

Tip

Validate an ABC pipeline on synthetic data generated with known parameters, and check that the posterior is stable as ϵ decreases and as summary statistics are added or removed.

Warning

ABC posteriors are approximations with two uncontrolled errors — tolerance and insufficiency. Always report ϵ , the summaries used and the acceptance rate.

13.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 14. Marin, Pudlo, Robert & Ryder, *Approximate Bayesian computational methods*, Stat. Comput. **22**, 1167 (2012). Sunnåker et al., *Approximate Bayesian computation*, PLoS Comput. Biol. **9**, e1002803 (2013). Marjoram, Molitor, Plagnol & Tavaré, *Markov chain Monte Carlo without likelihoods*, PNAS **100**, 15324 (2003). Beaumont, Zhang & Balding, *Genetics* **162**, 2025 (2002). Cranmer, Brehmer & Louppe, *The frontier of simulation-based inference*, PNAS **117**, 30055 (2020).

